

ASSESSMENT OF THE MEDITERRANEAN SWORDFISH STOCK BY MEANS OF EXTENDED SURVIVORS ANALYSIS (XSA)

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SUMMARY

Updated assessments of the Mediterranean swordfish stock are performed by means of Extended Survivors Analysis (XSA). They include a continuity run employing the model settings used in the 2016 ICCAT assessment, as well as candidate runs using modified model settings and two different assumptions regarding natural mortality-at-age. Additionally, scenarios assuming discards have been considered. All assessment runs indicate that the stock is overexploited and the current fishing mortality rates are higher than those corresponding to maximum sustainable yield (MSY) levels.

Introduction

Extended survival analysis (XSA) (Shepherd, 1992) is a variant of Virtual Population Analysis (VPA) which derives population abundance using estimates of terminal year numbers for each cohort, either the year in which it reaches the oldest age, or the final year of the assessment. The derived population numbers by year and independent Catch Per Unit Effort (CPUE) indices are used to estimate catchability for each CPUE series allowing predictions of N to be obtained by year, age and series. Weighted averages of the predicted values of N for a particular cohort are then used to estimate the terminal populations (the oldest age in each cohort) after adjustment for mortality. Inverse variance weighting is used to down-weight data from CPUE indices with poor fits. In addition shrinkage to the mean is performed, where the terminal N s also including a term related to recent F s or F s at younger ages (shrinkage to the mean F) and numbers-at-age for recruiting age classes are estimated from the geometric mean of recent recruitment (shrinkage to the mean n). Time series weights can be applied to discount past values. Since estimates of N are a function of the terminal populations, XSA iterates until the difference in estimates of fishing mortality between updates of the terminal populations is small. XSA has been used for formulating advice in previous assessments of the Mediterranean swordfish stock (ICCAT, 2105; 2017). In the present document, XSA assessments using different configurations have been performed and compared.

Materials and Methods

The assessments were conducted using XSA in R using the FLXSA package (part of the FLR-project, Kell et al., 2007; <http://www.flr-project.org/>). The available catch-at-size (CAS) data covered the period 1972-2018 and the catch at age (CAA) data were generated using a statistical approach similarly to the two previous assessments. In those past assessments, the statistical method was found to be preferable to the deterministic age slicing approach. The available CAS did not justify splitting CAS into ages greater than 6 and in line with the previous assessment XSA runs were conducted with a plus group of 6. It should be noted, however, that serious updates have been performed in the historical CAS data since the previous assessment.

Four standardized CPUE series extending up to the latest years were considered: Greek longliners (SCRS 2020/021), Moroccan longliners (SCRS 2020/026), Spanish longliners (SCRS 2020/043) and Ligurian longliners (SCRS 2020/027). The standardized CPUE indices were not differentiated by age and were considered to be representative of the 2-4 age-group abundances. Fleet catchability was assumed to be independent of year-class size for all terminal years. In the 2016 assessment it was assumed that fish first mature at age 3 (when 50% are mature) and are fully mature at older ages. Considering new information (ICCAT, 2020), it was currently assumed 15% and 65% maturation at ages 2 and 3 respectively. Full maturity was considered from age 4 onward. Weights-at-age were derived from the mixture analysis and were consistent with the CAA. The F range ($\bar{f}$) was set to ages 2-4, as in the previous assessments. Figures 1 and 2 illustrate the available CAA data by age and year.

Specific details regarding the configuration of the different XSA runs are provided below:

a. Continuity run

This run was applied on the CAA data from 1985 onward by adopting the same assumptions with the 2016 continuity assessment: (a) natural mortality equal to 0.2 for all ages and years, (b) tuning based on three standardized CPUE series (Greek, Spanish and Moroccan longliners and the following XSA control settings: `shk.n=TRUE`, `shk.yrs=10`, `rage=1`, `maxit=100`, `shk.f=TRUE`, `shk.ages=2`, `fse=0.5`, `min.nse=0.3`, `qage=6`, `window=100`, `tsrange=10`, `tspower=1`, `vpa=TRUE`, F range: `minfbar=2`, `maxfbar=4`).

b. Candidate runs

Preliminary trials with the full CAA dataset revealed unusual estimates for the earliest period. Given that, and the fact that the earliest CPUE series starts in 1987, it was decided to apply the candidate runs from 1985 onward, similarly to the previous assessments (ICCAT, 2015; 2017). Various candidate models were tested using all available CPUE series (four in total) and most of them provided comparable results, at least for the most recent years. Given the very irregular pattern regarding the presence of older fish in the

catch, particularly in the latest years, it was decided to adopt in the final runs a shrinkage factor of 4 for the terminal age, instead of 2 used in the continuity run, and also increase the “tsrange” rate (number of years to be used in the time series weighting) to 20 instead of 10. These adjustments have also improved the robustness of the model estimates as it was revealed from the retrospective plots. All other settings remained the same with those used in the continuity run. In line with the 2016 assessments two different assumptions were applied regarding natural mortality in the candidate runs: either constant $M=0.2$ or an M by age vector, estimated according to the Lorenzen formulation followed in the 2016 assessment.

The same candidate runs were also applied to a CAA data set including discards, based on the scenario developed in the SCRS 2020/028 document.

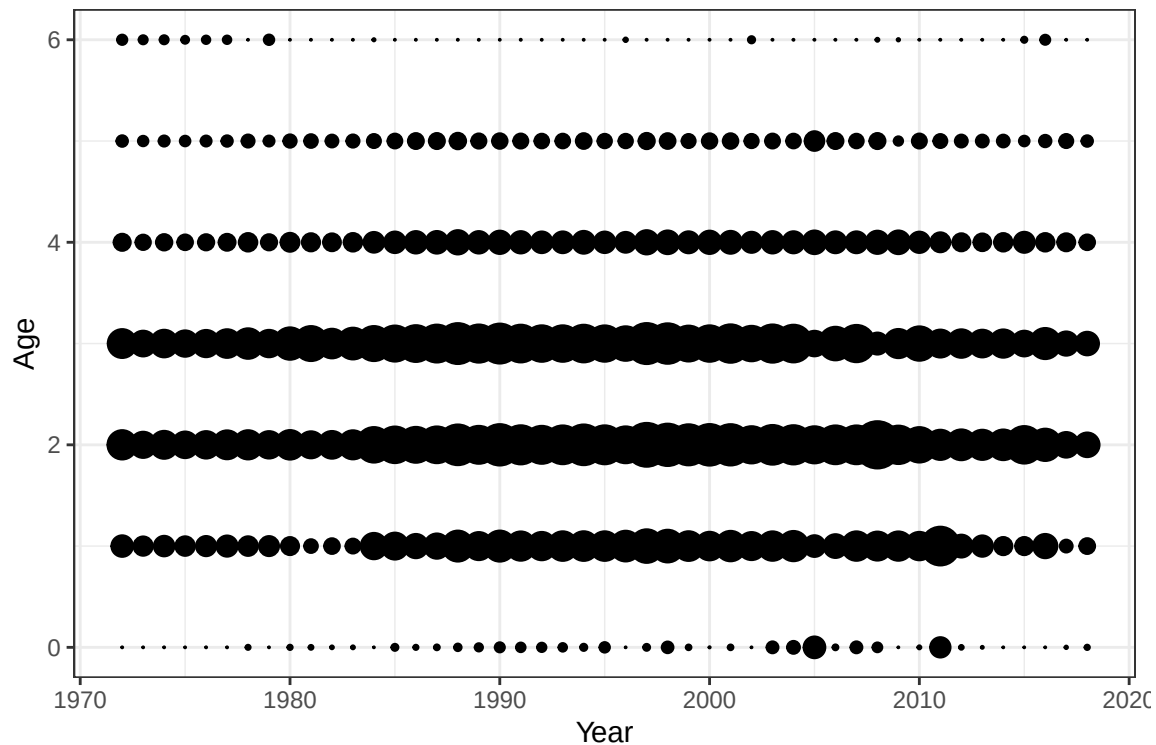


Figure 1. Catch-at-age by year. Bubbles are proportional to abundance.

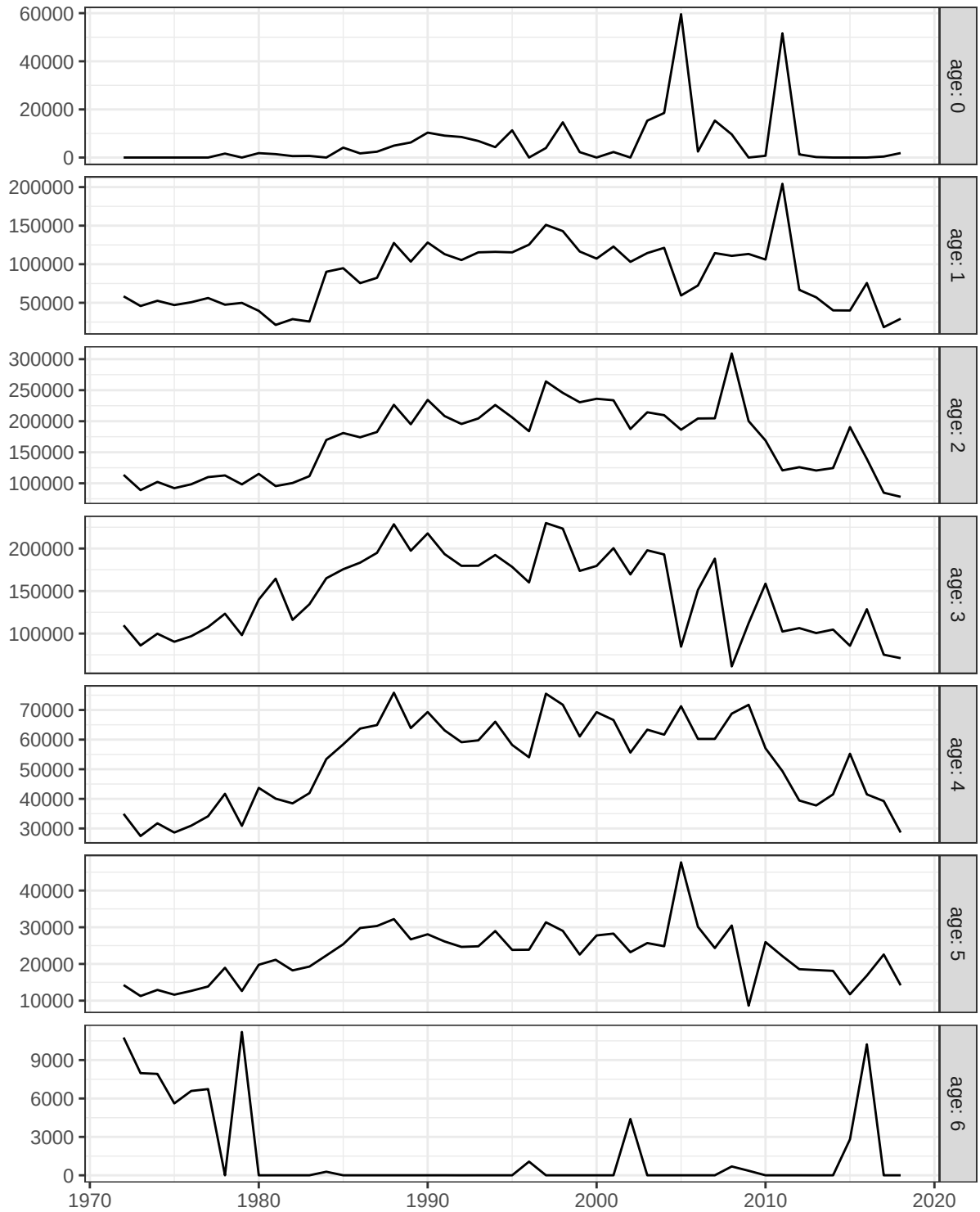


Figure 2. Time series of catch-at-age.

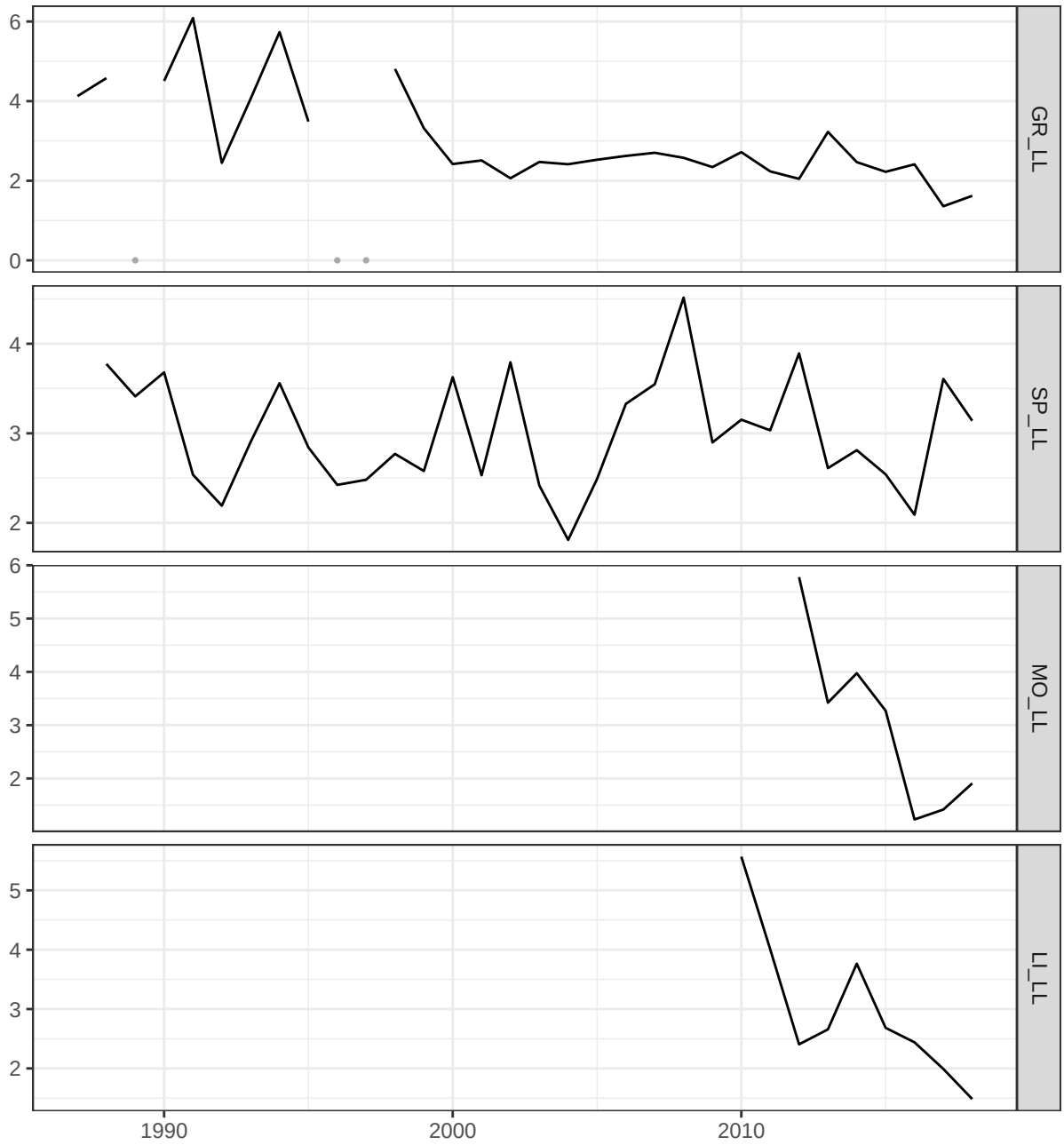


Figure 3. Time series of scaled standardized CPUE indexes by fleet: GR_LL= Greek longline, SP_LL=Spanish longline, MO_LL=Moroccan longline, LI_LL=Ligurian Longline.

Results

a. Continuity run

Model estimates are plotted in **Figures 4-5**. The various diagnostic plots are shown in **Figures 6-14**, while **Figure 15** shows the retrospective plot.

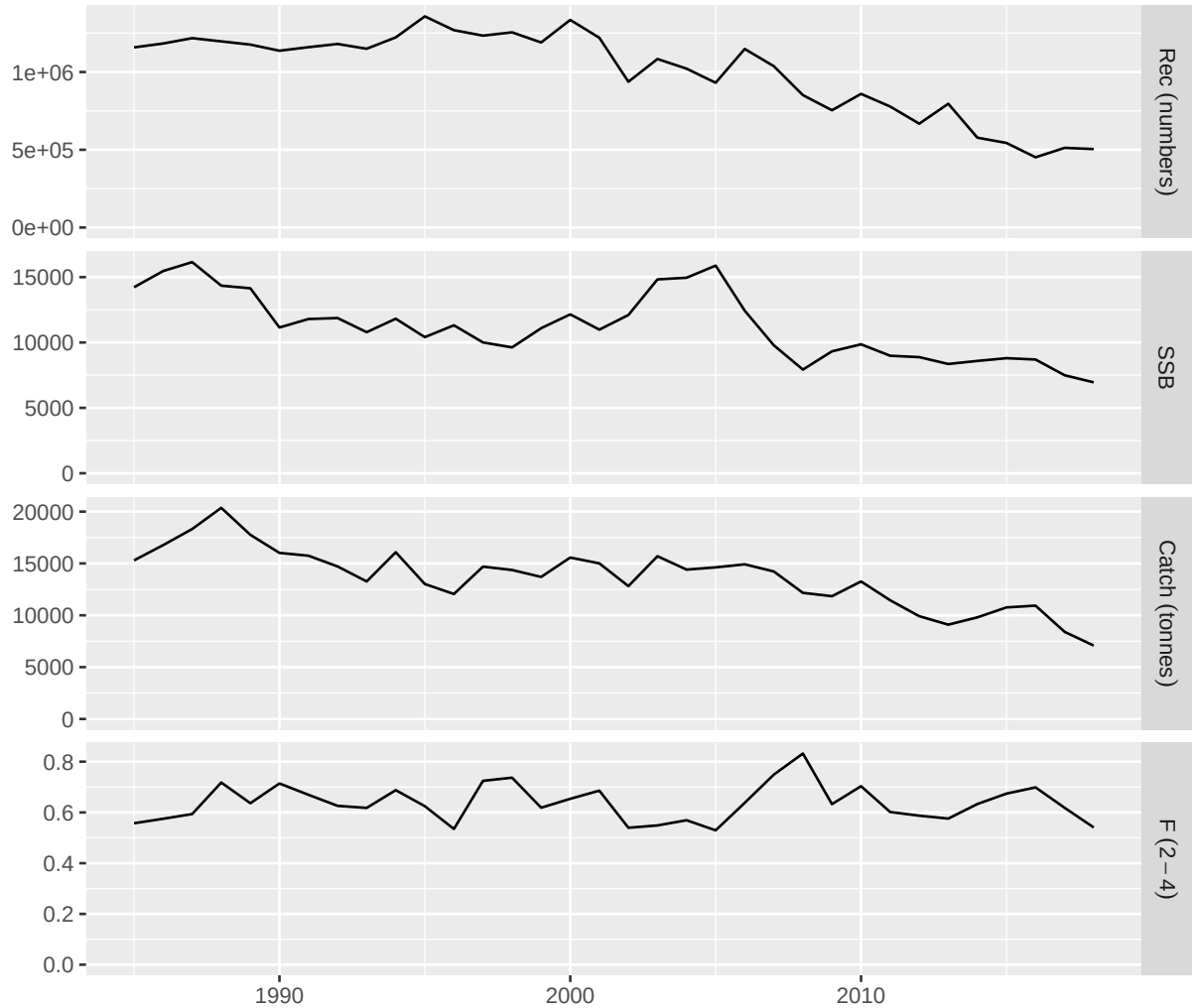


Figure 4. XSA time series estimates for the continuity run.

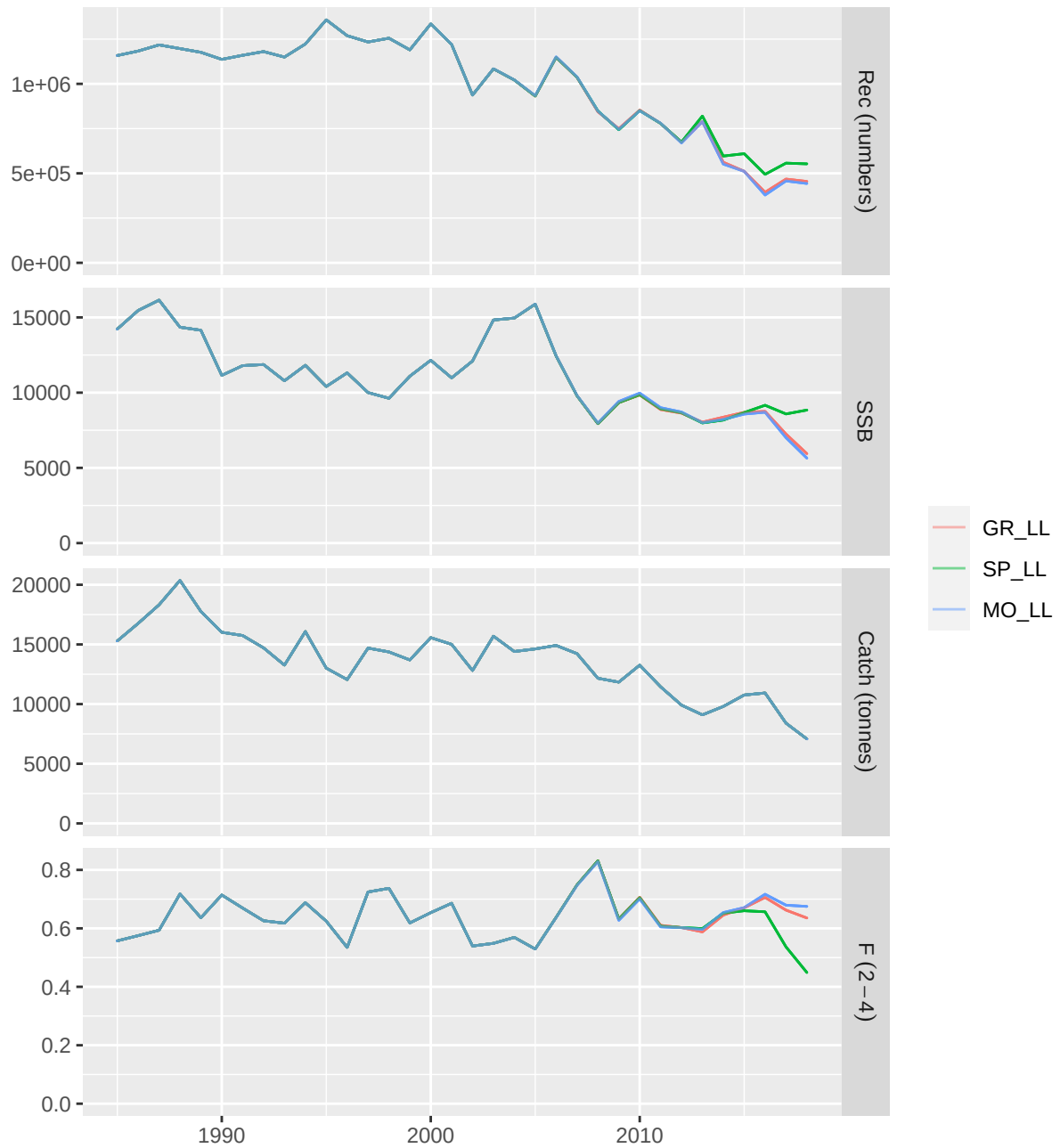


Figure 5. XSA time series estimates by CPUE for the continuity run

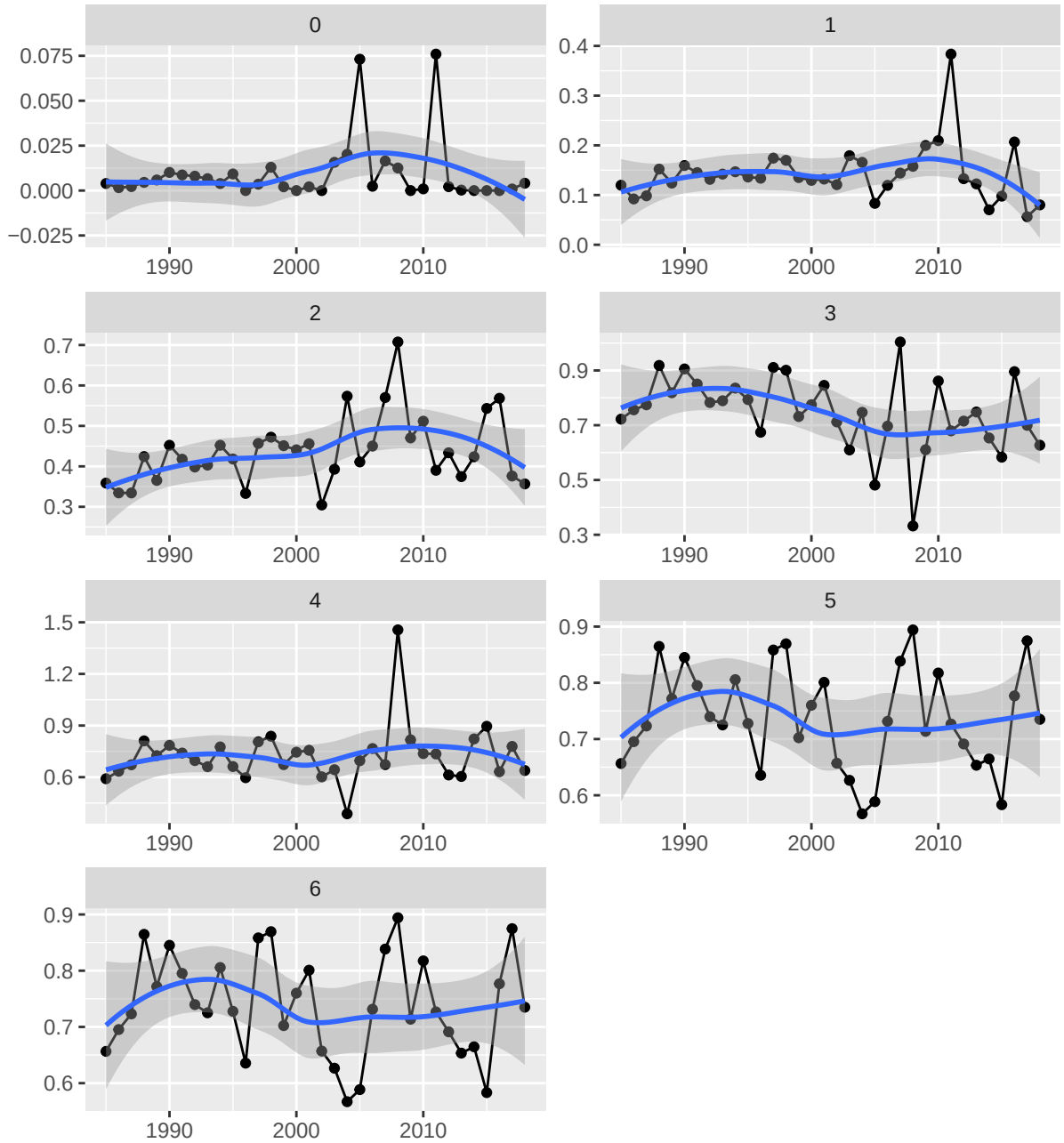
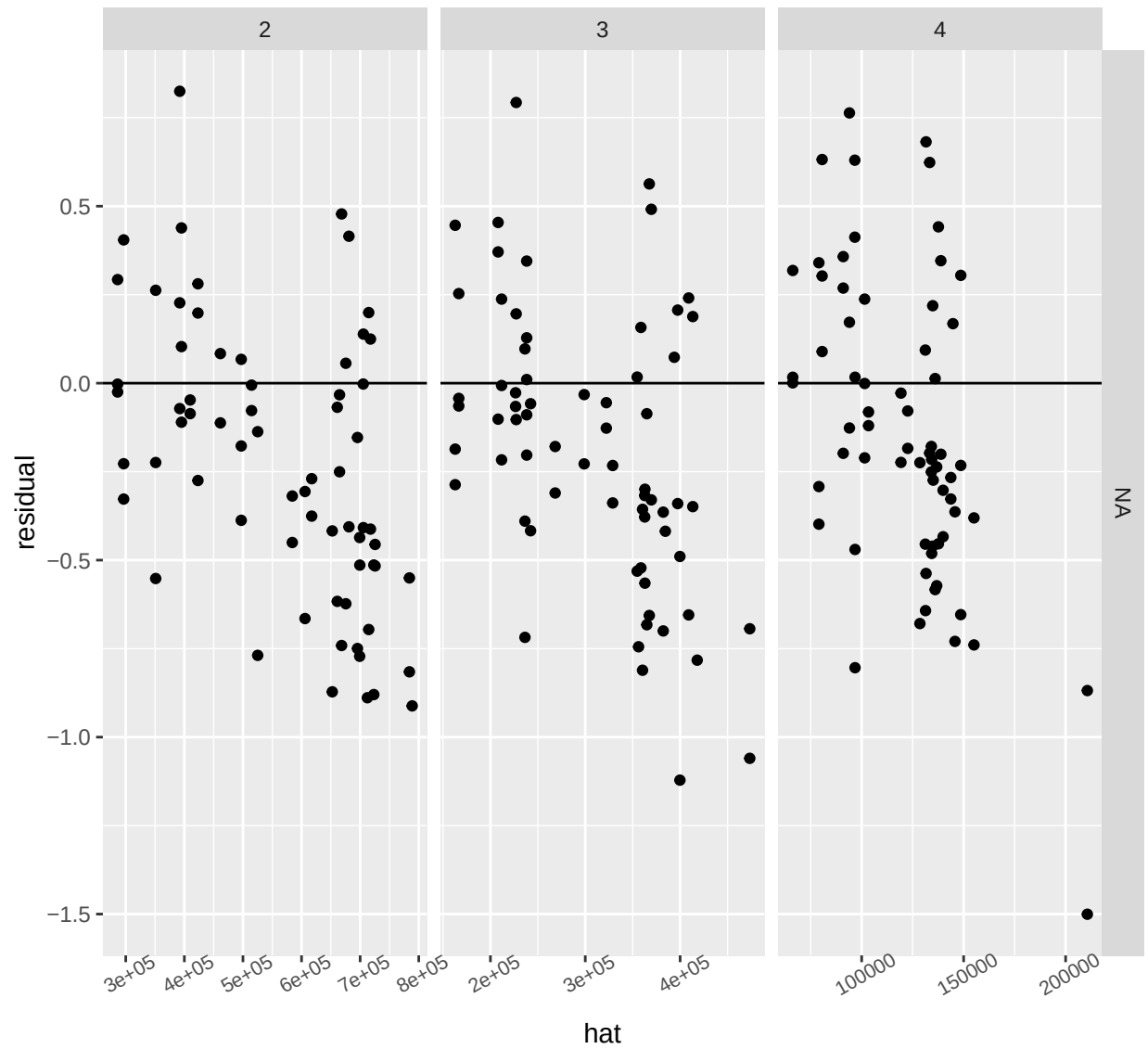


Figure 6. Estimates of harvest-at-age by year for the continuity run; continuous lines represent lowest smoothers.

Diagnostic plots**Figure 7.** Residuals against fitted value for the continuity run.

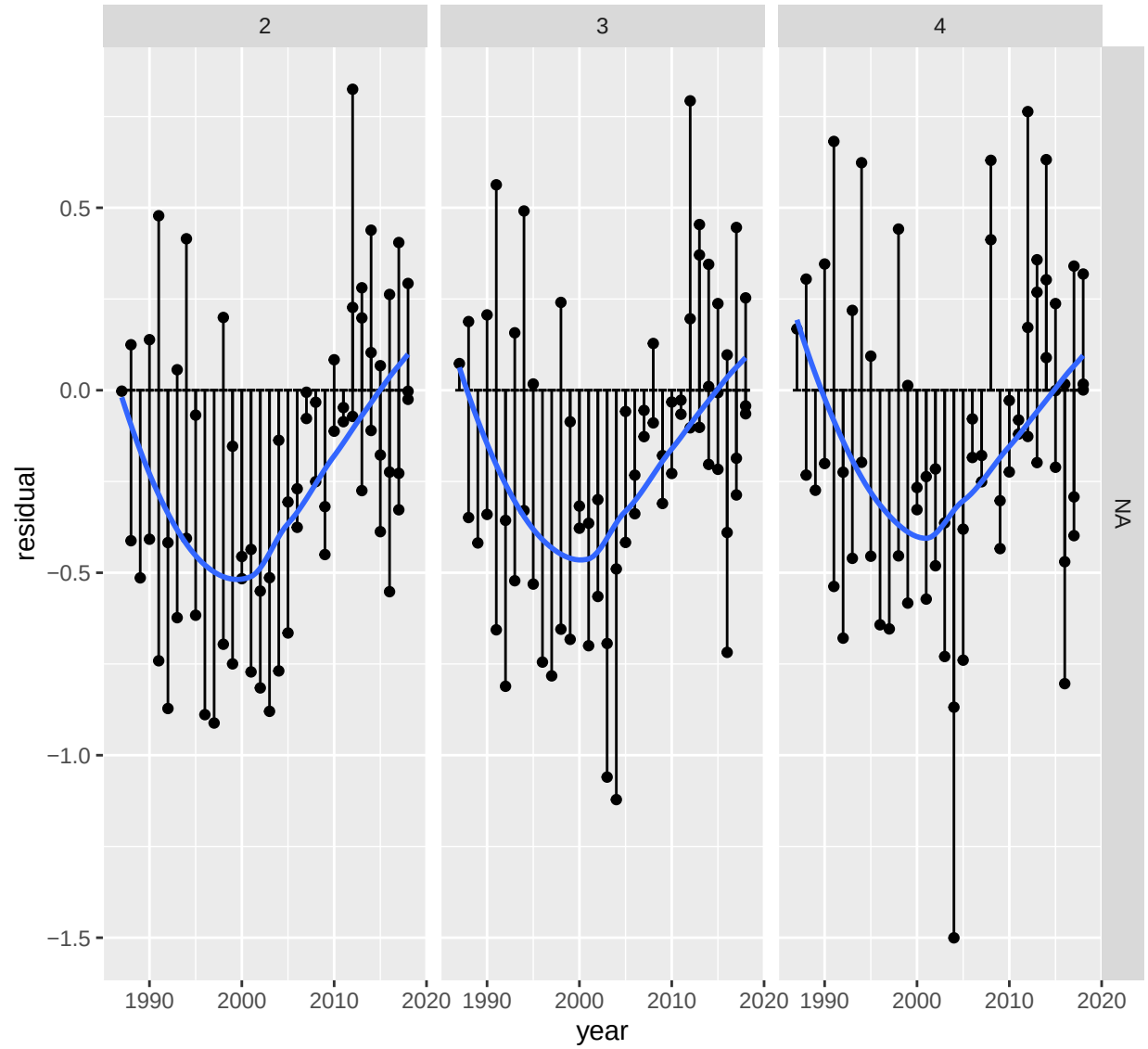


Figure 8. Residuals by fleet against year for the continuity run.

```
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```

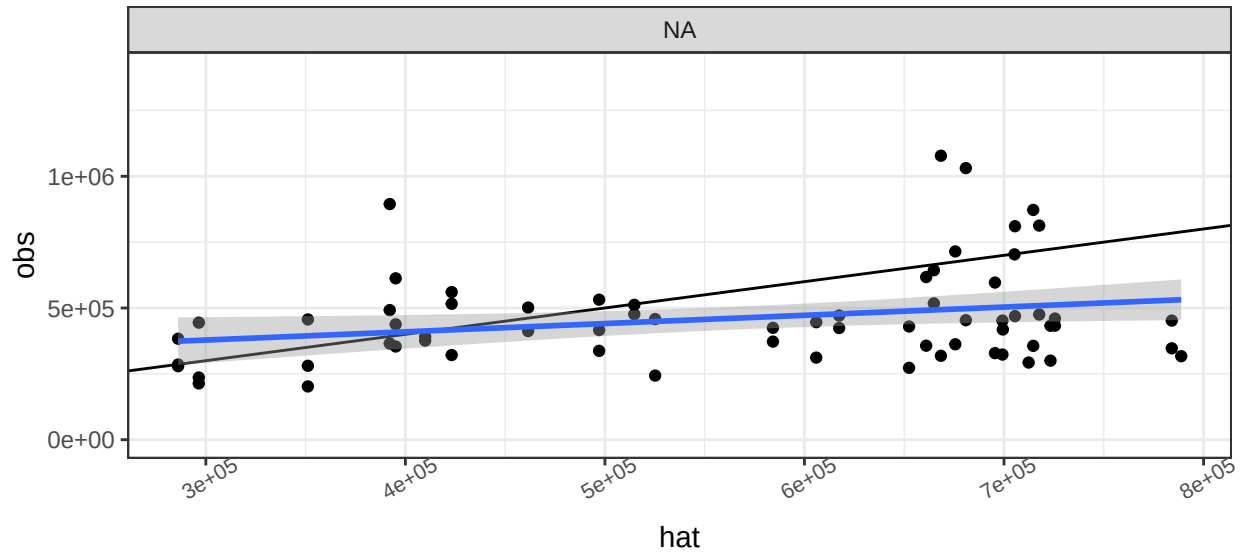


Figure 9. Calibration regression plots for age 2 for the continuity run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
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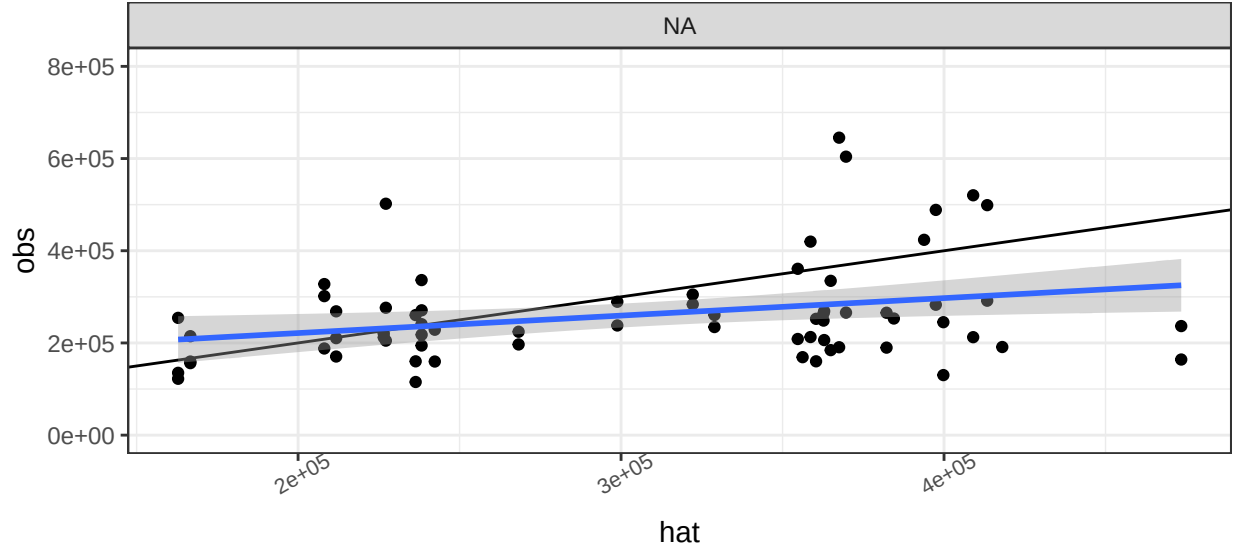


Figure 10. Calibration regression plots for age 3 for the continuity run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
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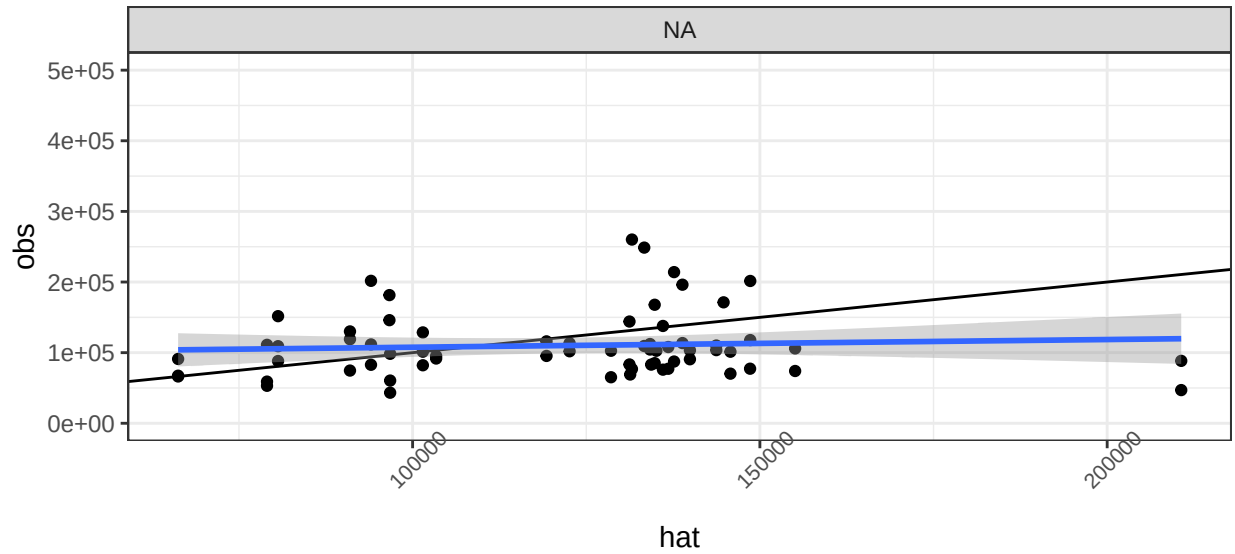


Figure 11. Calibration regression plots for age 4 for the continuity run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
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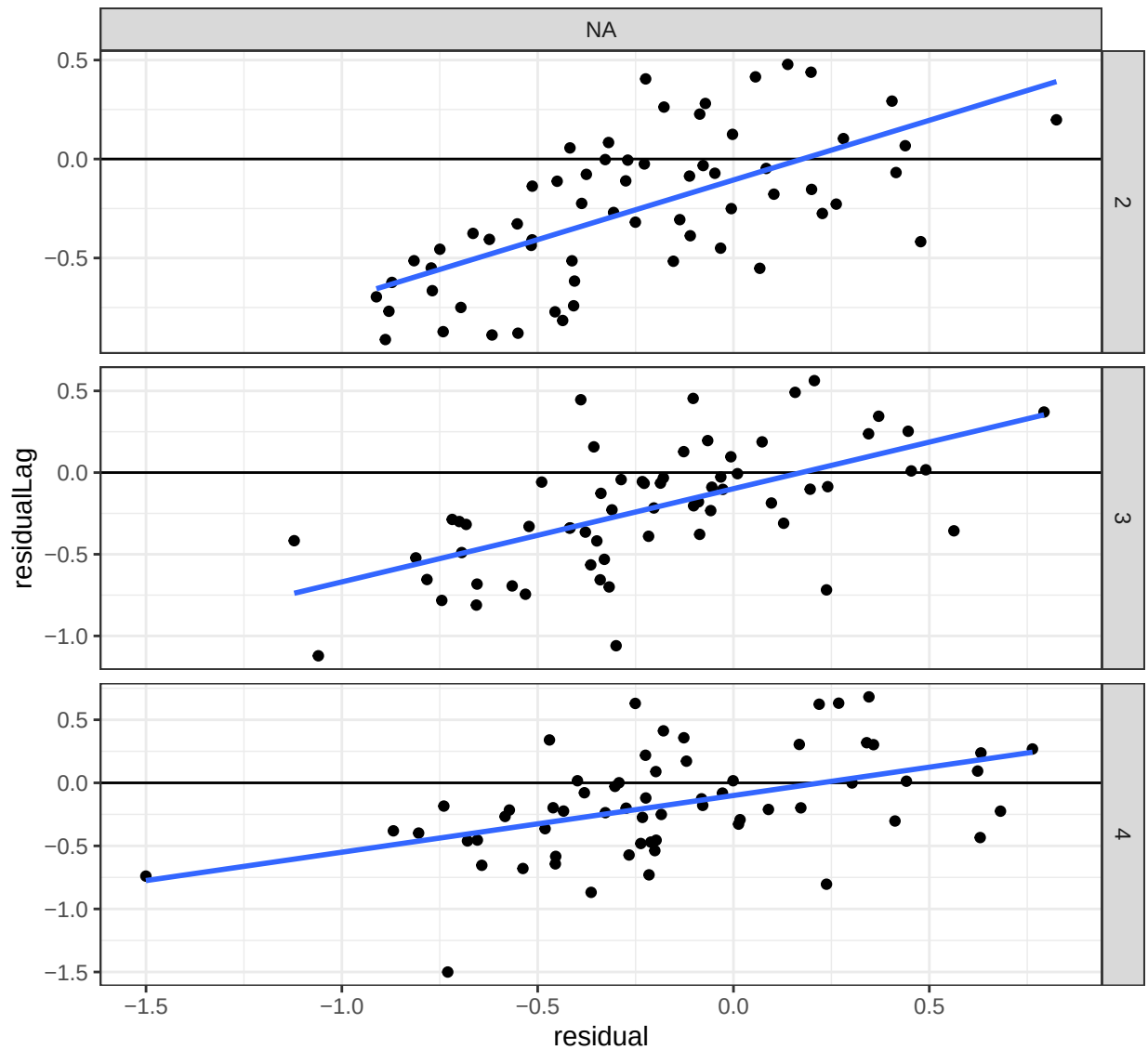


Figure 12. AR plots of lagged residuals for the continuity run. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

```
## `geom_smooth()` using formula 'y ~ x'
```

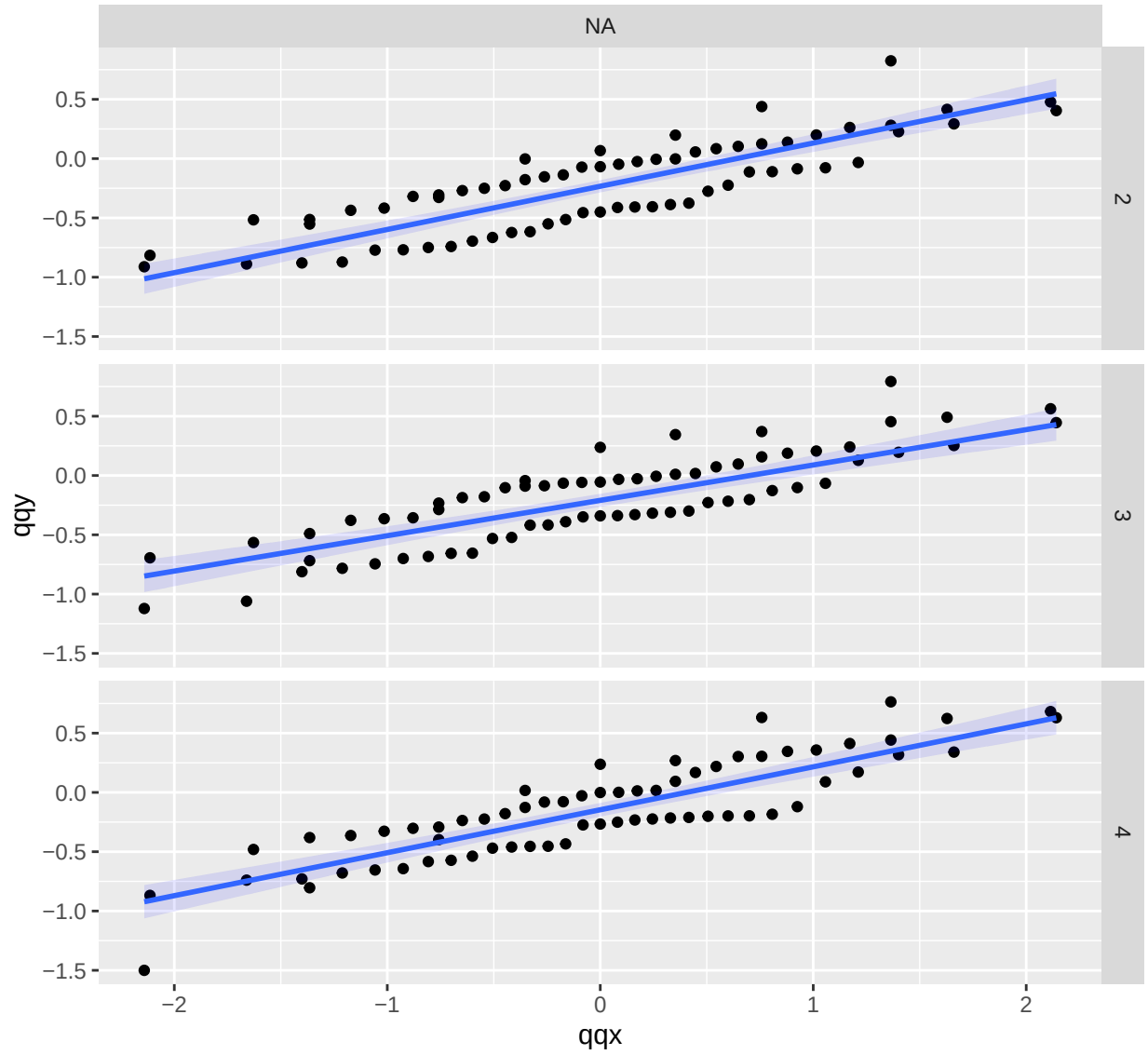


Figure 13. QQ plots to check for log normality (i.e. points and line should coincide) for the continuity run.

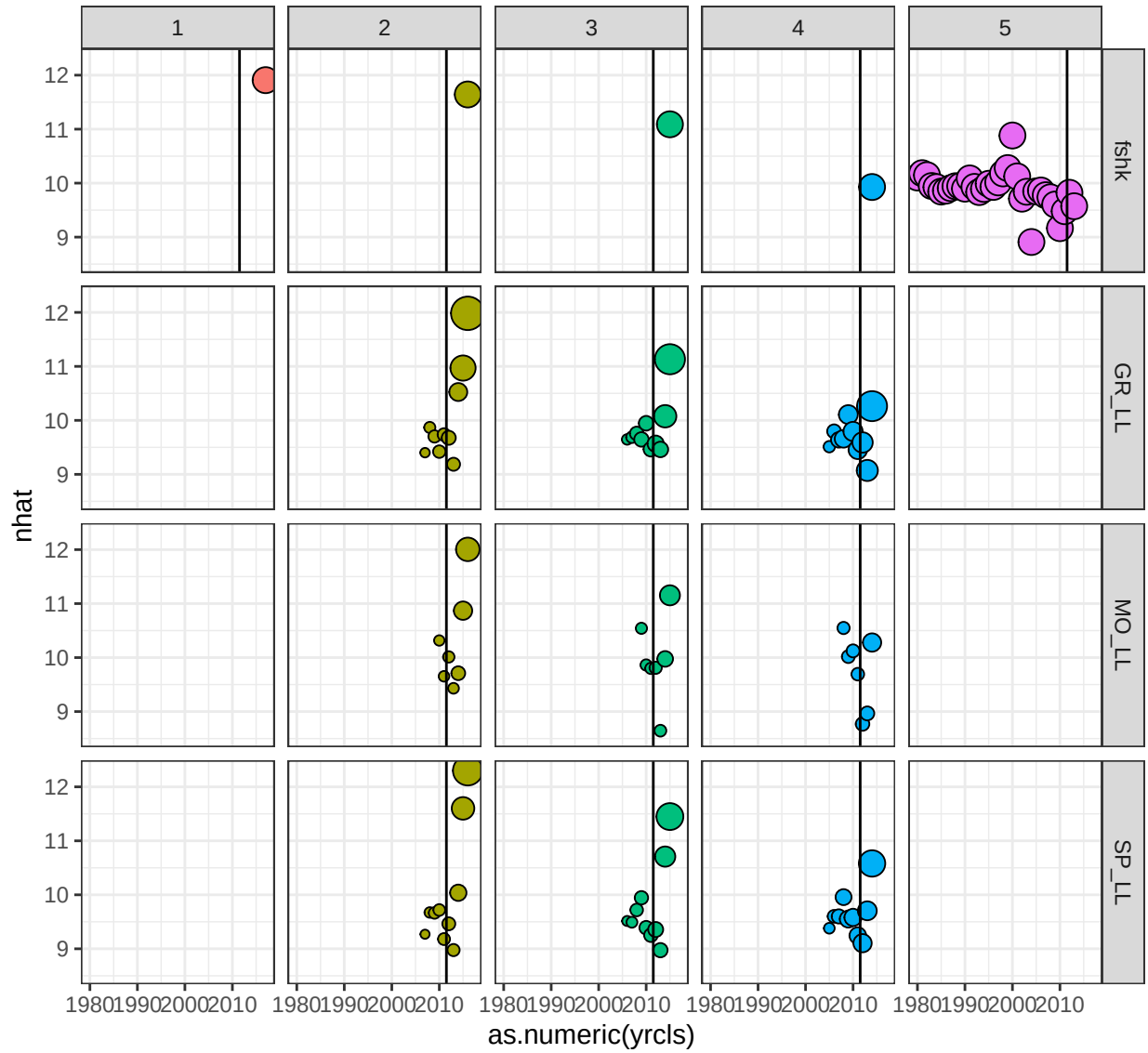


Figure 14. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(fshk)$ for the continuity run. Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

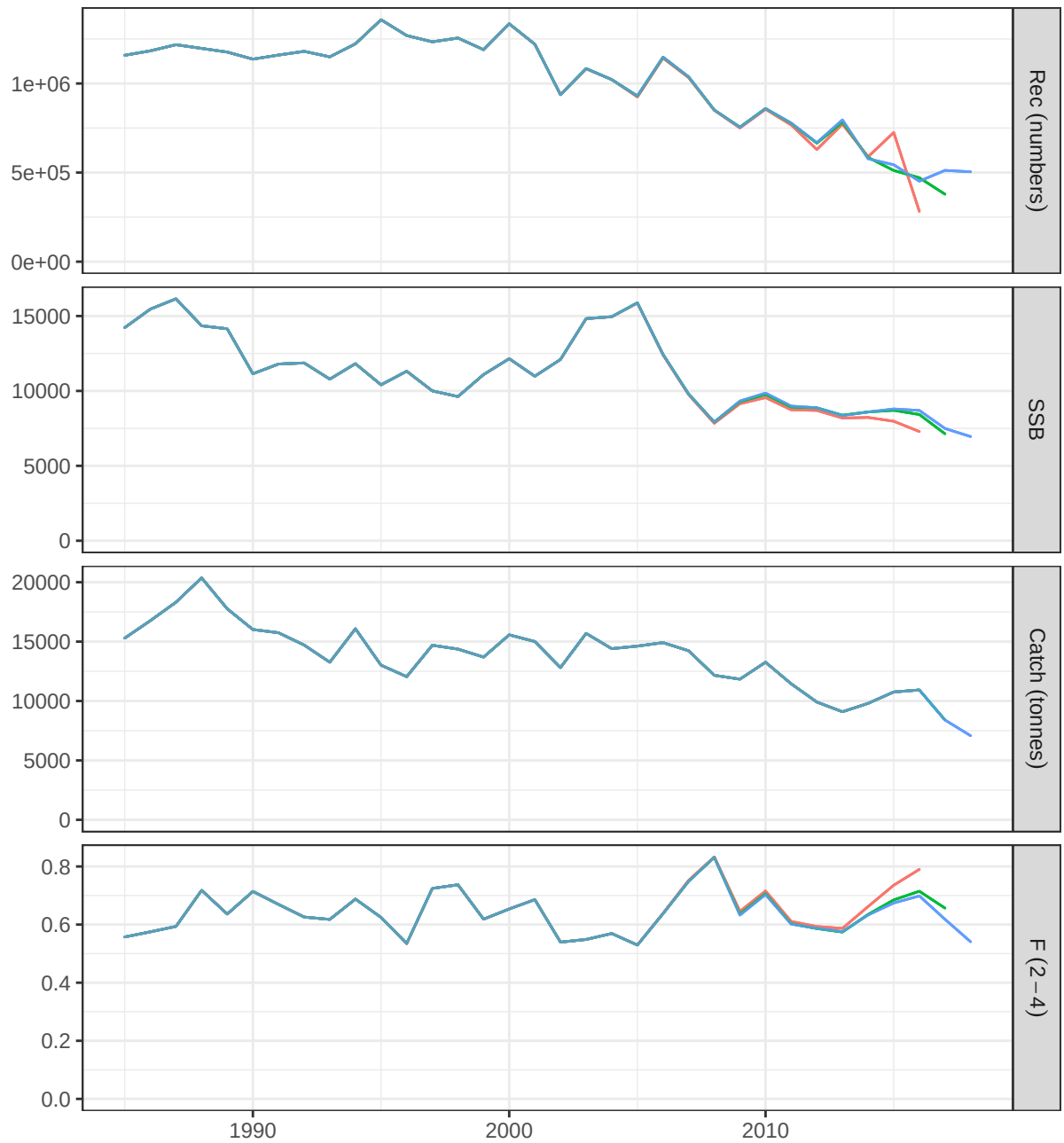


Figure 15. Retrospective plot for the continuity run.

b. Candidate run - constant M

Model estimates are plotted in **Figures 16-17**. The various diagnostic plots are shown in **Figures 18-26**, while **Figure 27** shows the retrospective plot.

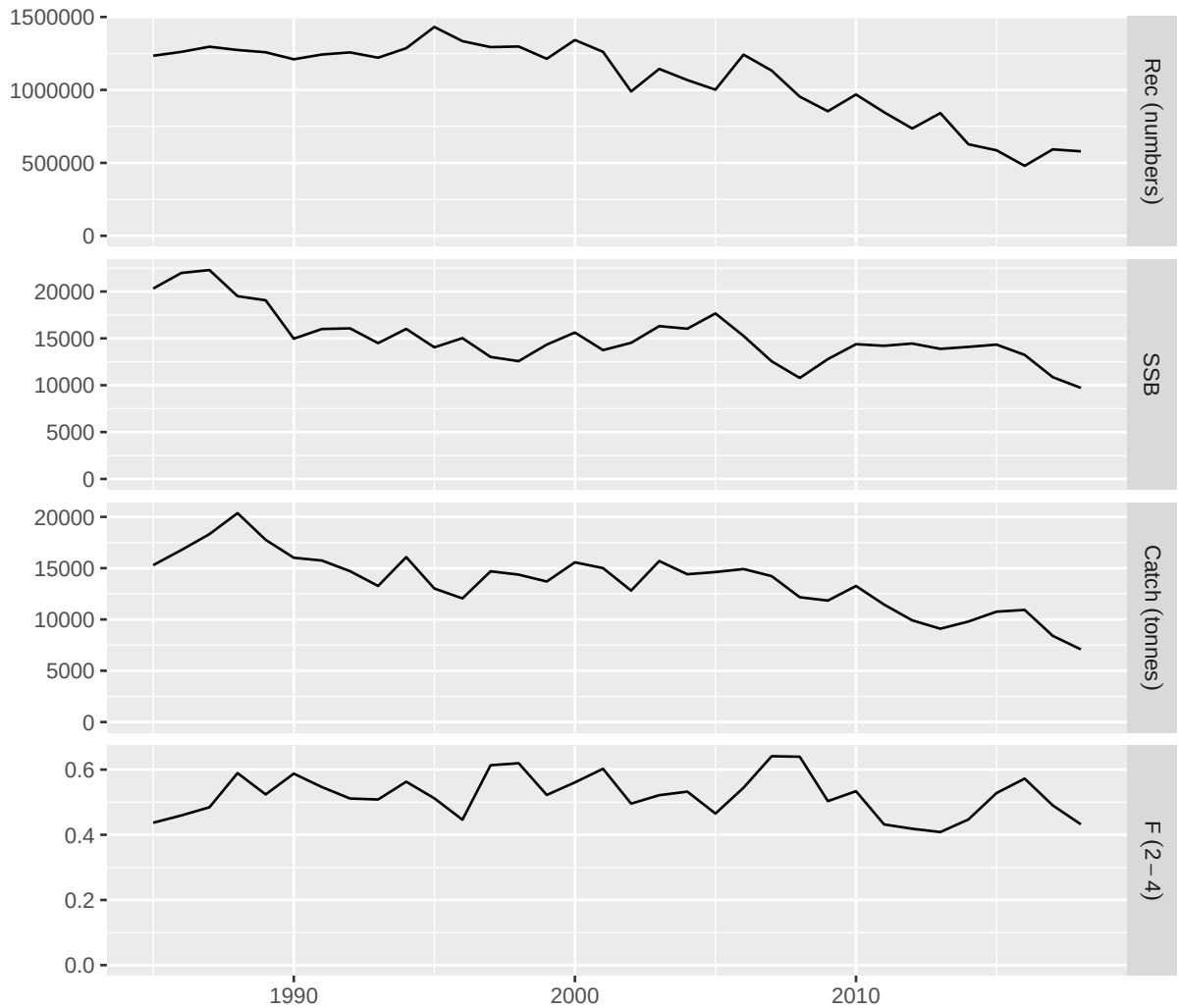


Figure 16. XSA time series estimates for the candidate run assuming constant M.

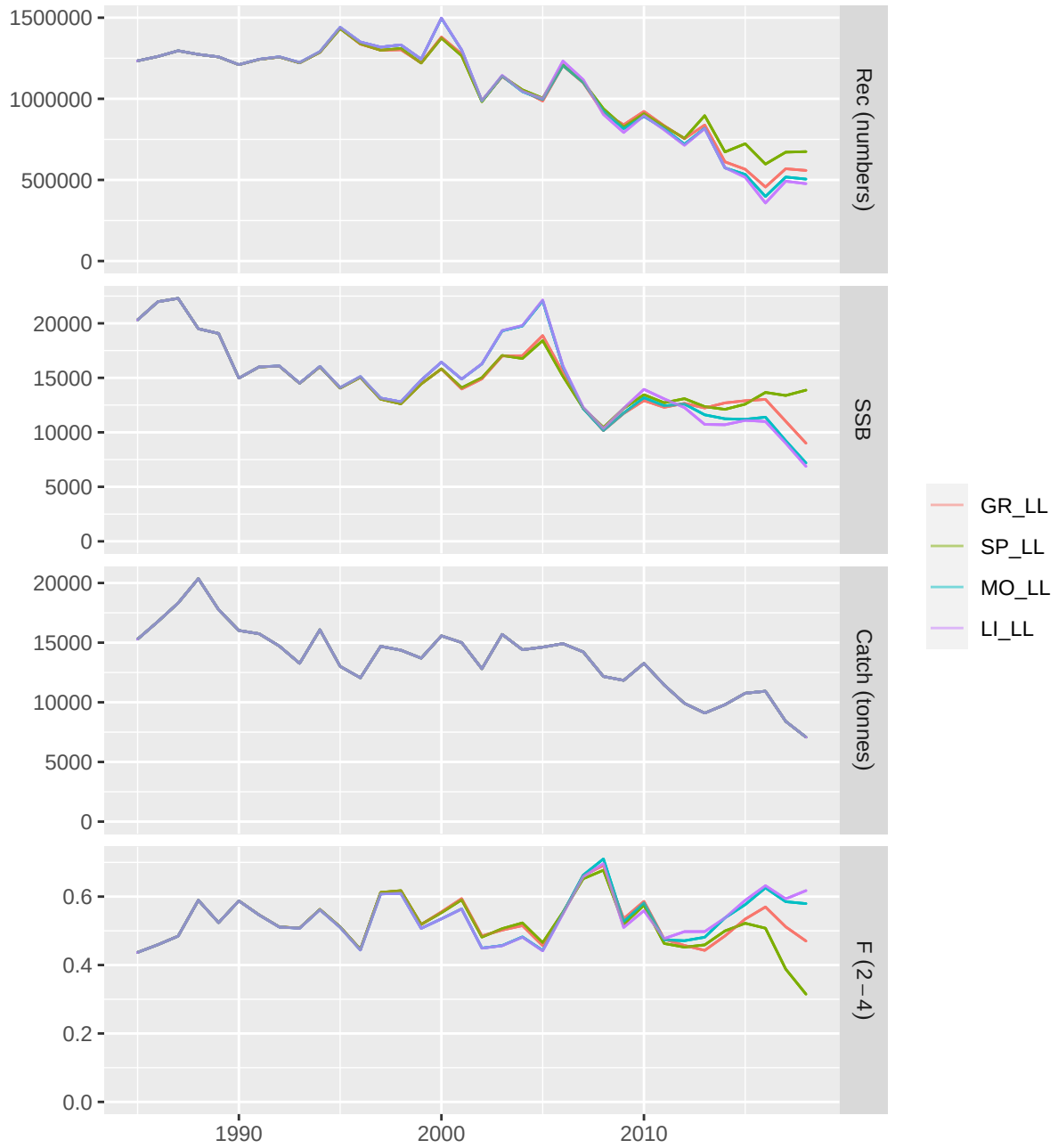


Figure 17. XSA time series estimates by CPUE, for the candidate run assuming constant M.

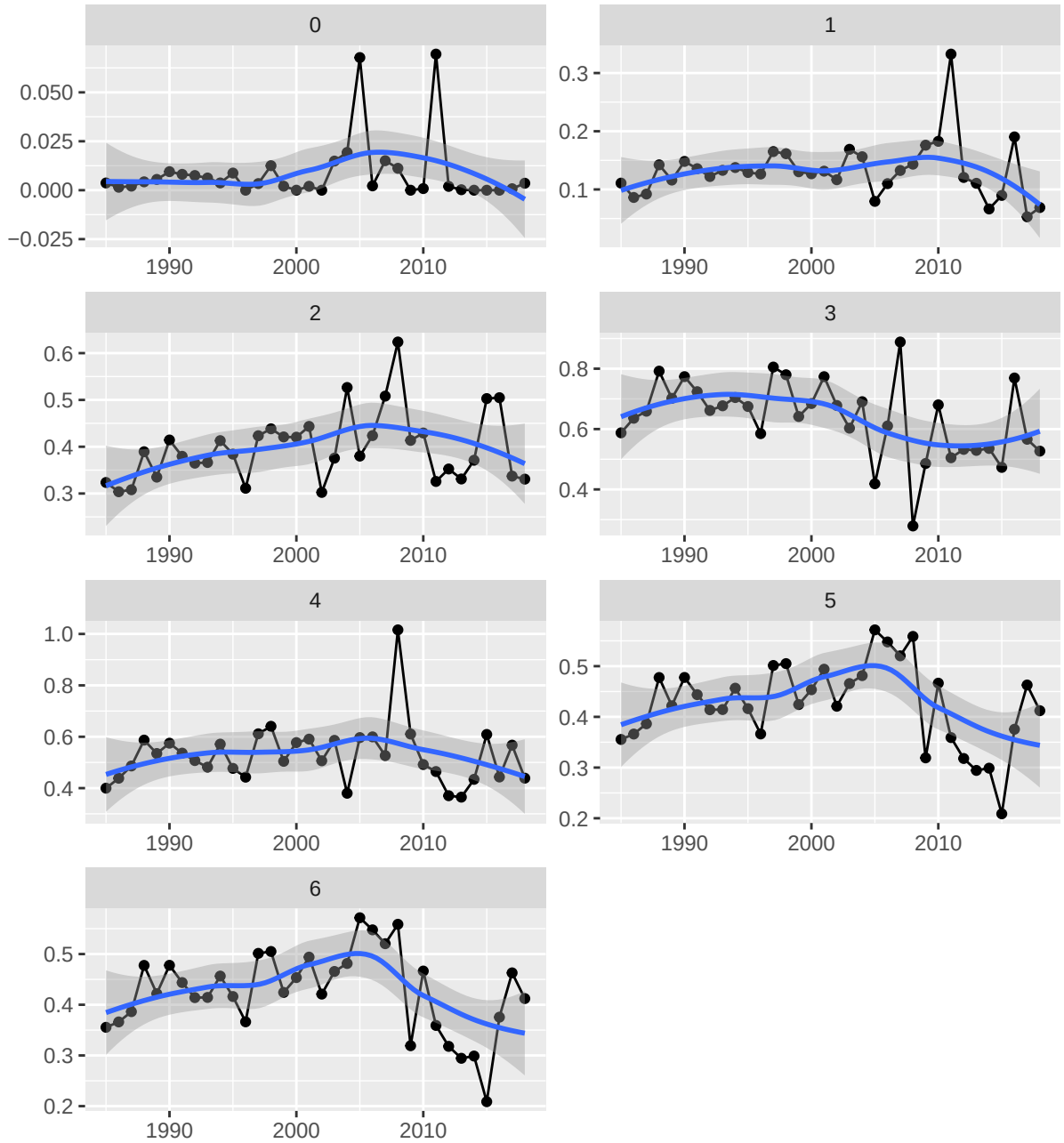


Figure 18. Estimates of harvest-at-age by year for the candidate run assuming constant M ; continuous lines represent lowest smoothers.

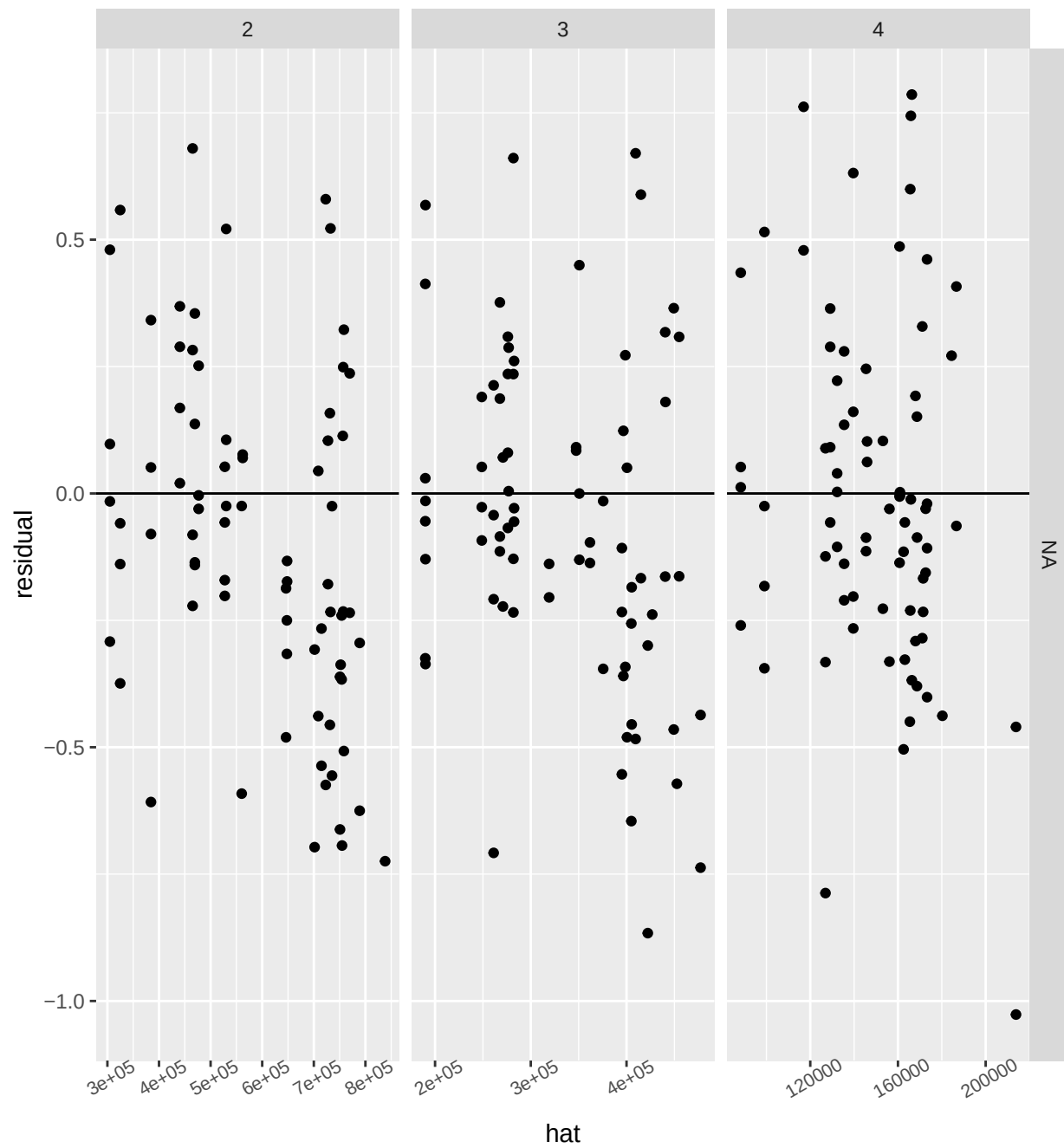
Diagnostic plots

Figure 19. Residuals against fitted value for the candidate run assuming constant M .

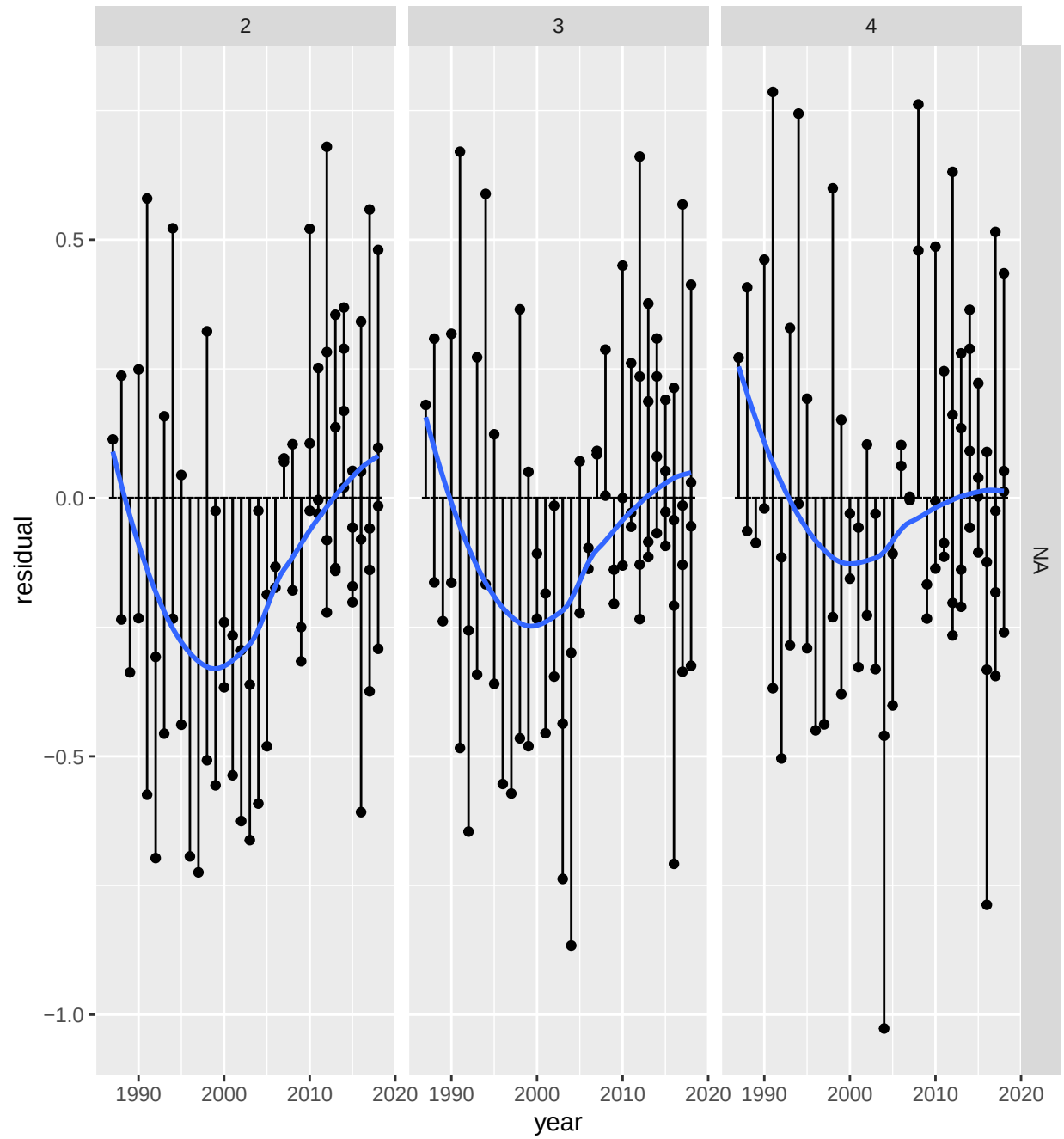


Figure 20. Residuals by fleet against year for the candidate run assuming constant M.

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## `geom_smooth()` using formula 'y ~ x'
```

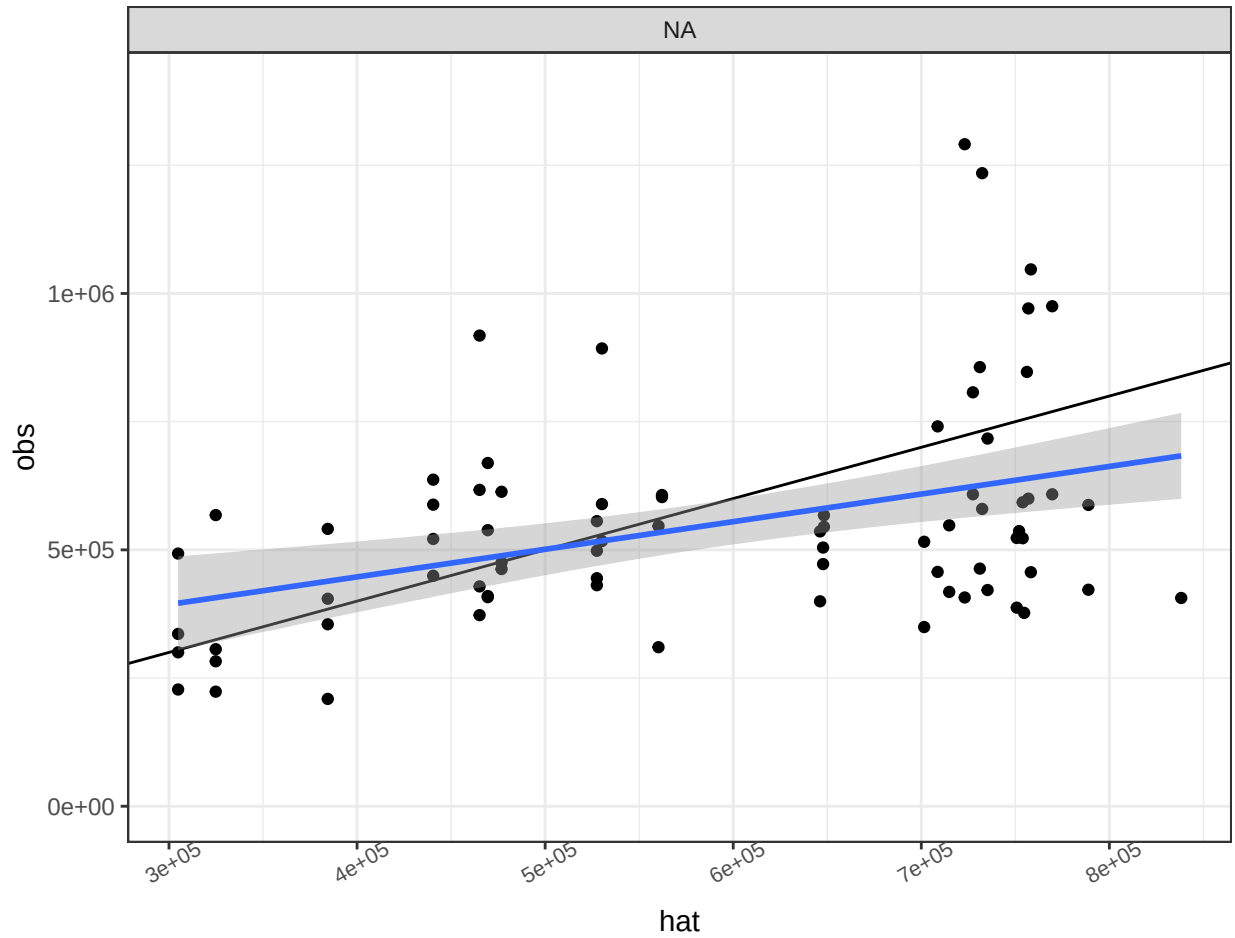


Figure 21. Calibration regression plots for age 2 for the candidate run assuming constant M . Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

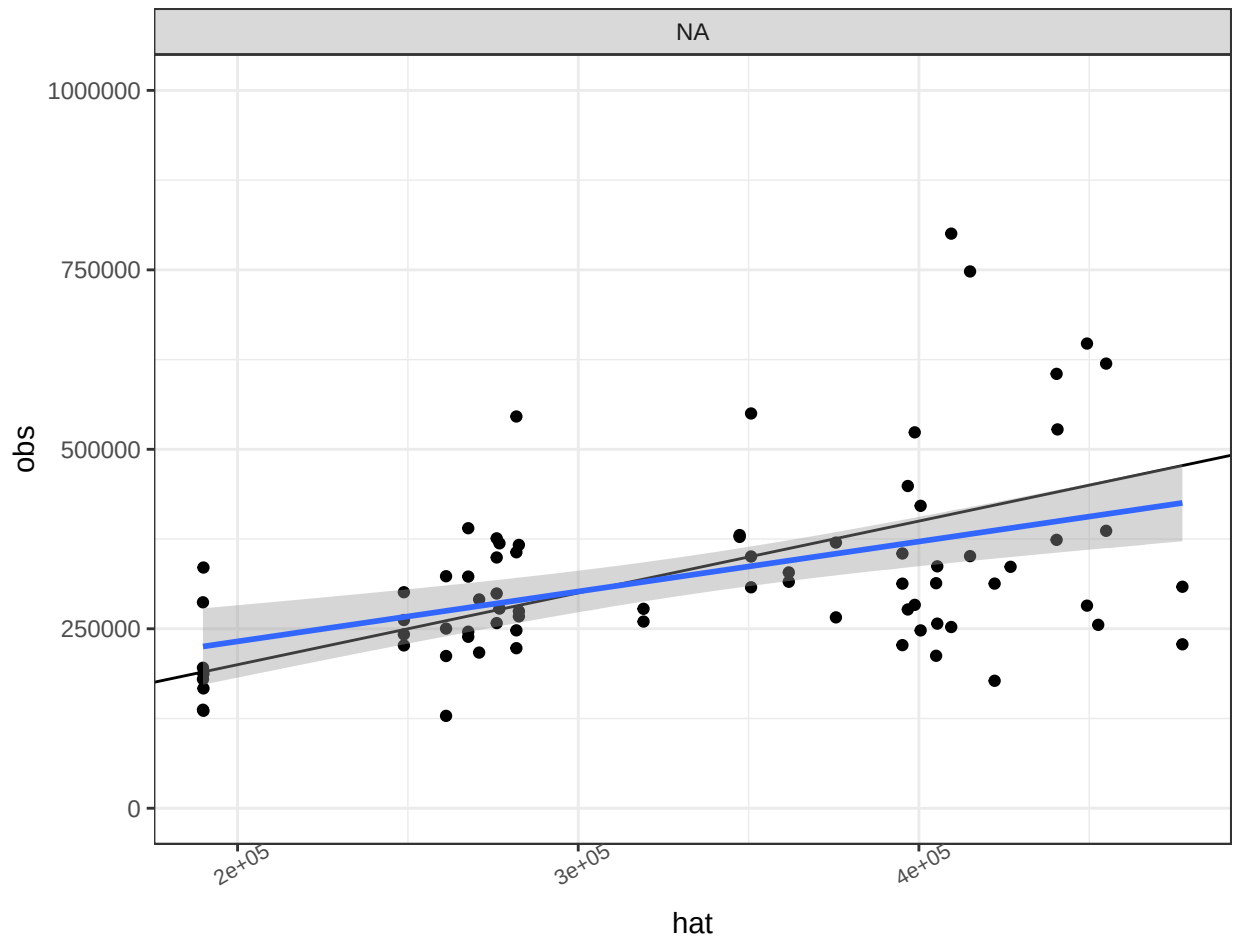


Figure 22. Calibration regression plots for age 3 for the candidate run assuming constant M . Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
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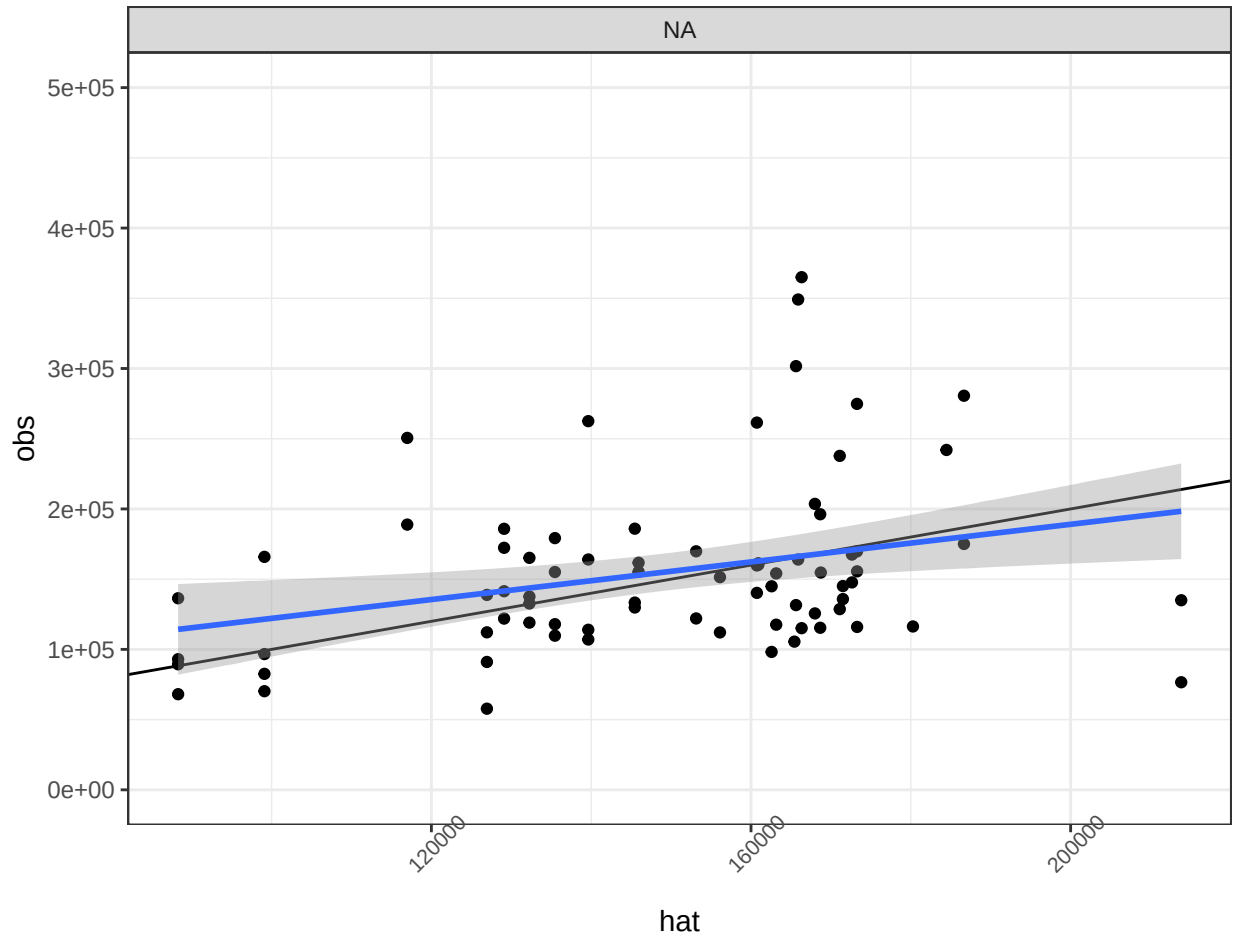


Figure 23. Calibration regression plots for age 4 for the candidate run assuming constant M . Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

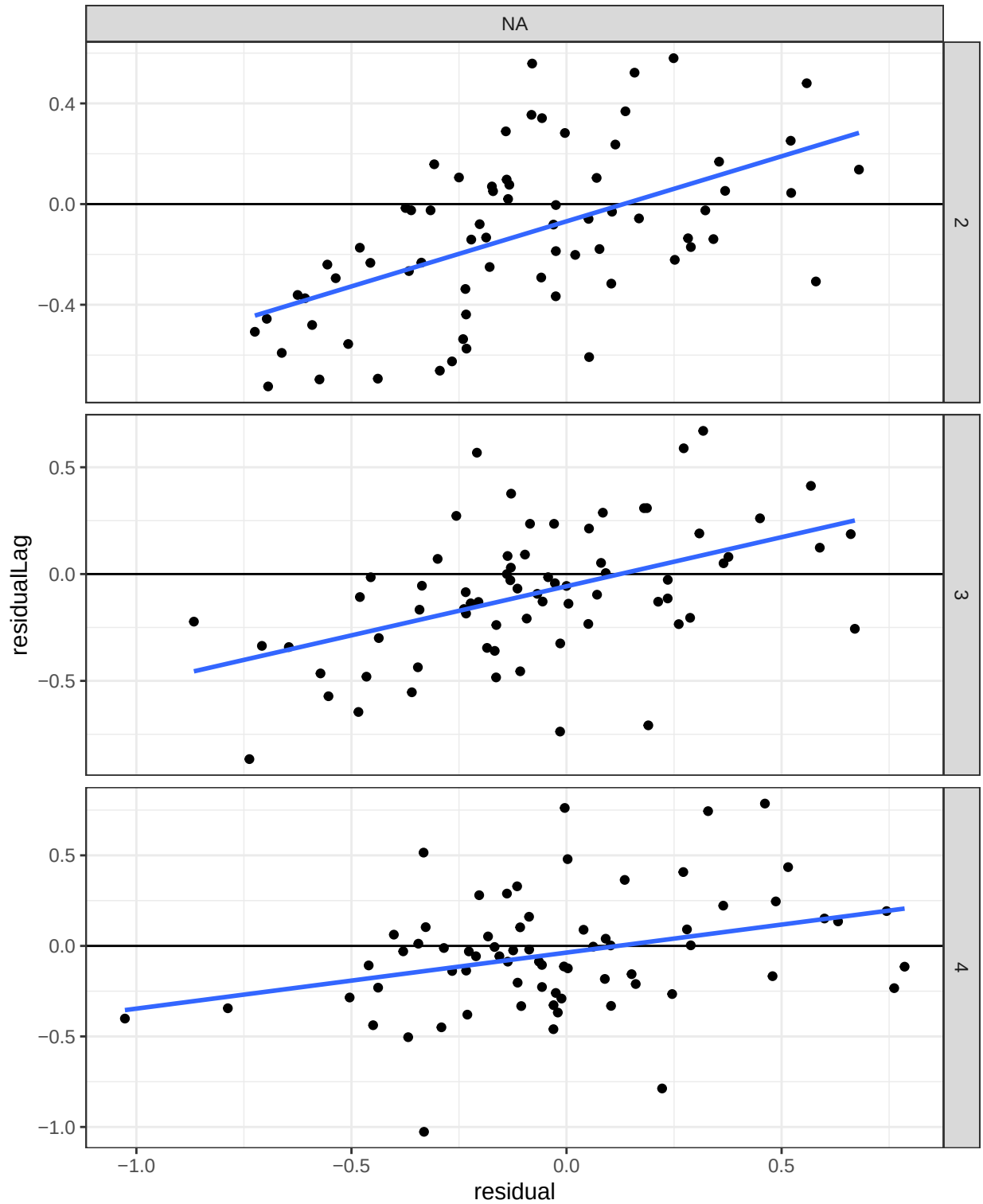


Figure 24. AR plots of lagged residuals for the candidate run assuming constant M. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

```
## `geom_smooth()` using formula 'y ~ x'
```

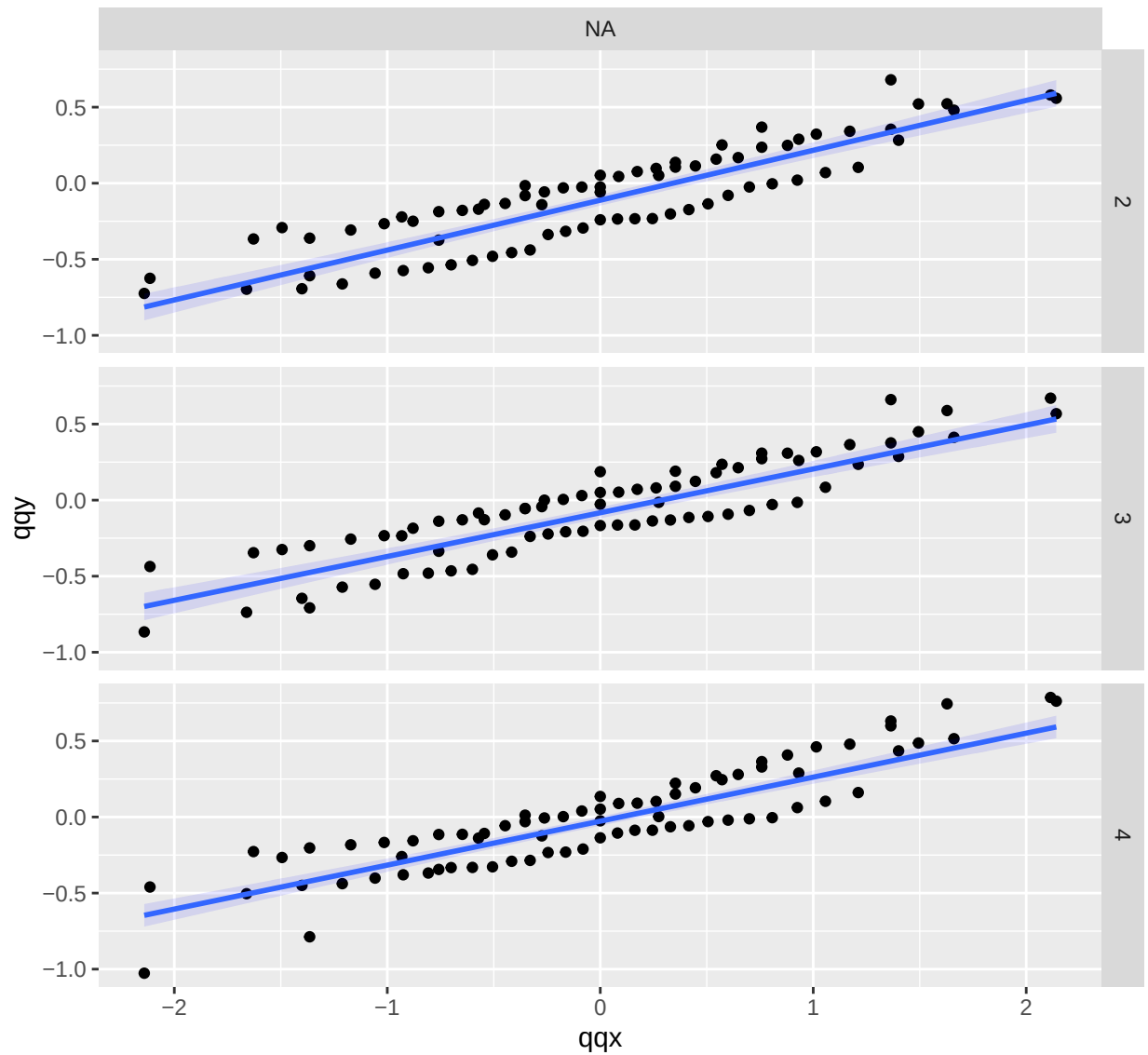


Figure 25. QQ plots to check for log normality (i.e. points and line should coincide) for the candidate run assuming constant M.

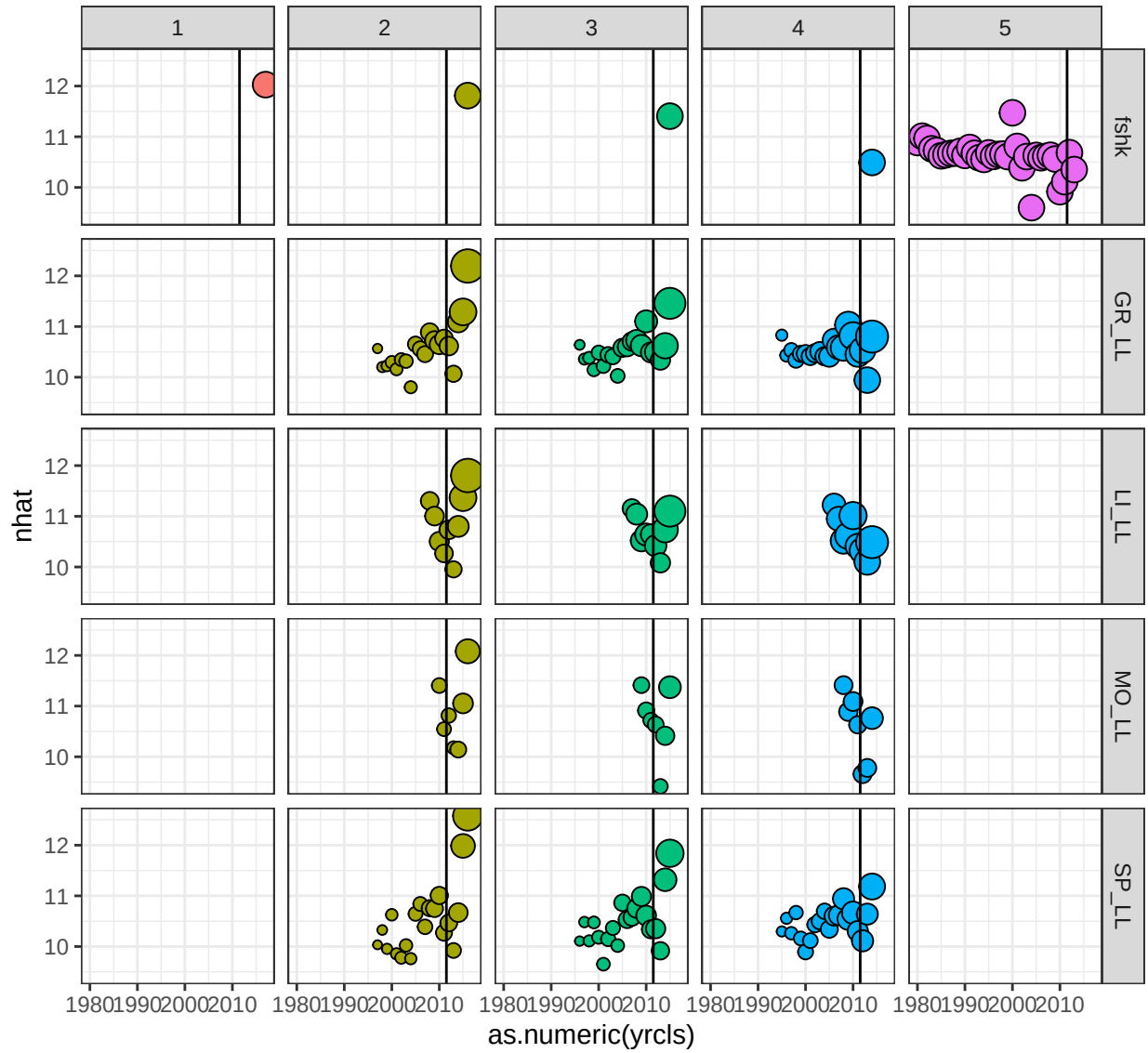


Figure 26. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(fshk)$ for the candidate run assuming constant M . Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

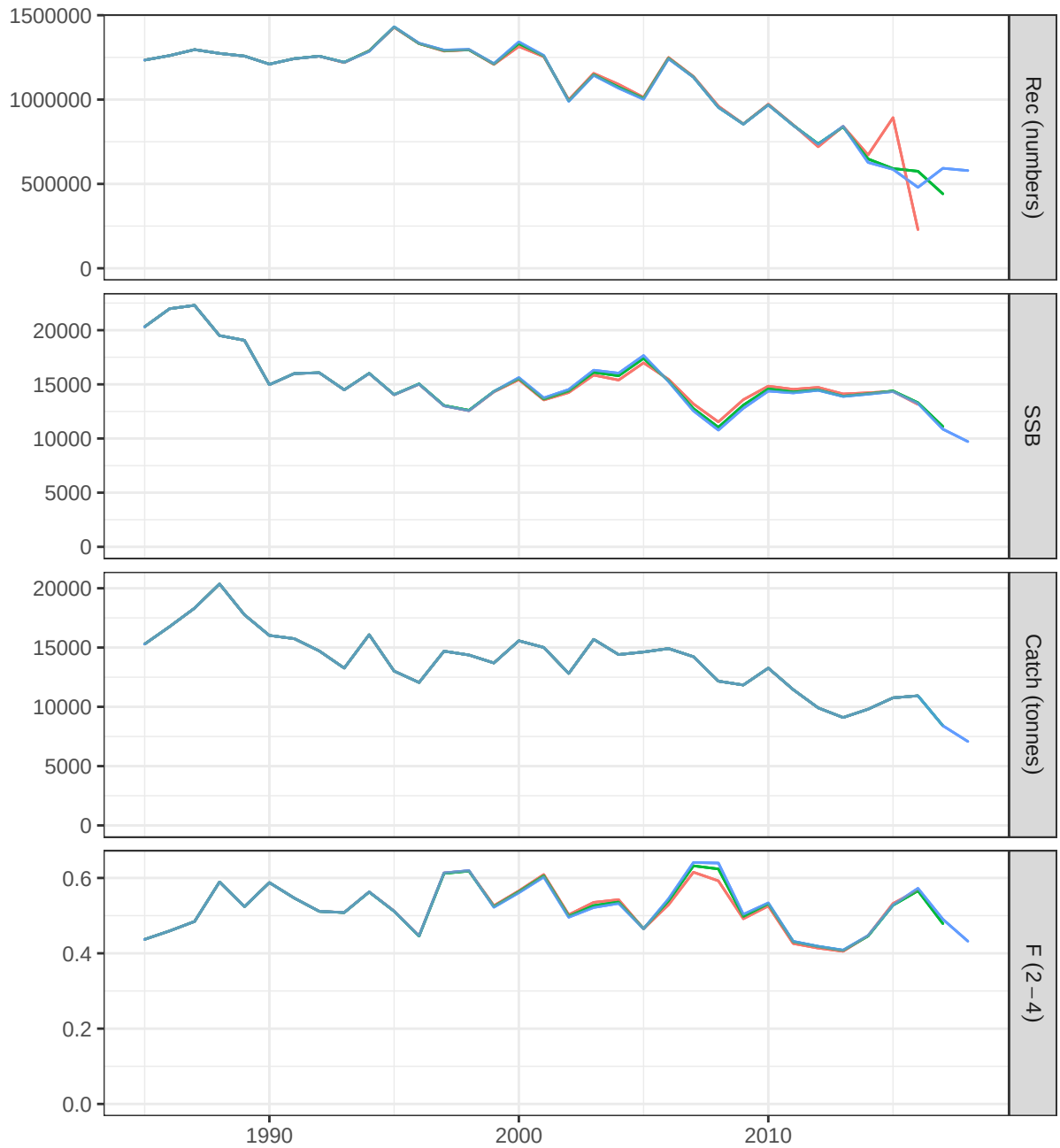


Figure 27. Retrospective plot for the candidate run assuming constant M.

c. Candidate run - M vector

Model estimates are plotted in **Figures 28-29**. The various diagnostic plots are shown in **Figures 30-38**, while **Figure 39** shows the retrospective plot.

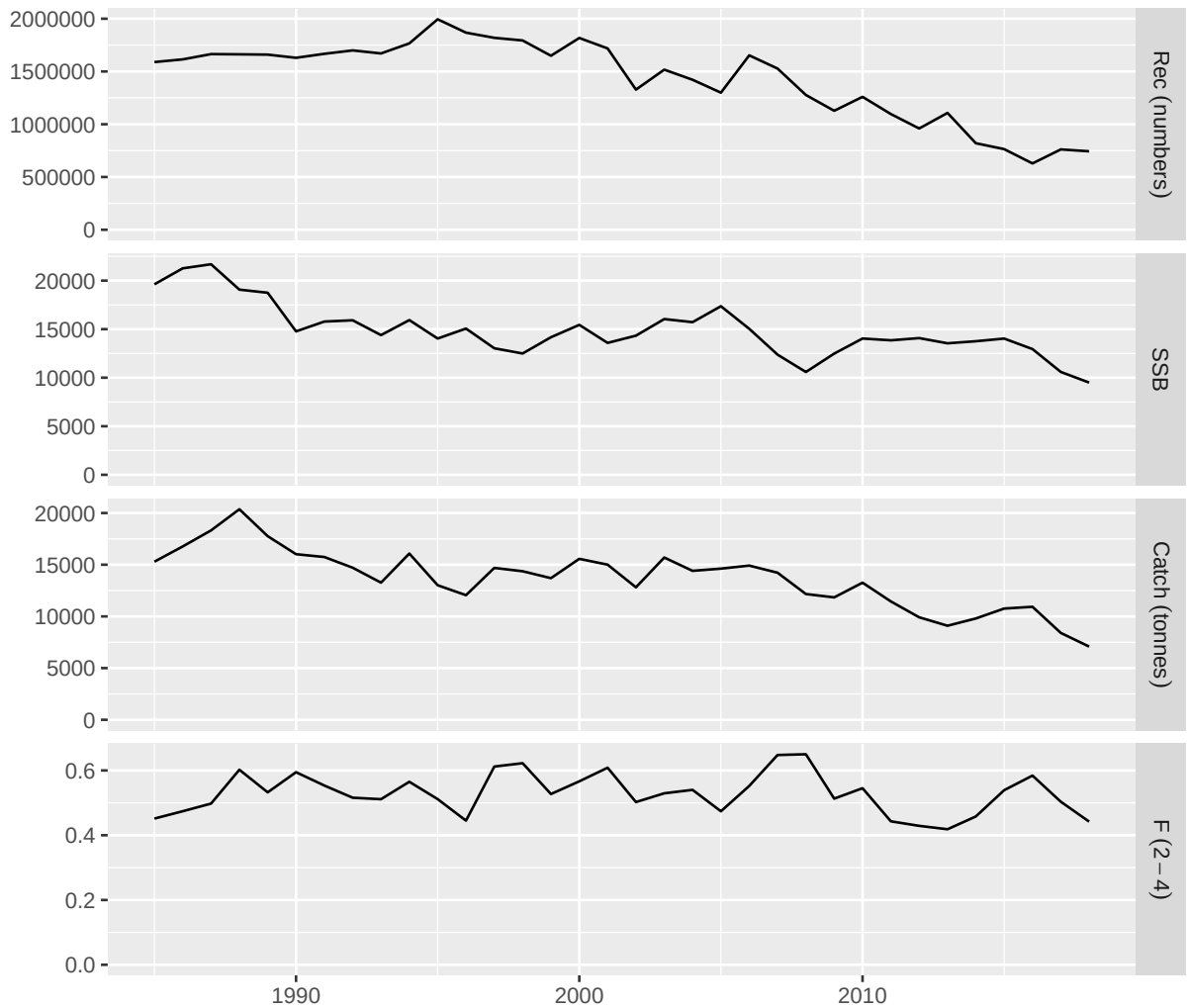


Figure 28. XSA time series estimates for the candidate run assuming an M vector by age

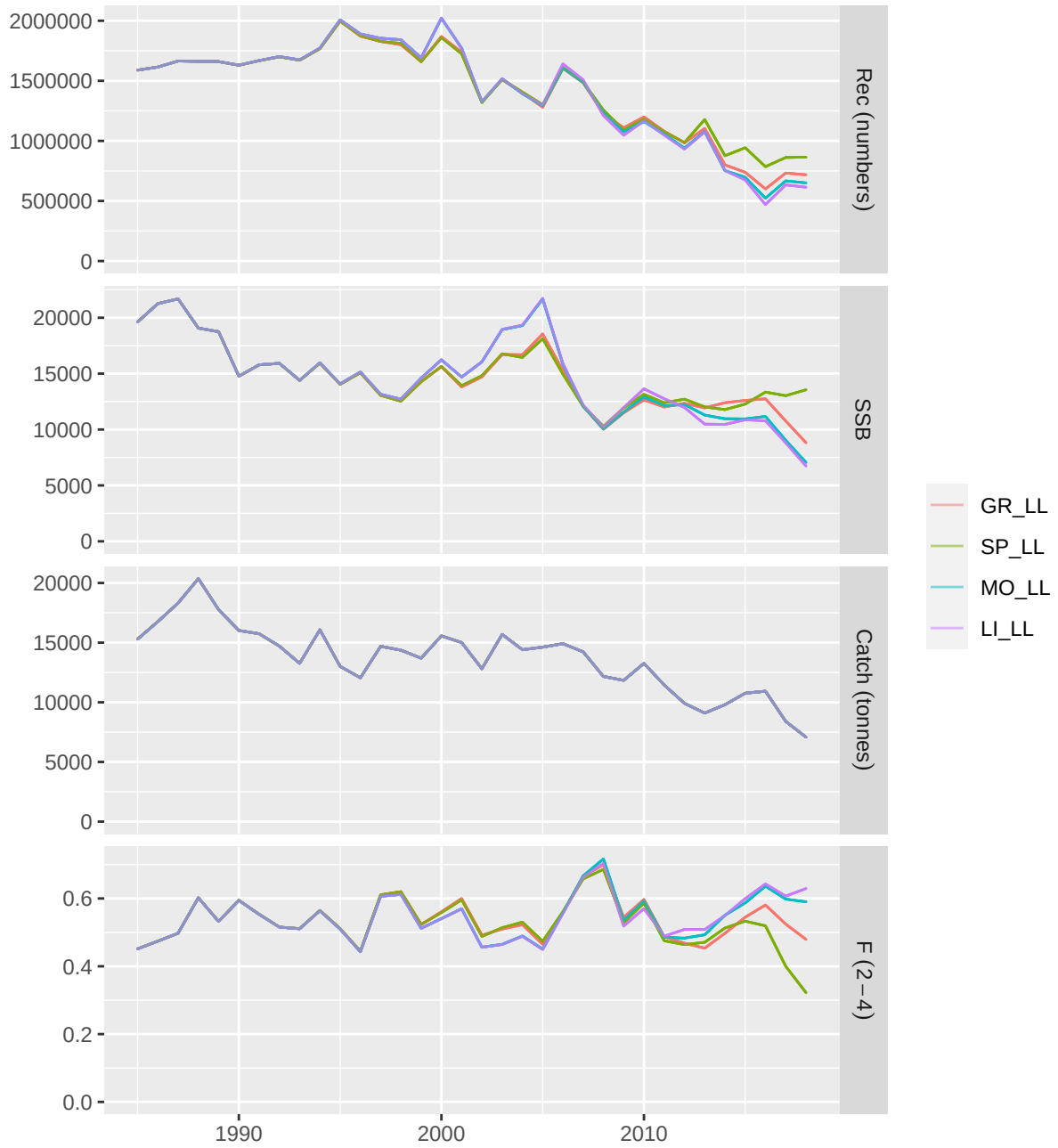


Figure 29. XSA time series estimates by CPUE, for the candidate run assuming an M vector by age.

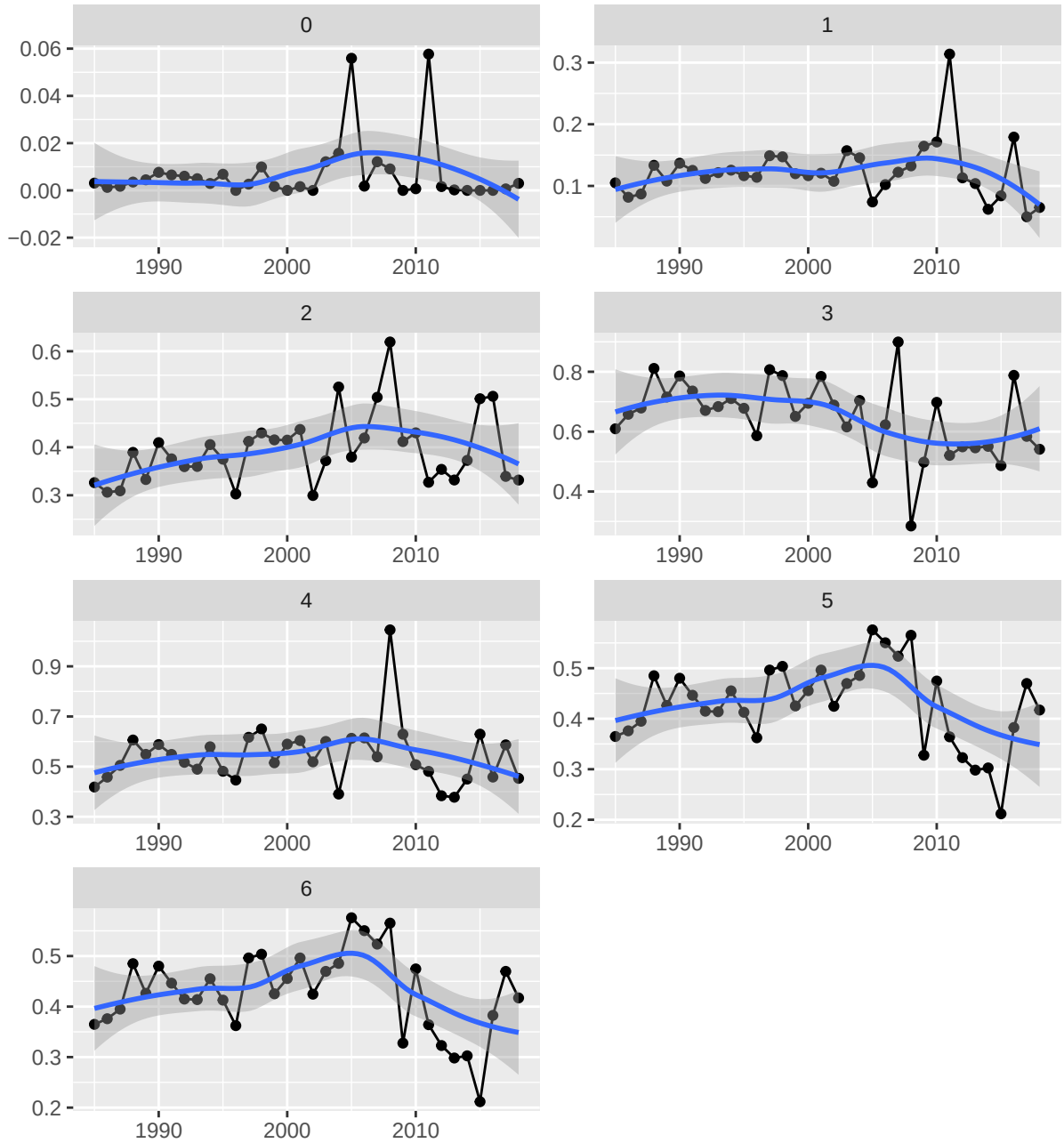


Figure 30. Estimates of harvest-at-age by year for the candidate run assuming an M vector by age; continuous lines represent lowess smoothers.

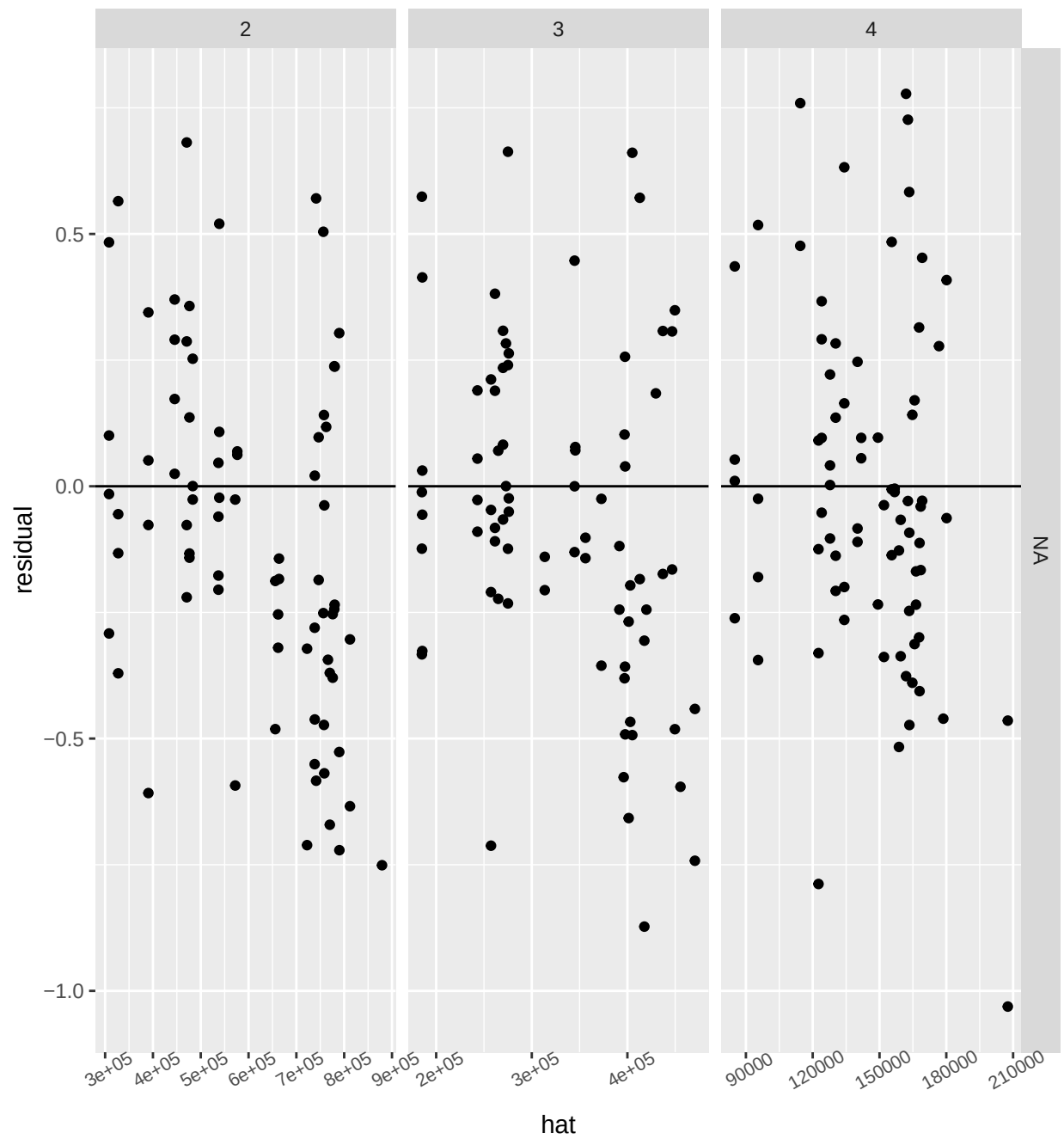
Diagnostic plots

Figure 31. Residuals against fitted value for the candidate run assuming an M vector by age.

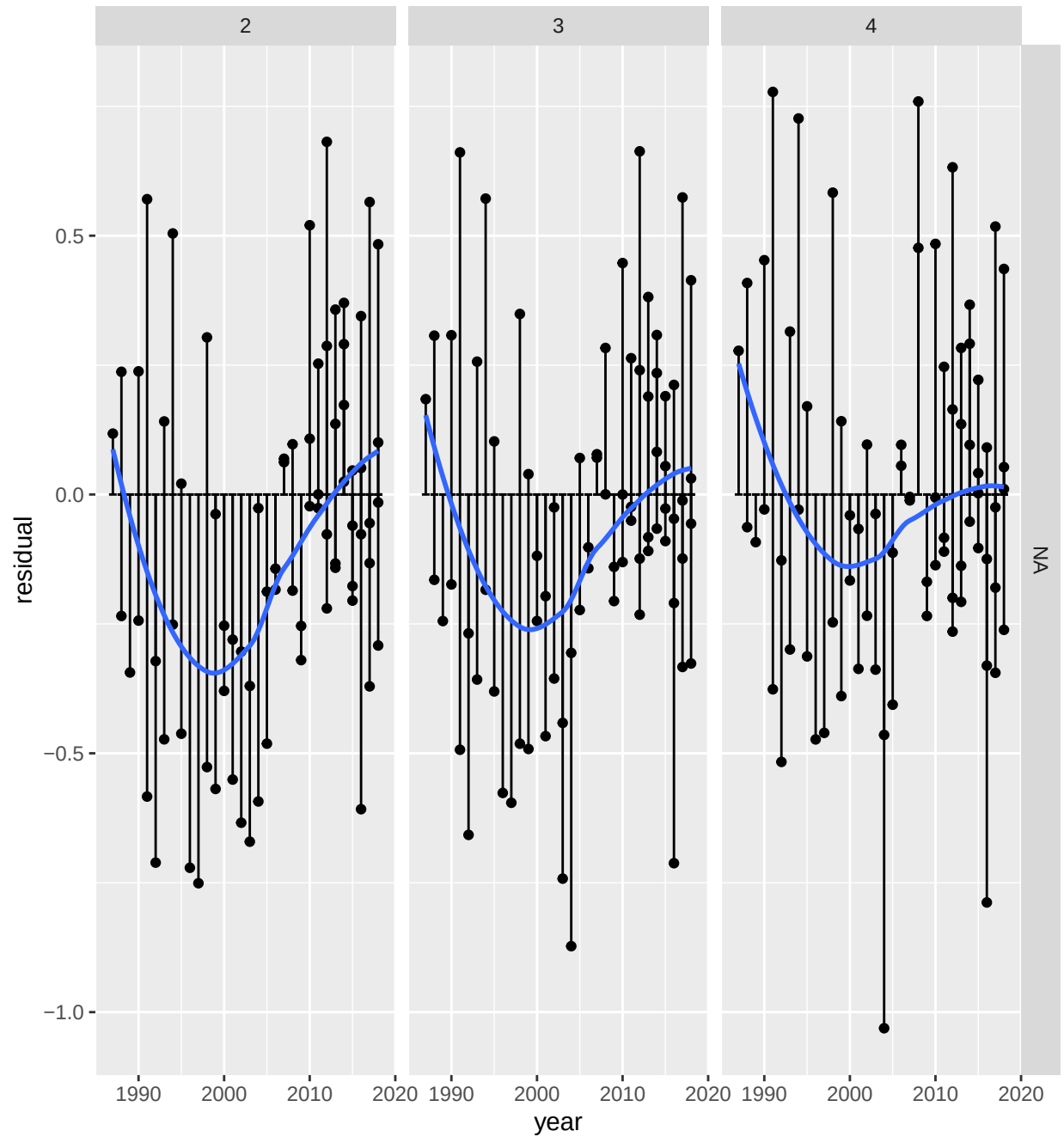


Figure 32. Residuals by fleet against year for the candidate run assuming an M vector by age.

```
## `geom_smooth()` using formula 'y ~ x'
```

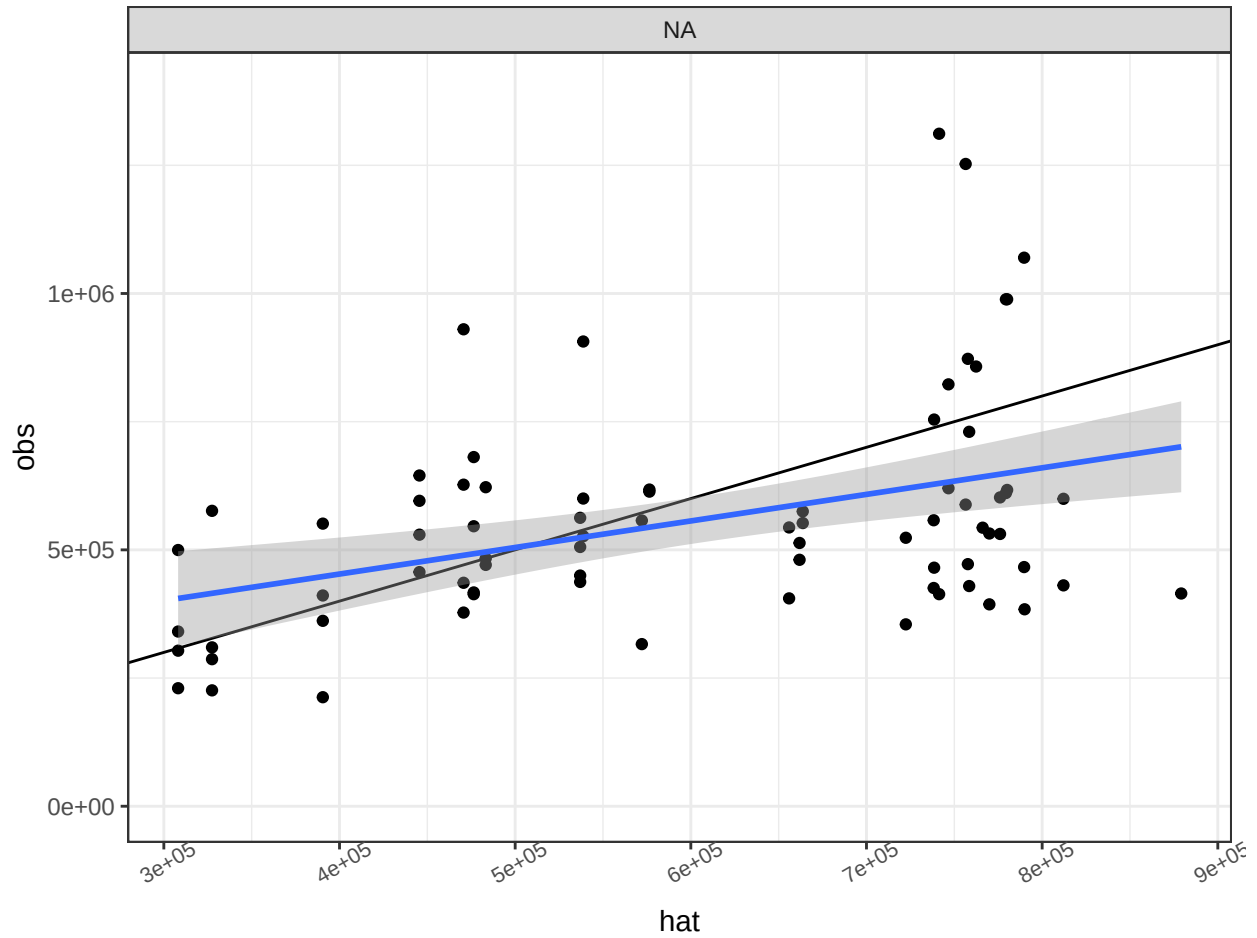


Figure 33. Calibration regression plots for age 2 for the candidate run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

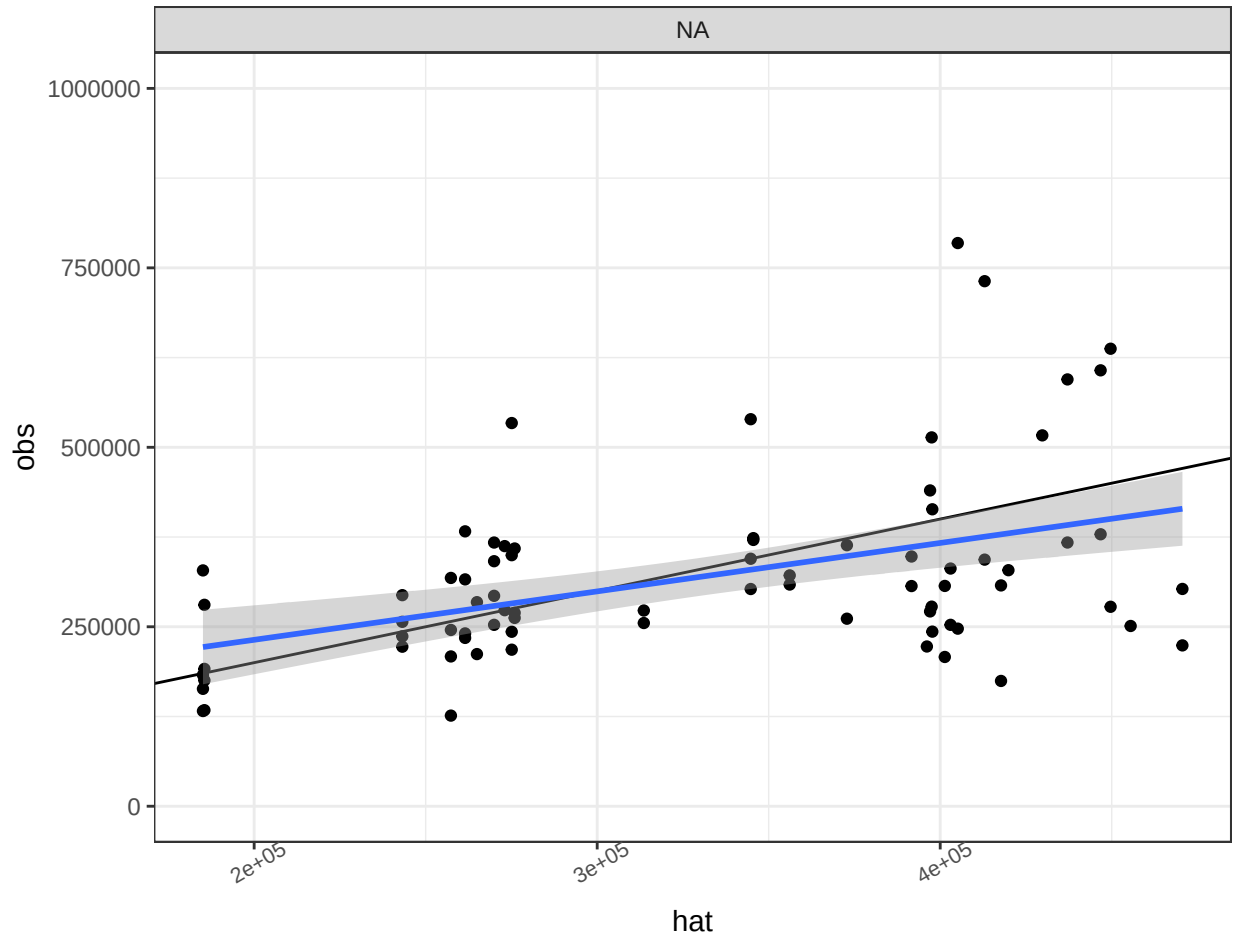


Figure 34. Calibration regression plots for age 3 for the candidate run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

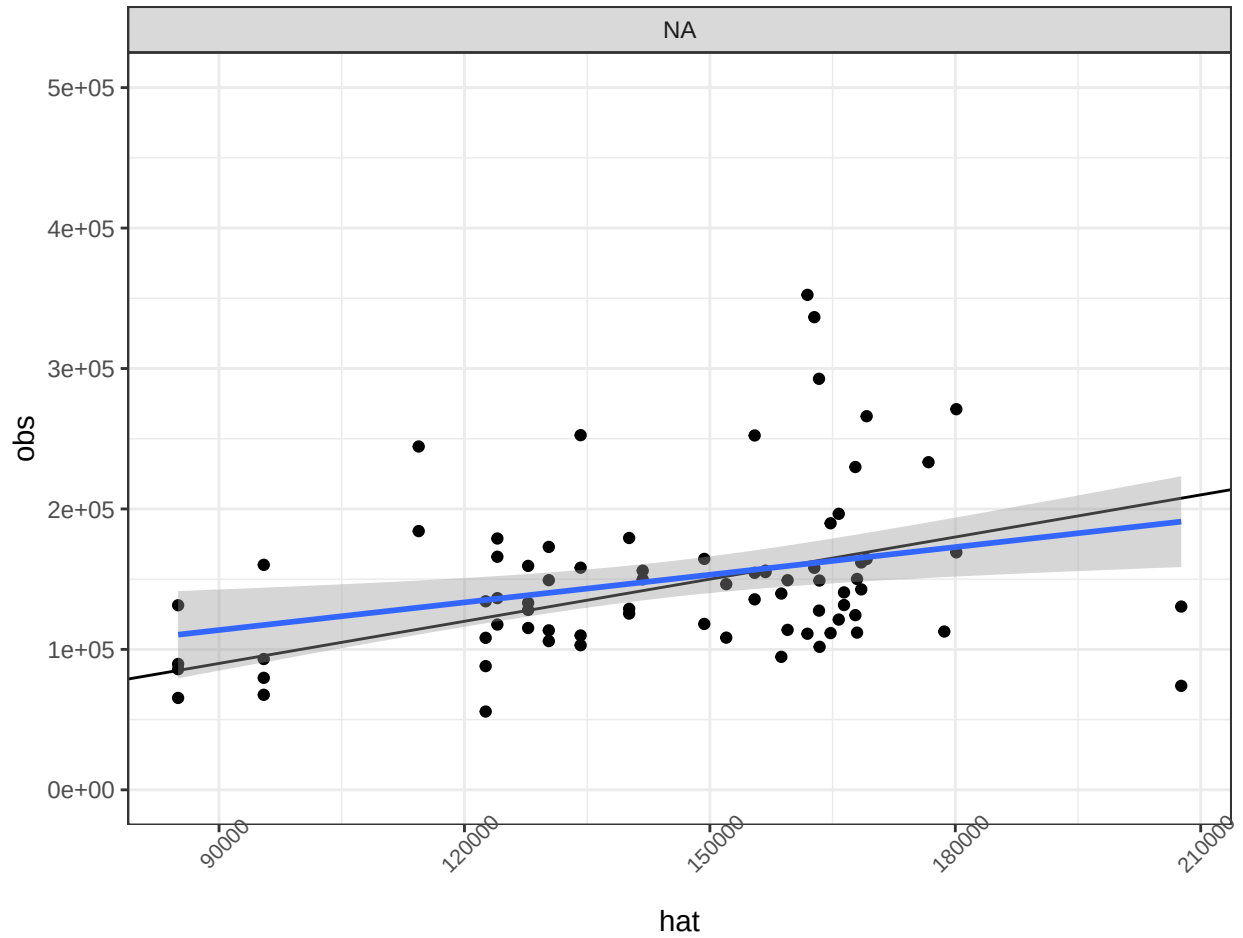


Figure 35. Calibration regression plots for age 4 for the candidate run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

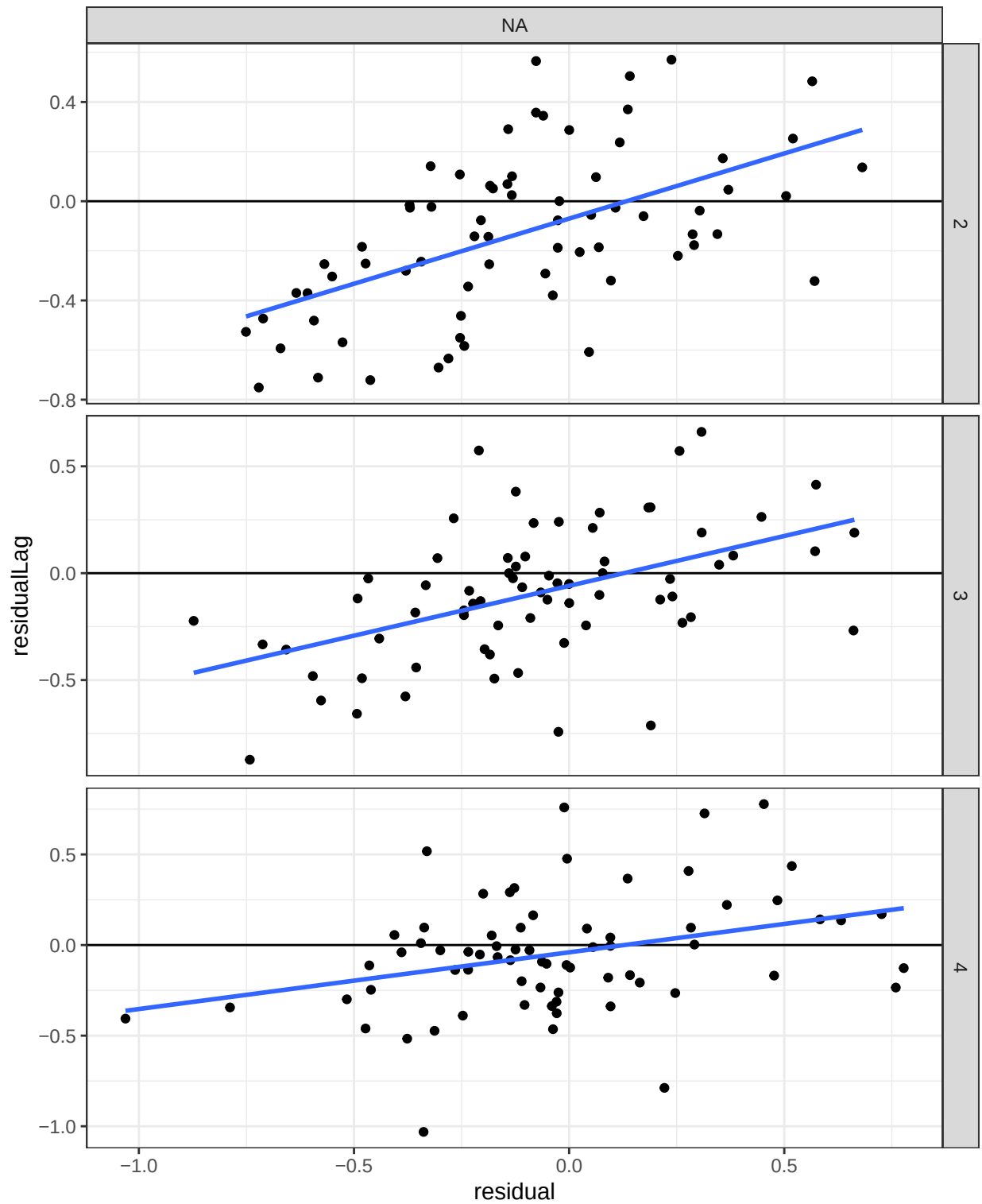


Figure 36. AR plots of lagged residuals for the candidate run assuming an M vector by age. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

```
## `geom_smooth()` using formula 'y ~ x'
```

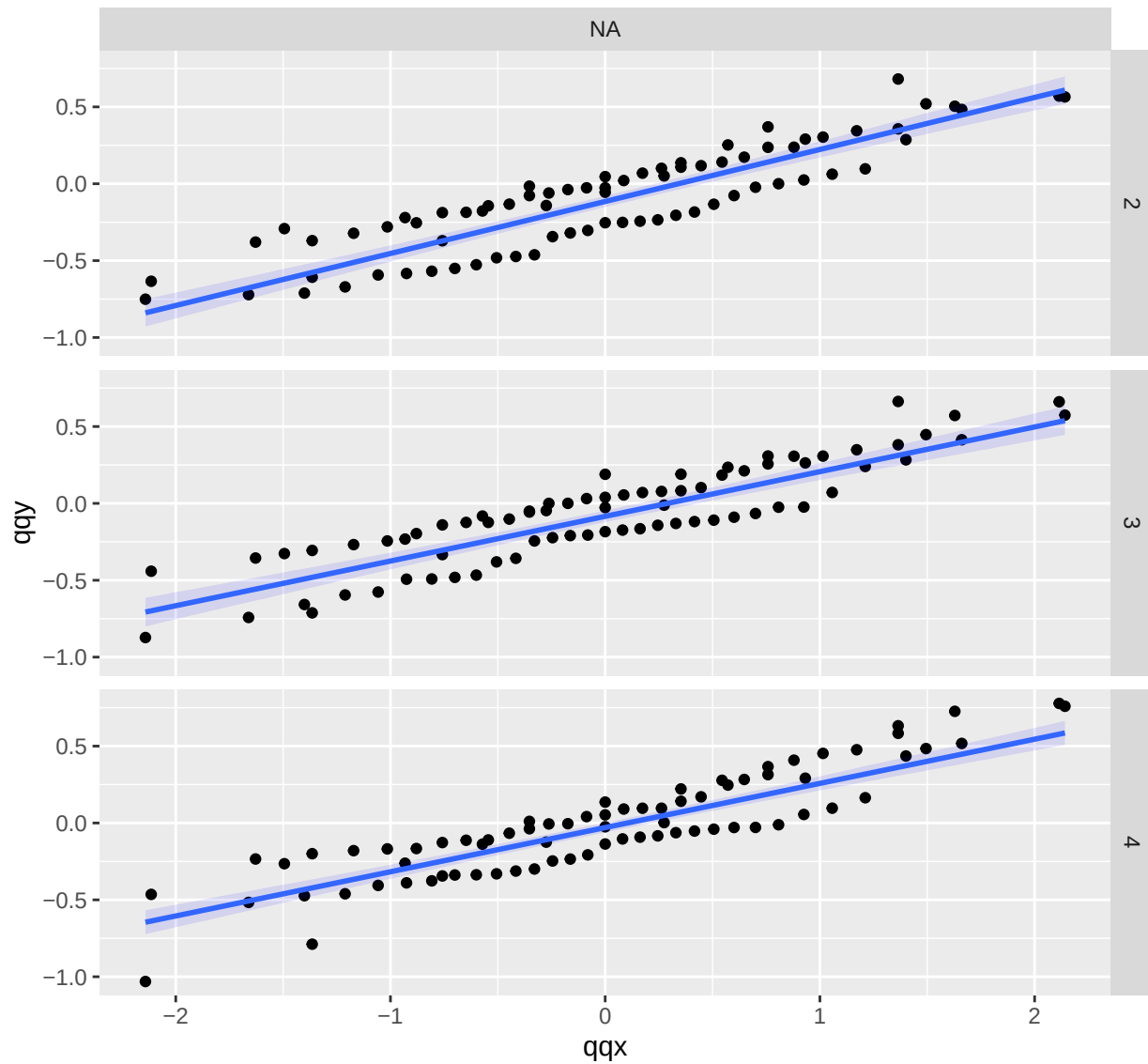


Figure 37. QQ plots to check for log normality (i.e. points and line should coincide) for the candidate run assuming an M vector by age.

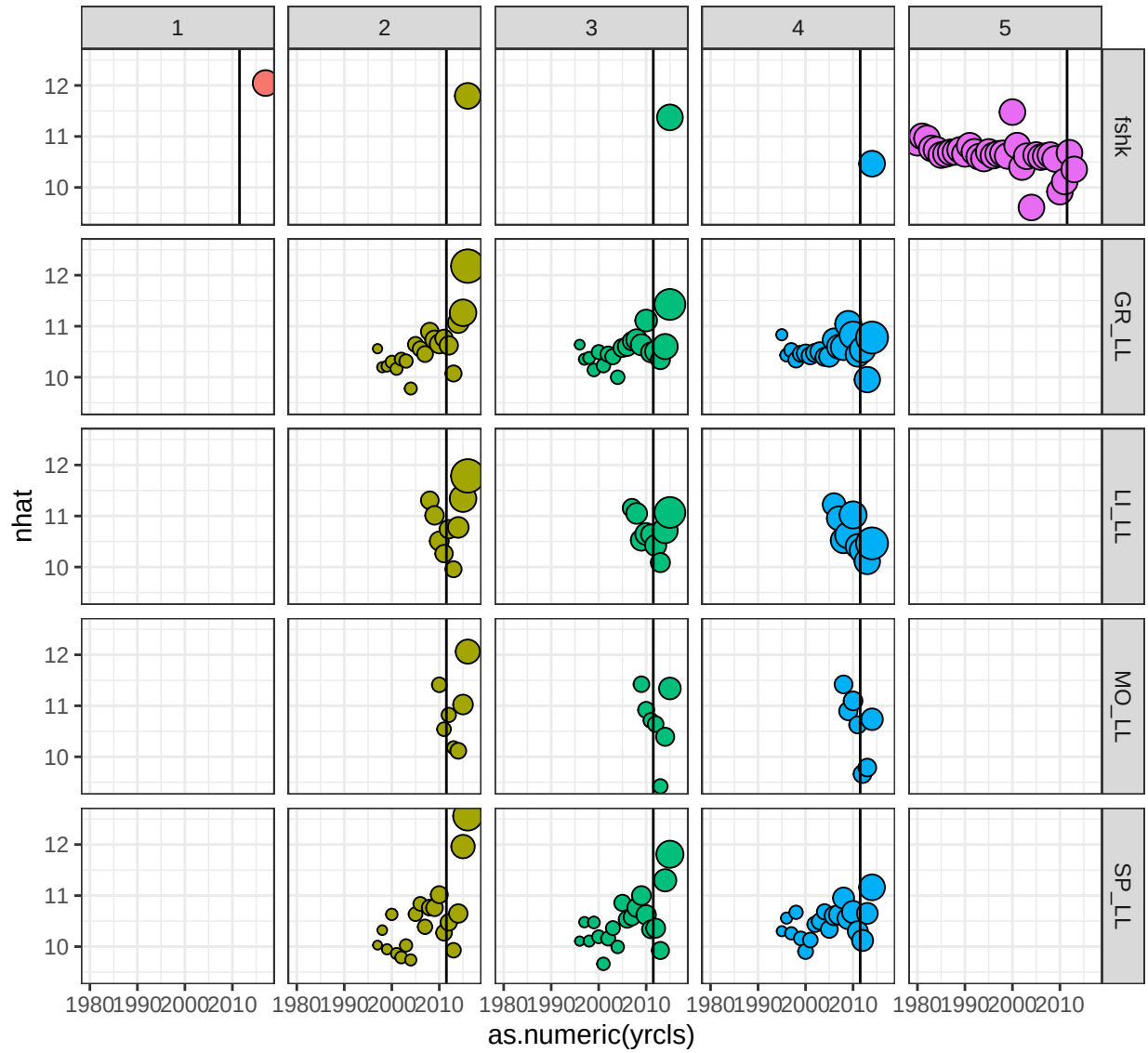


Figure 38. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(fshk)$ for the candidate run assuming an M vector by age. Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

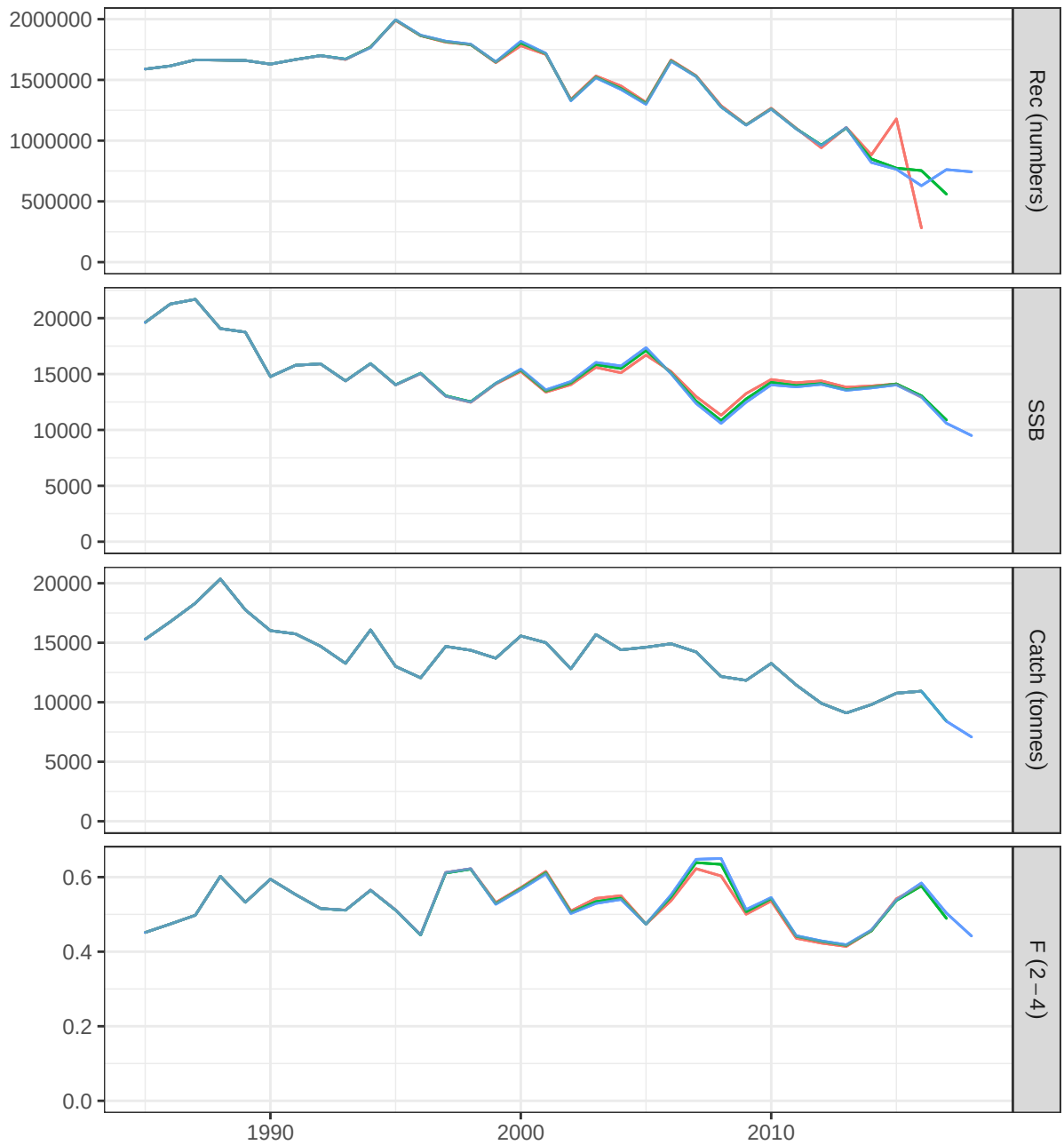


Figure 39. Retrospective plot for the candidate run assuming an M vector by age.

d. Discard run - constant M

Model estimates are plotted in **Figures 16-17**. The various diagnostic plots are shown in **Figures 18-26**, while **Figure 27** shows the retrospective plot.

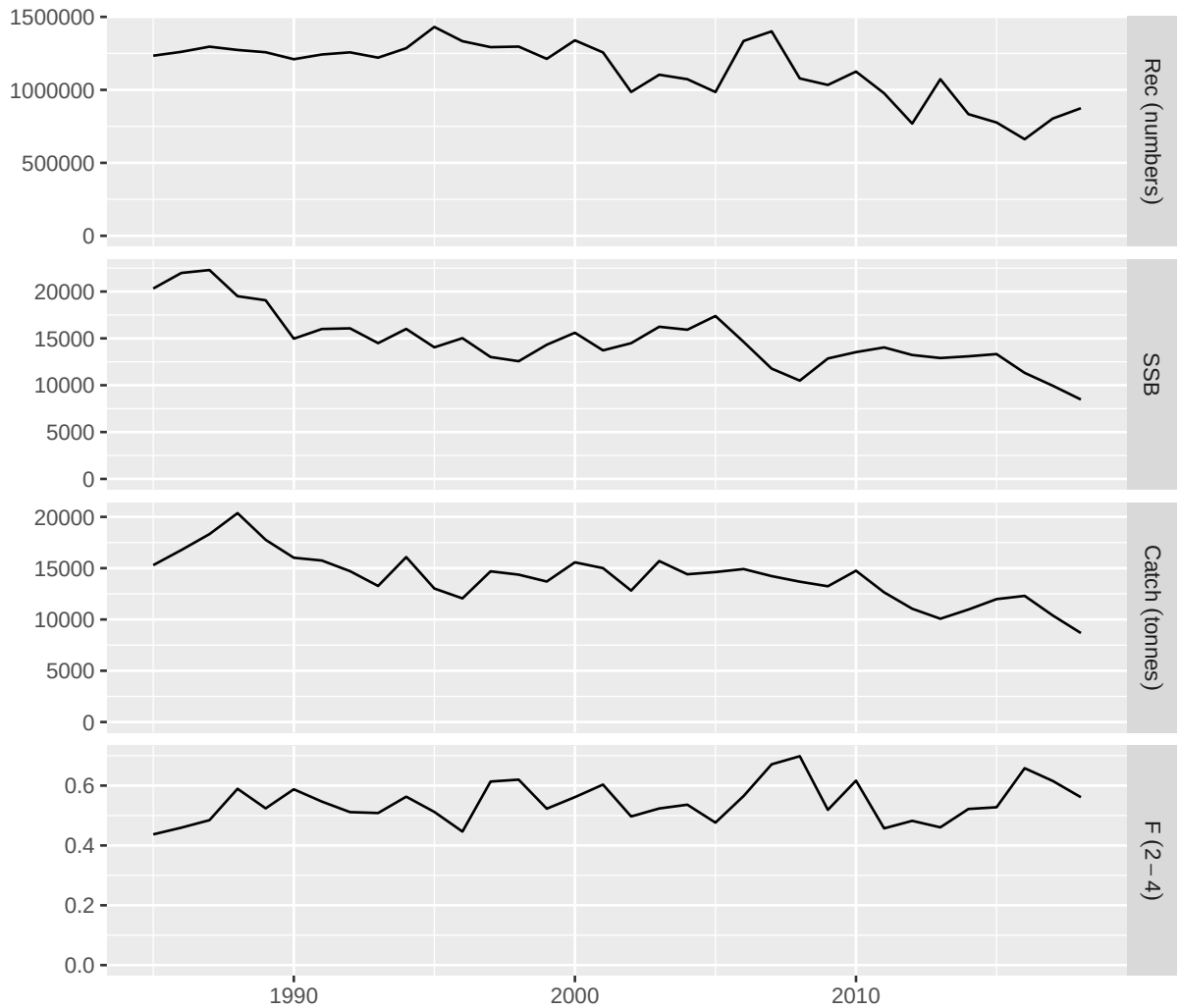


Figure 40. XSA time series estimates for the discard run assuming constant M.

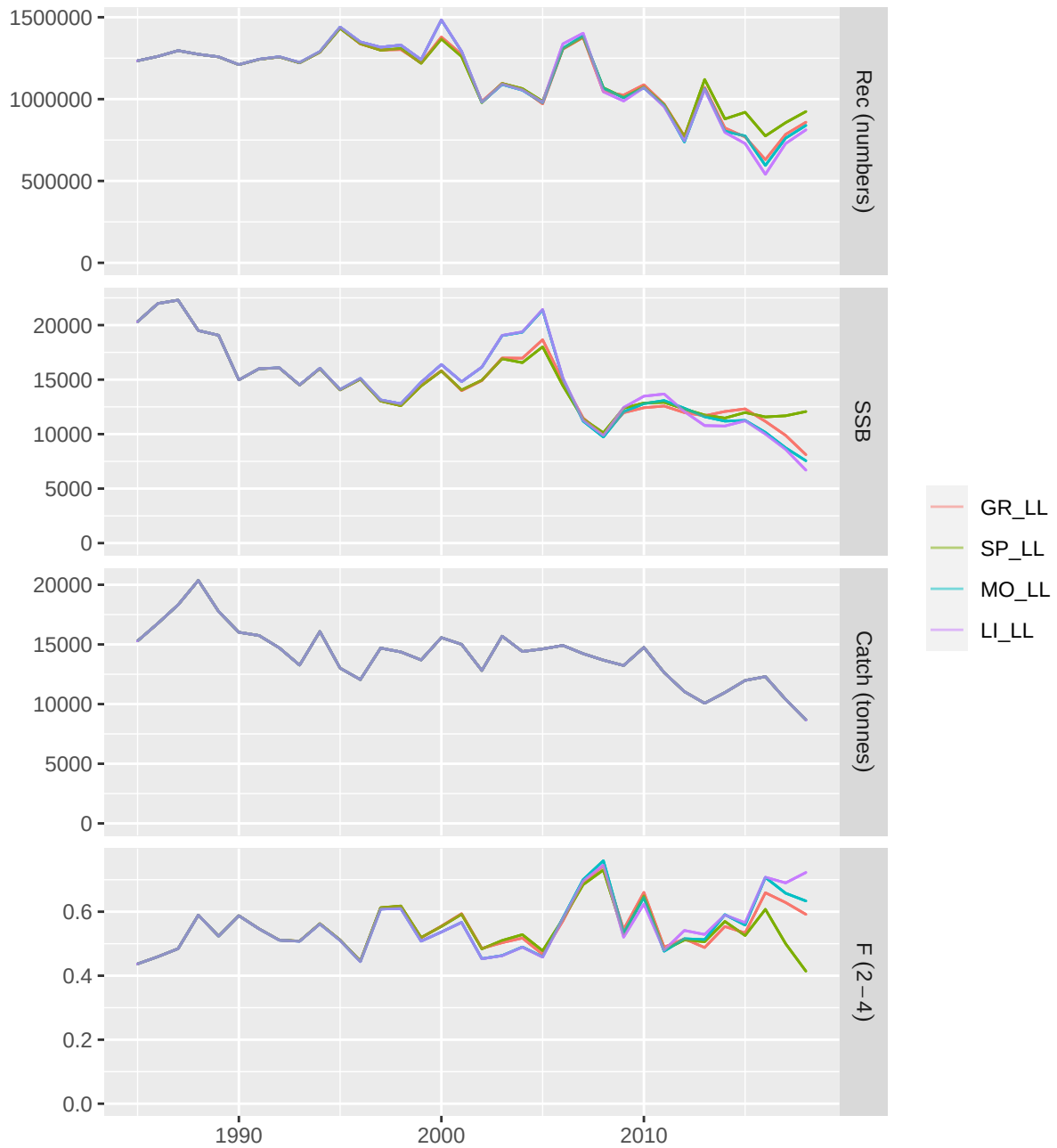


Figure 41. XSA time series estimates by CPUE, for the discard run assuming constant M.

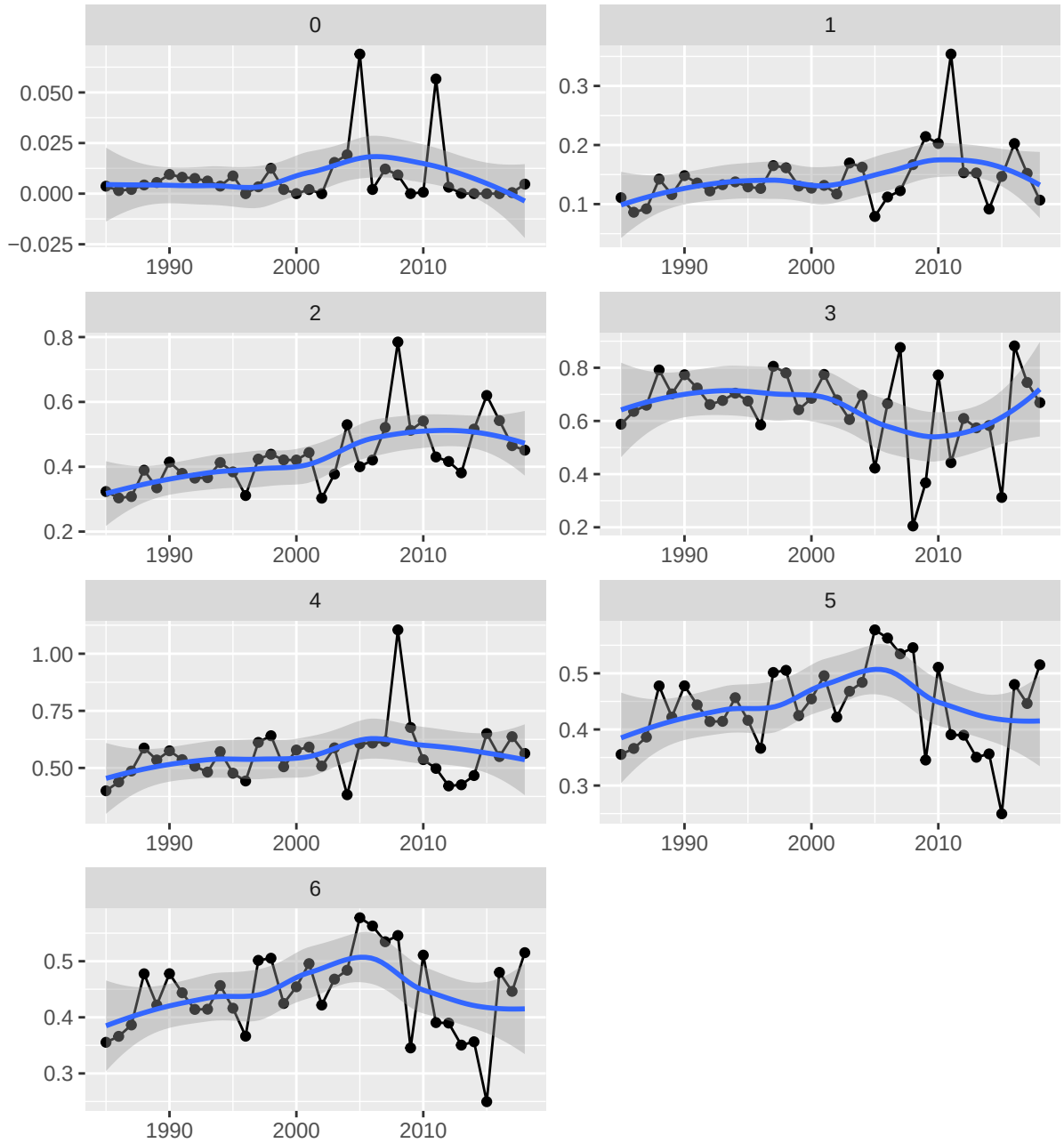


Figure 42. Estimates of harvest-at-age by year for the discard run assuming constant M ; continuous lines represent lowest smoothers.

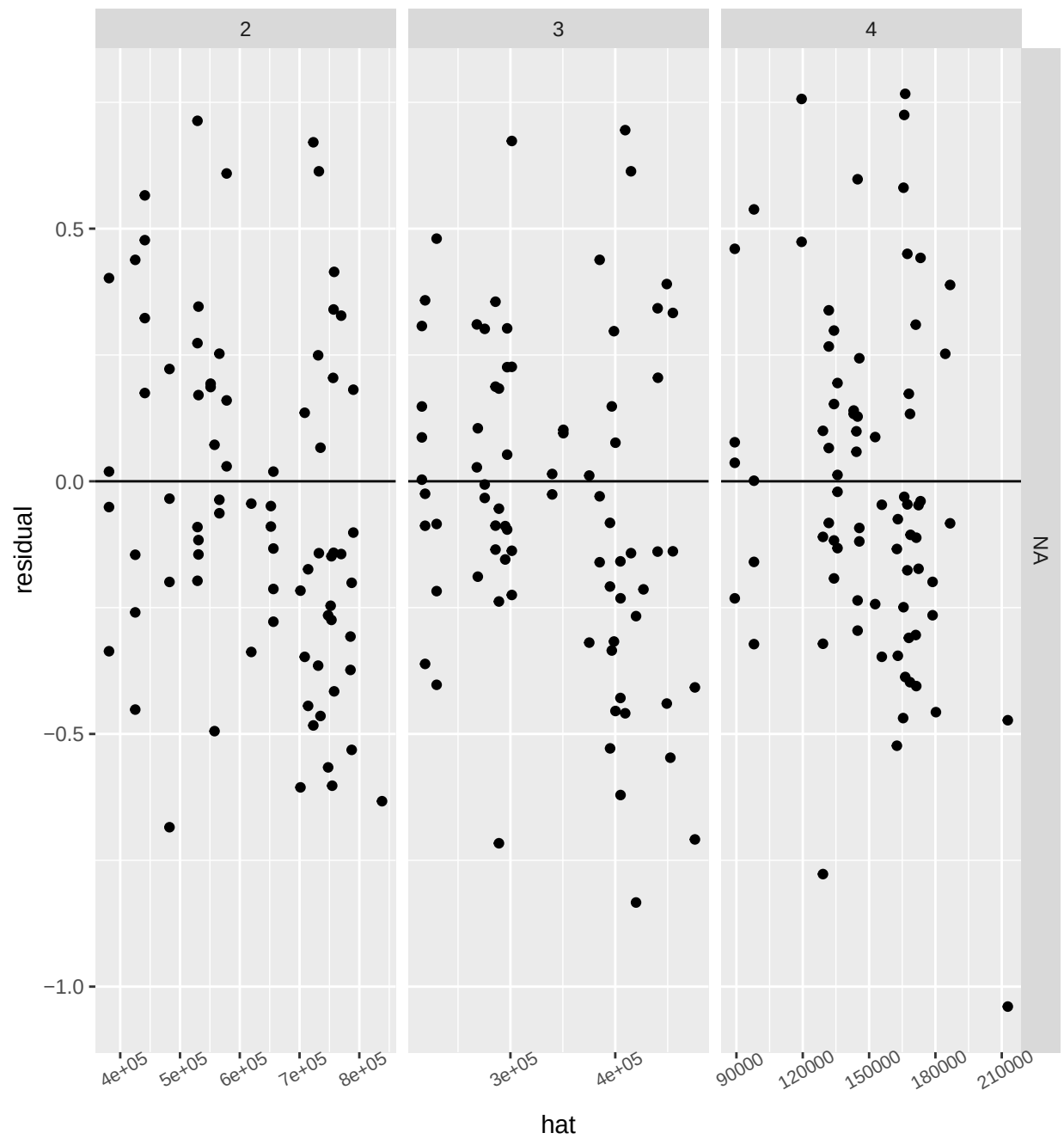
Diagnostic plots

Figure 43. Residuals against fitted value for the discard run assuming constant M .

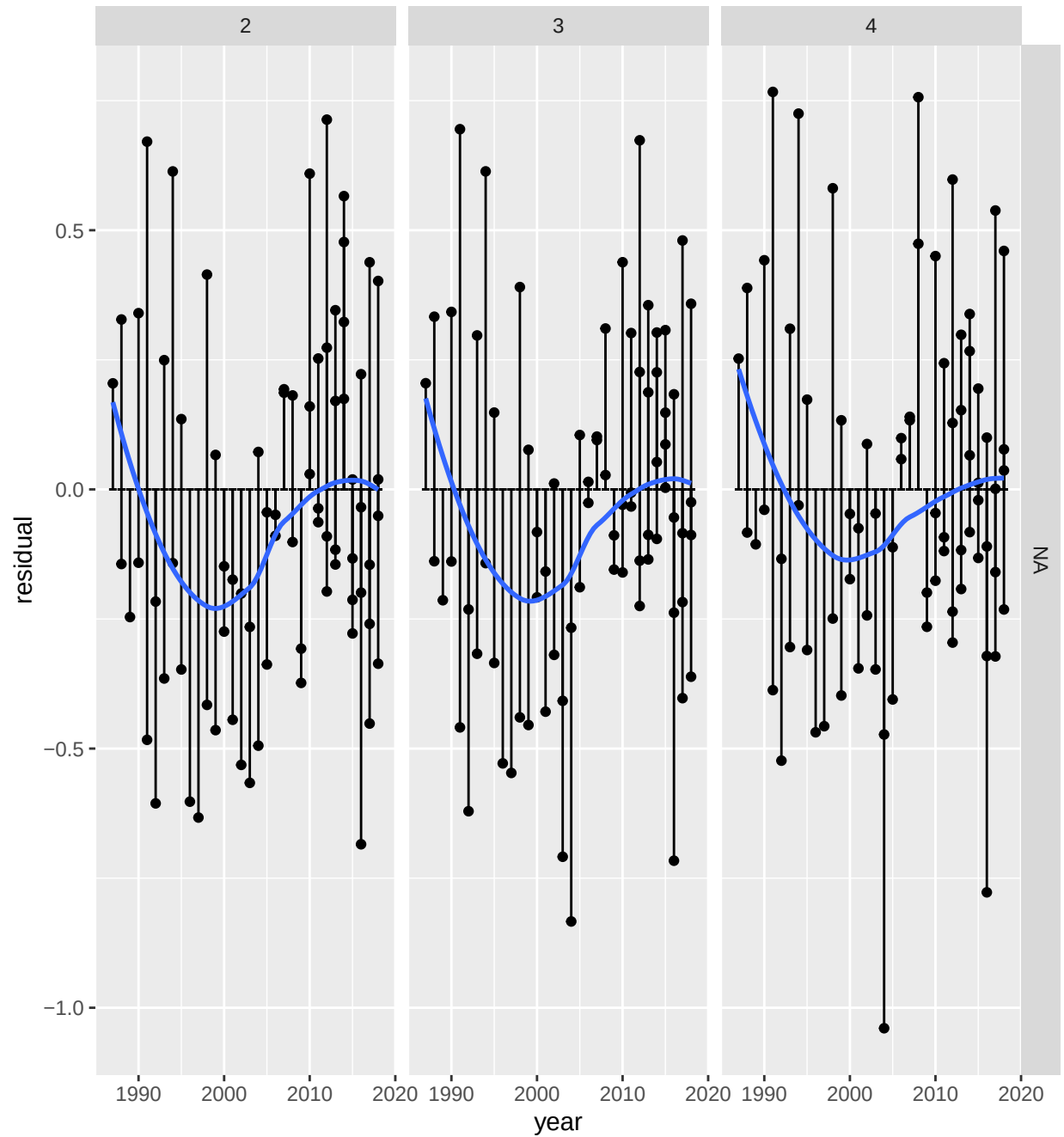


Figure 44. Residuals by fleet against year for the discard run assuming constant M .

```
## `geom_smooth()` using formula 'y ~ x'
```

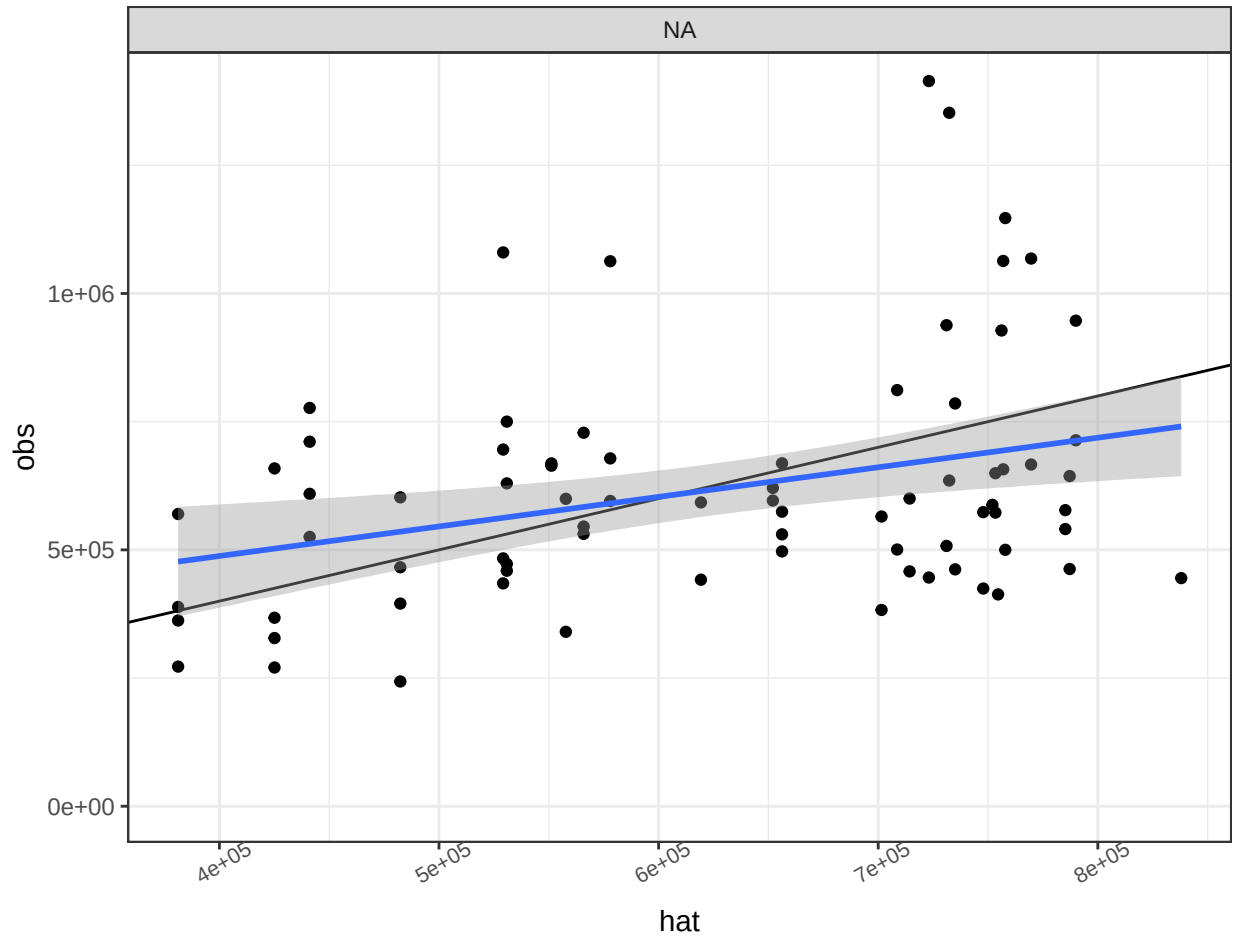


Figure 45. Calibration regression plots for age 2 for the discard run assuming constant M . Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

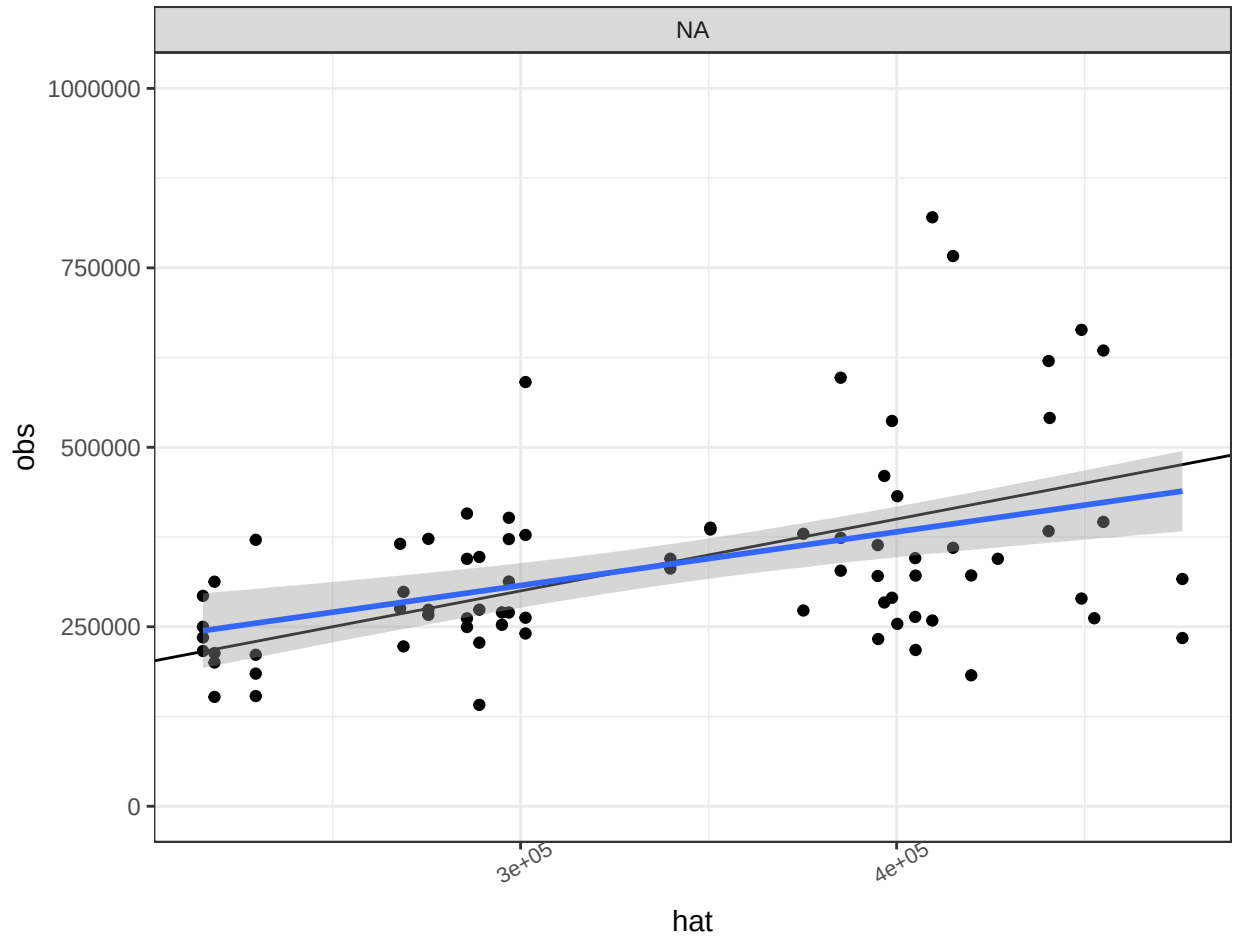


Figure 46. Calibration regression plots for age 3 for the discard run assuming constant M . Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

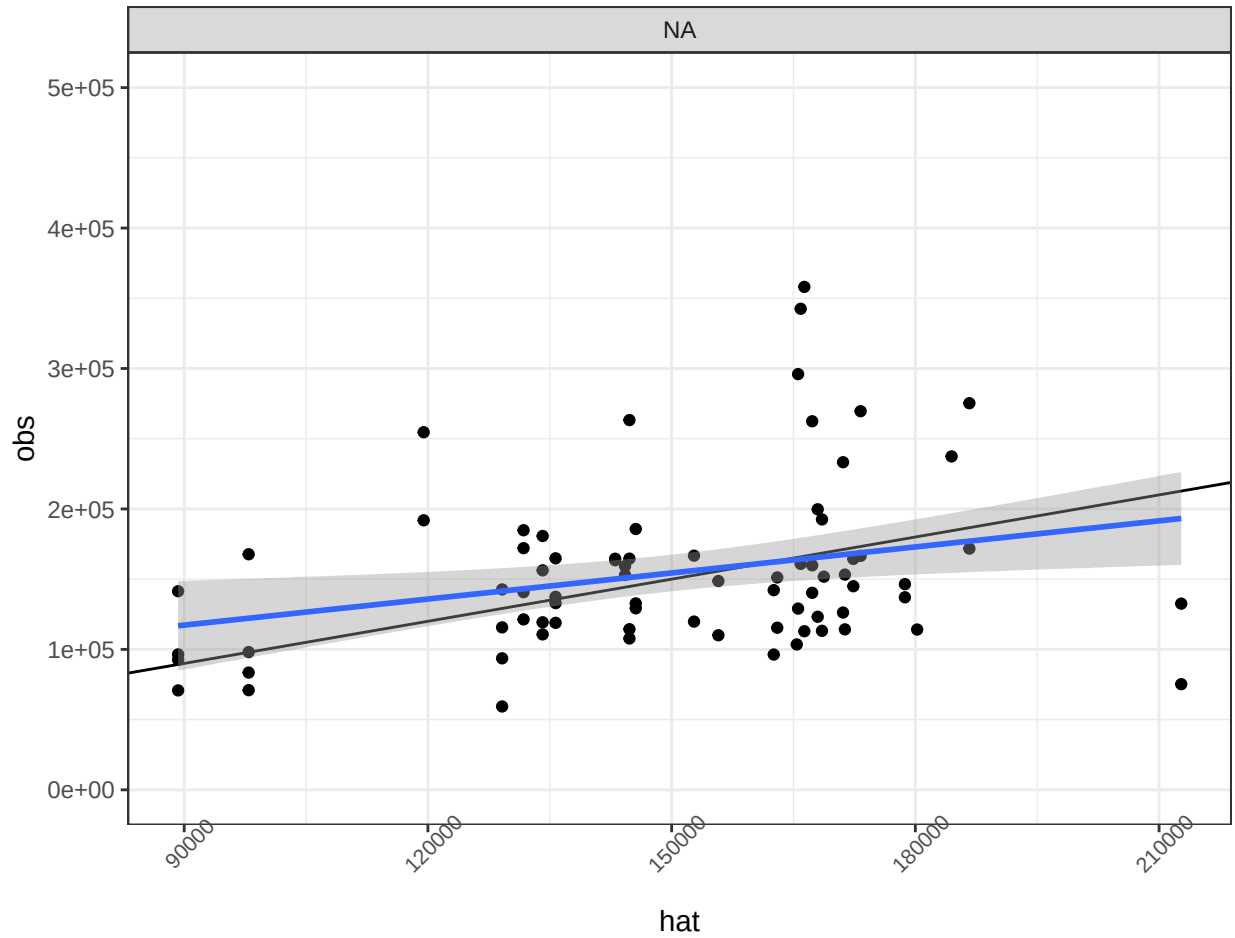


Figure 47. Calibration regression plots for age 4 for the discard run assuming constant M. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

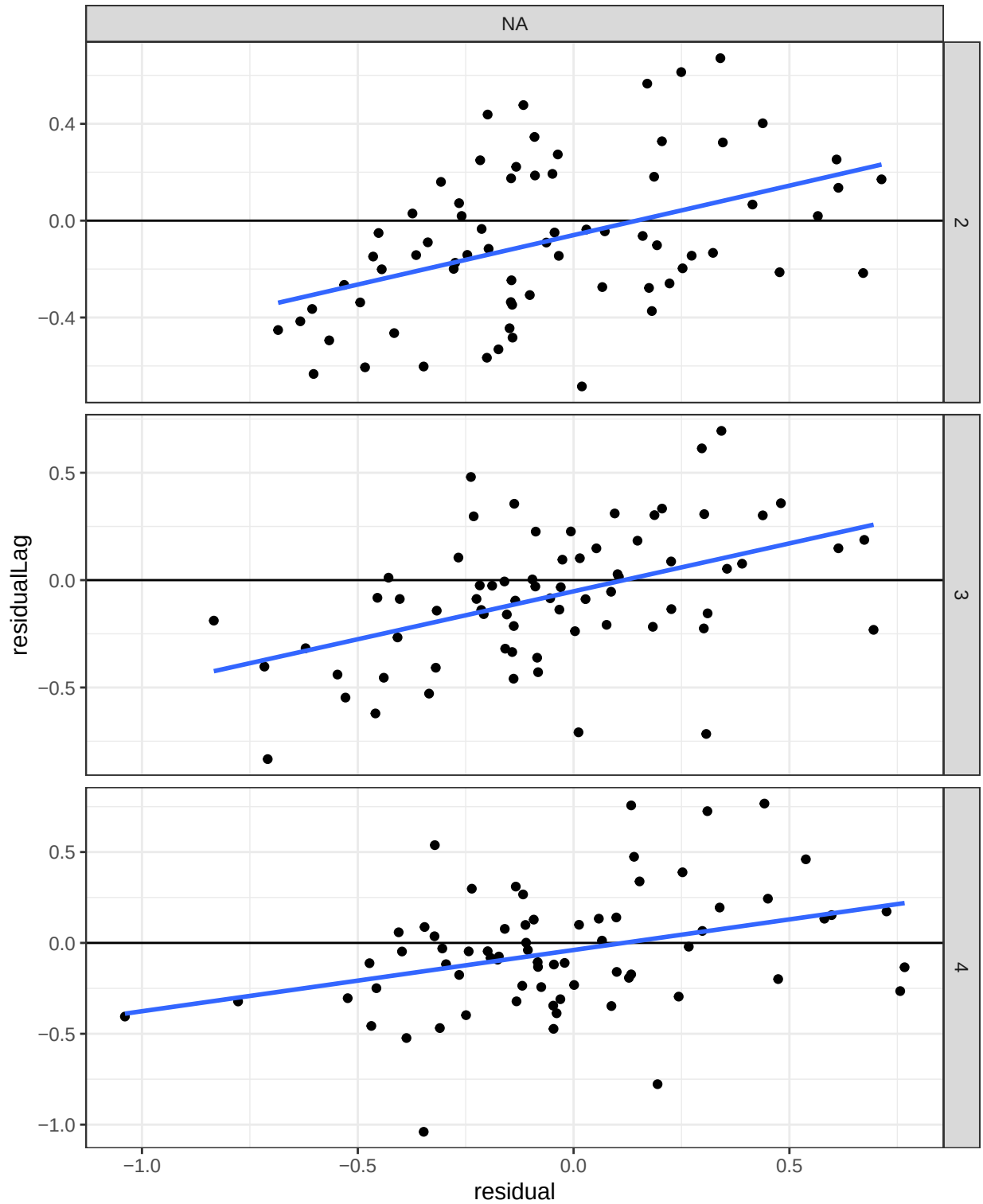


Figure 48. AR plots of lagged residuals for the discard run assuming constant M. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

```
## `geom_smooth()` using formula 'y ~ x'
```

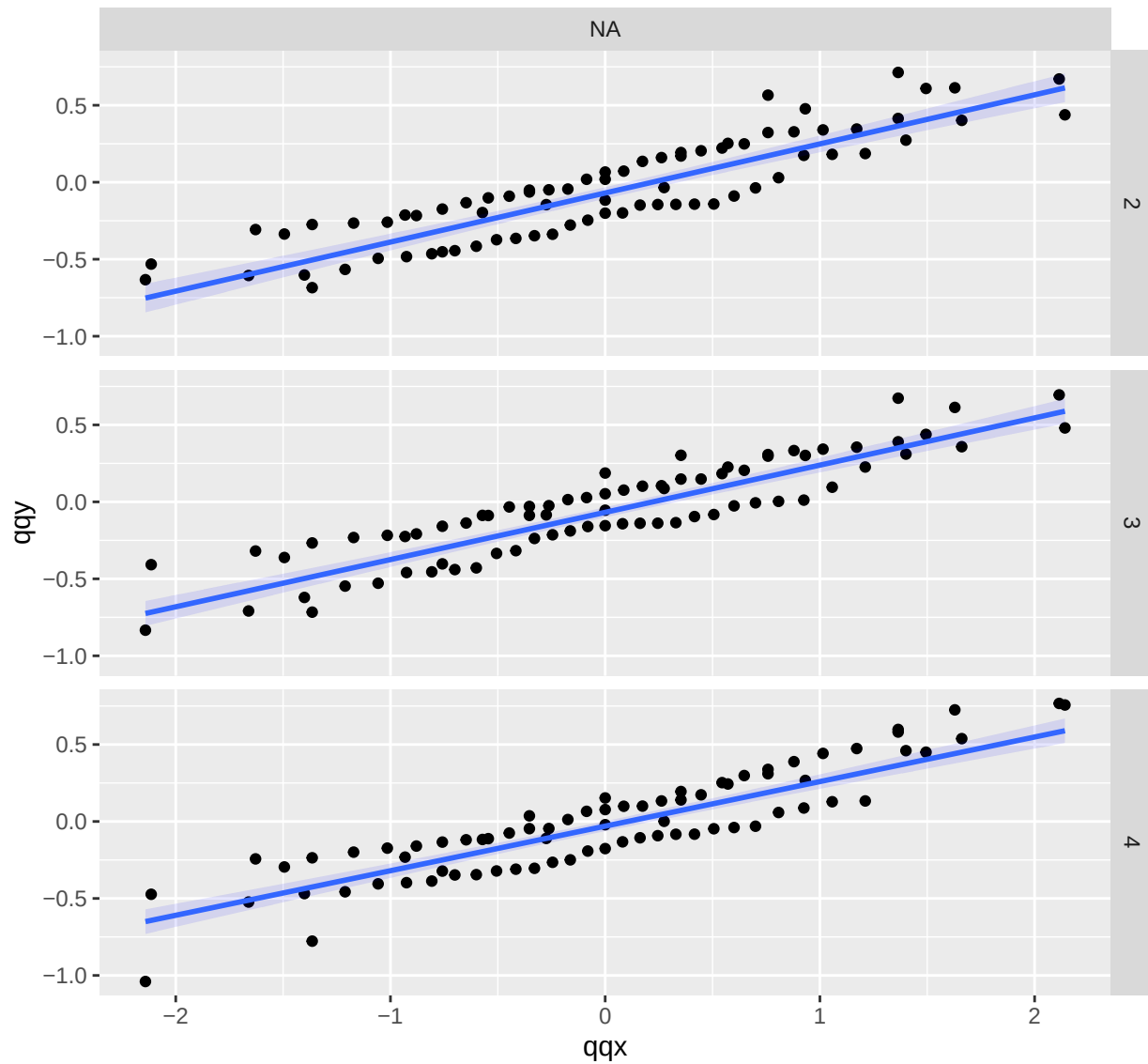


Figure 49. QQ plots to check for log normality (i.e. points and line should coincide) for the discard run assuming constant M.

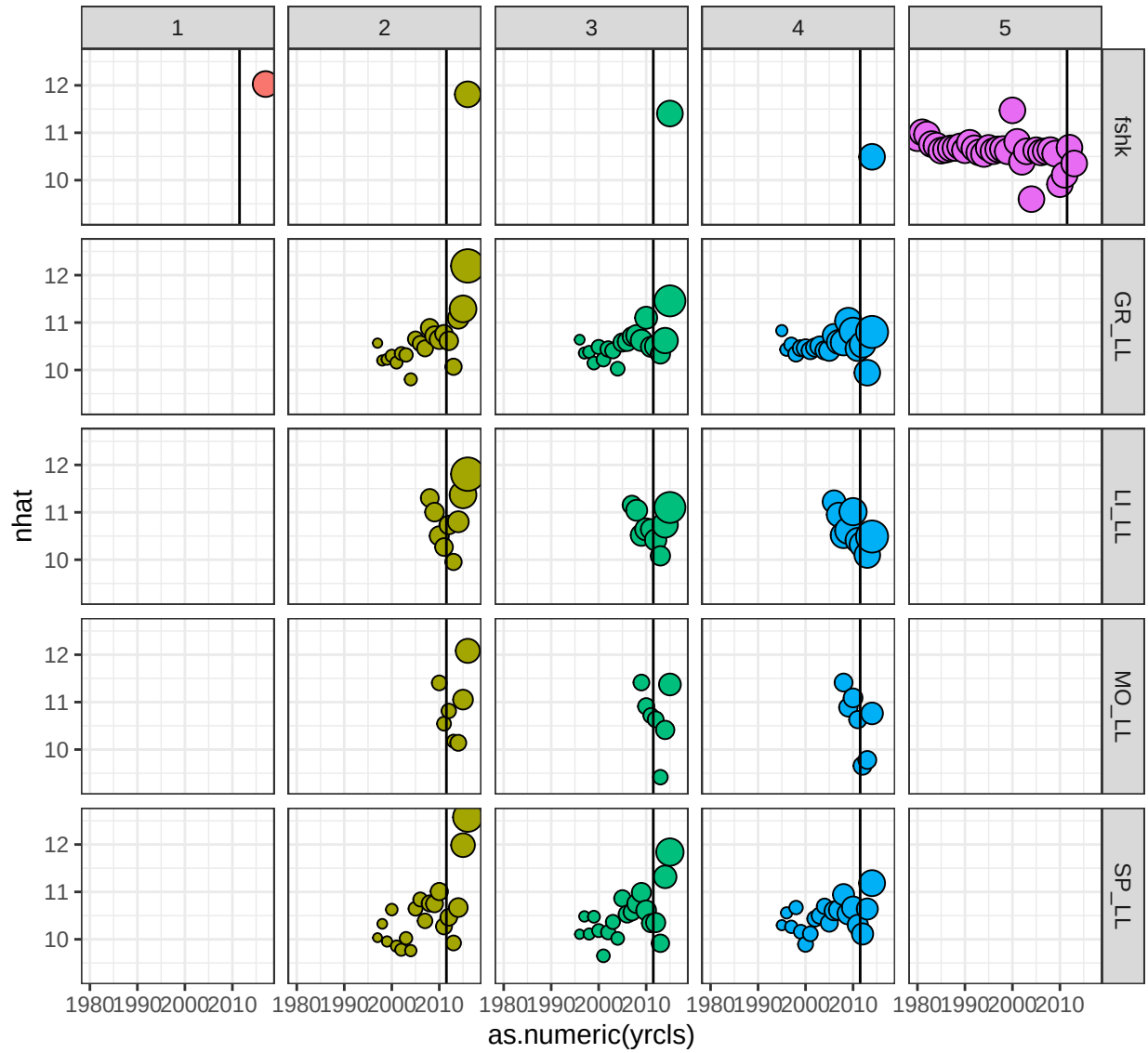


Figure 50. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(fshk)$ for the discard run assuming constant M . Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

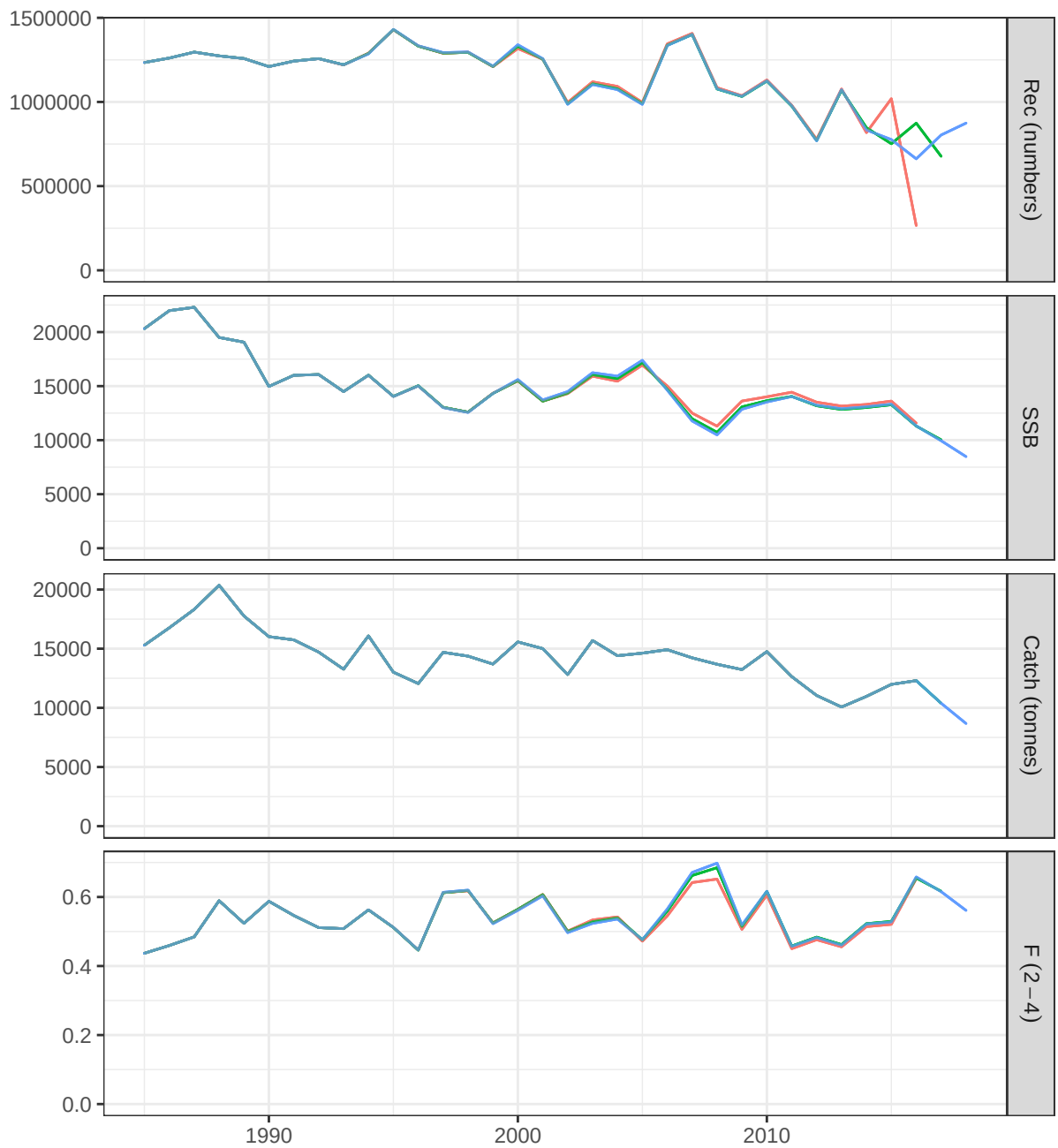


Figure 51. Retrospective plot for the discard run assuming constant M.

e. Discard run - M vector

Model estimates are plotted in **Figures 28-29**. The various diagnostic plots are shown in **Figures 30-38**, while **Figure 39** shows the retrospective plot.

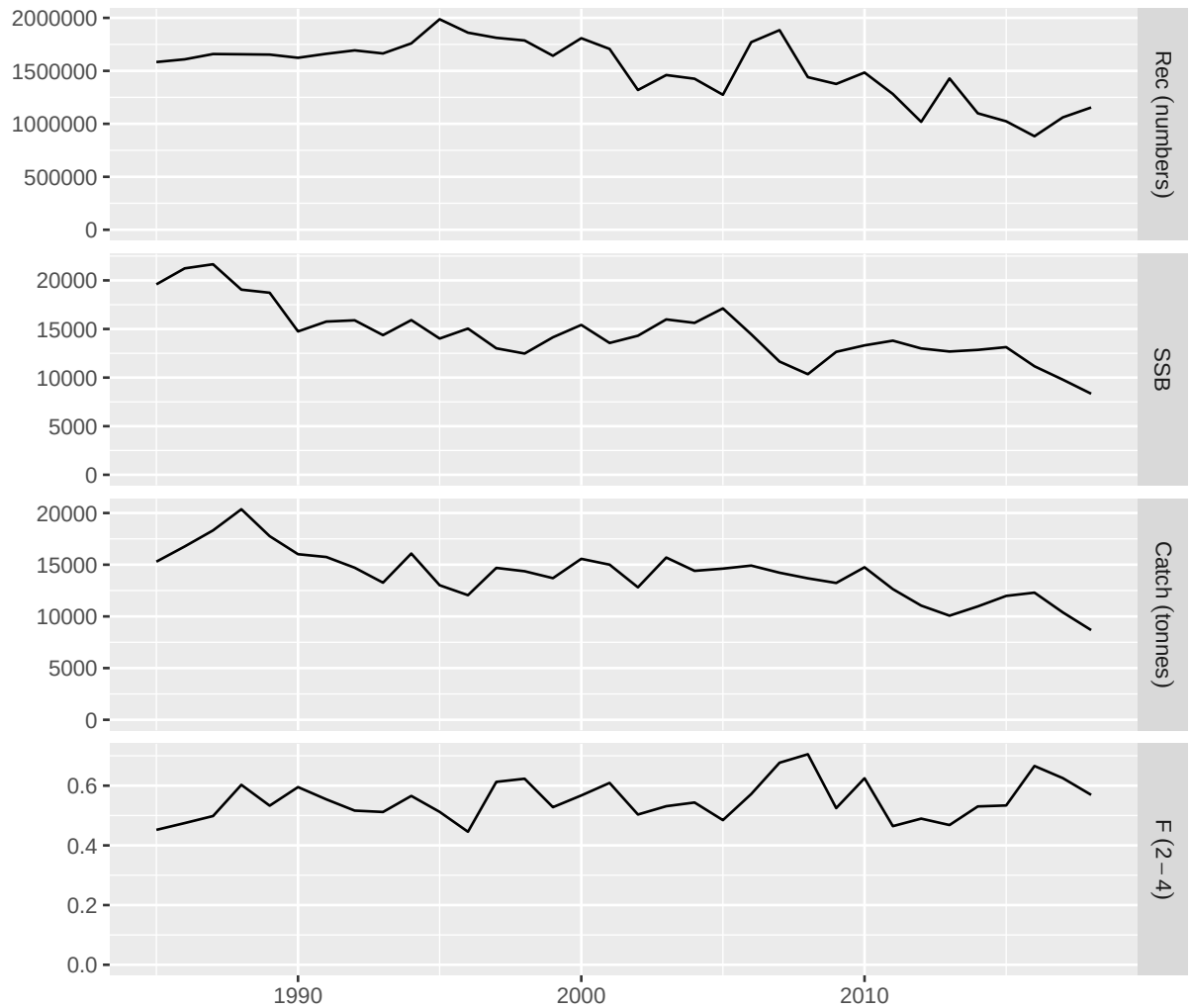


Figure 52. XSA time series estimates for the discard run assuming an M vector by age

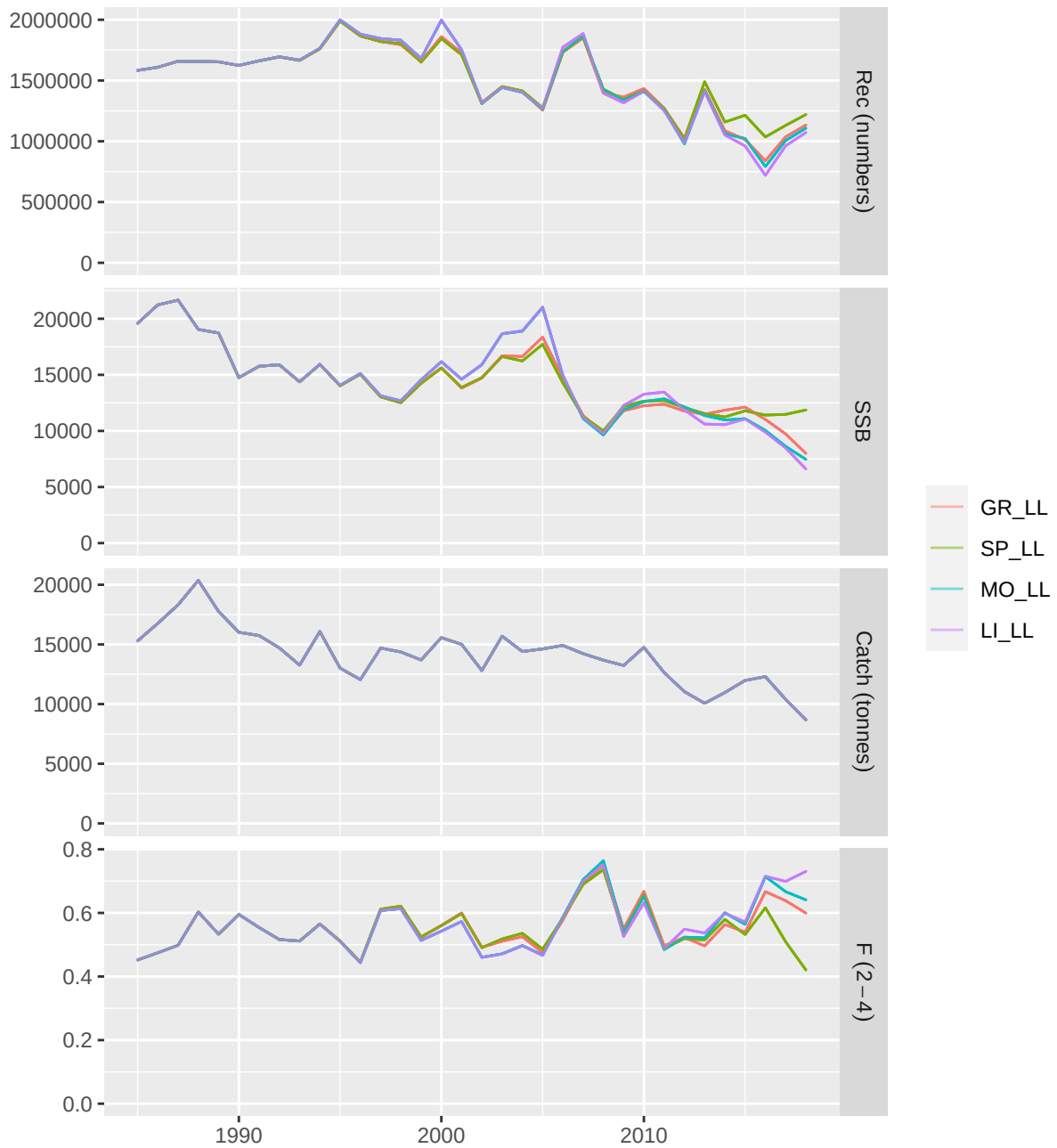


Figure 53. XSA time series estimates by CPUE, for the discard run assuming an M vector by age.

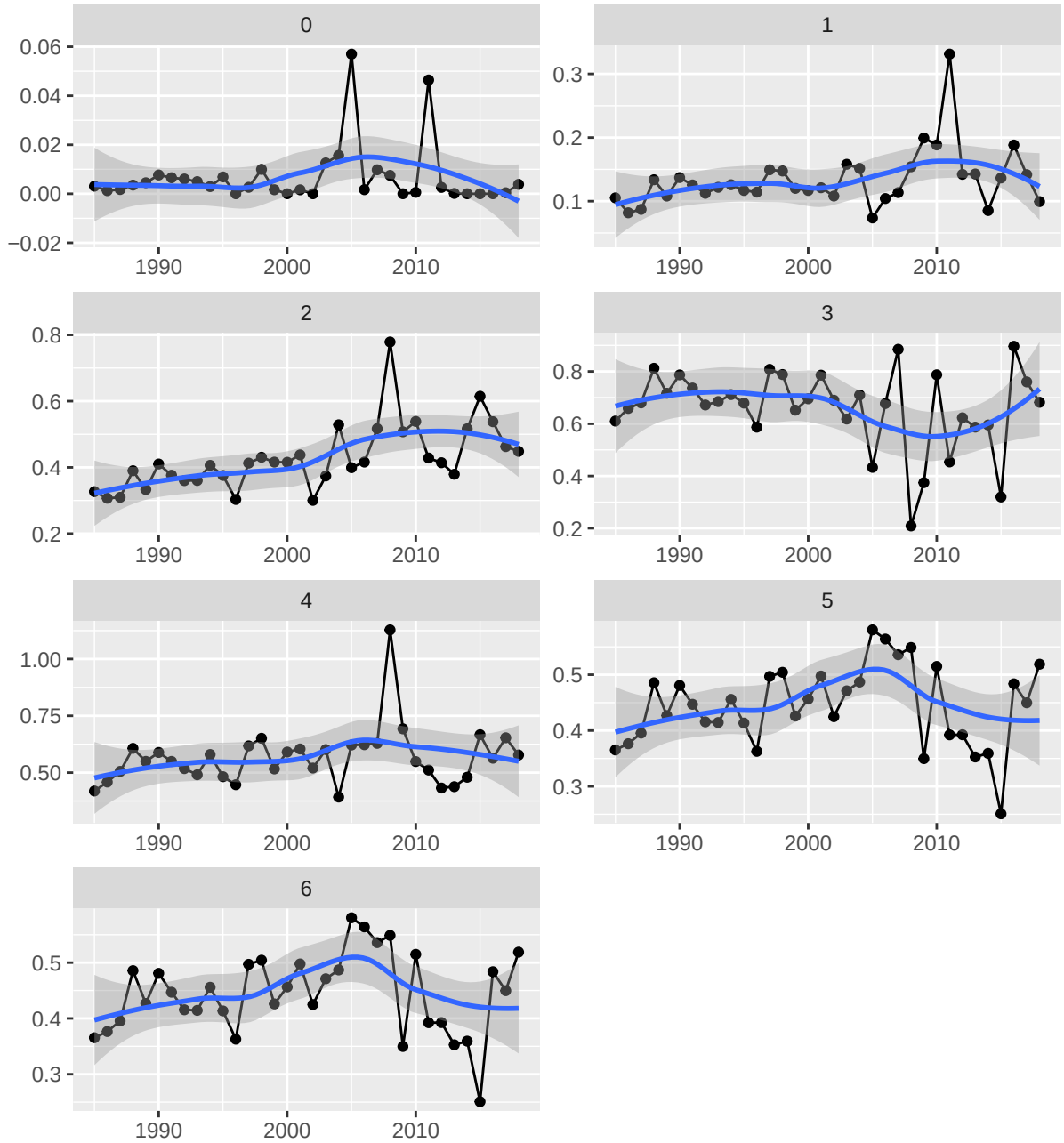


Figure 54. Estimates of harvest-at-age by year for the discard run assuming an M vector by age; continuous lines represent lowess smoothers.

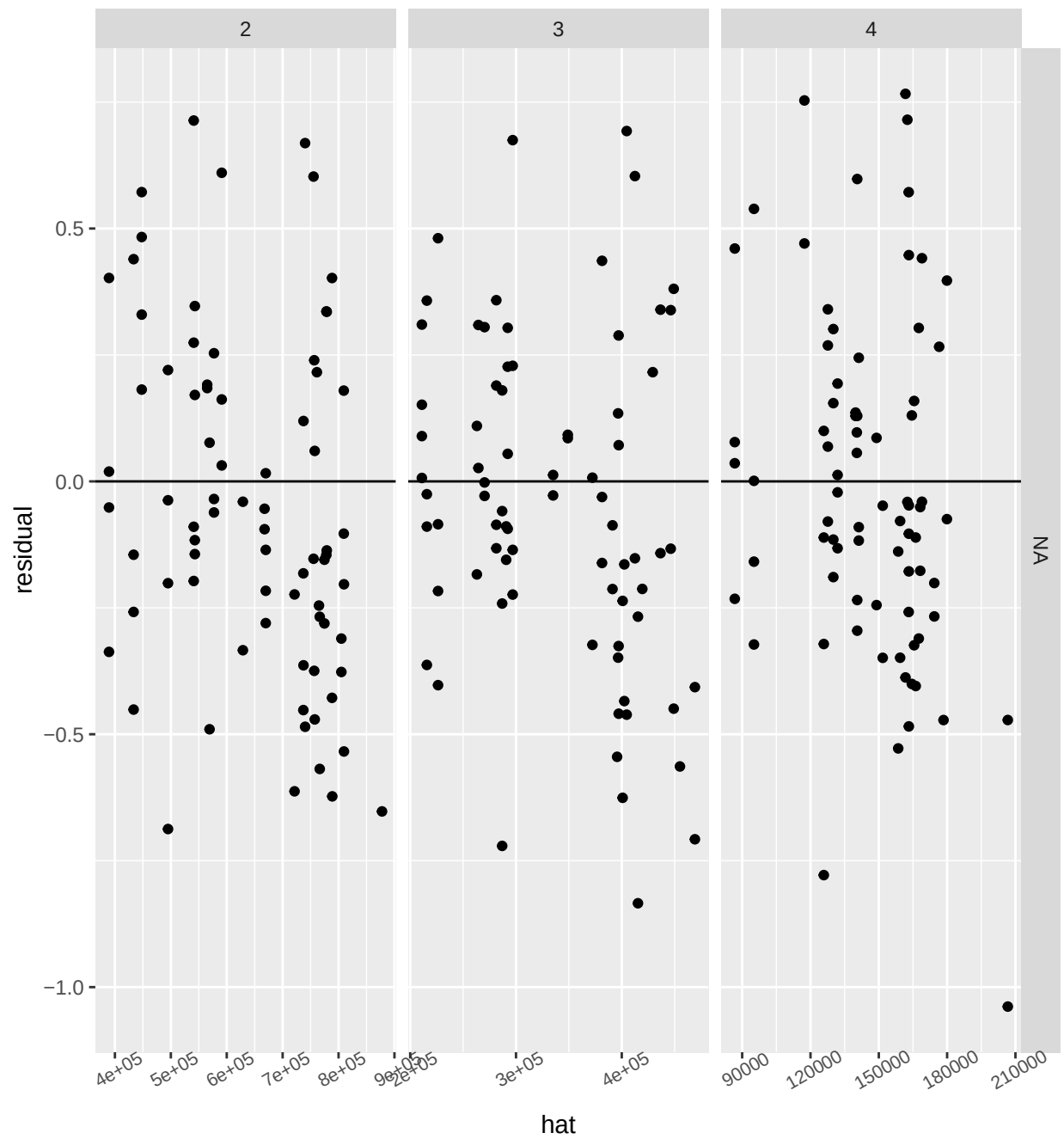
Diagnostic plots

Figure 55. Residuals against fitted value for the discard run assuming an M vector by age.

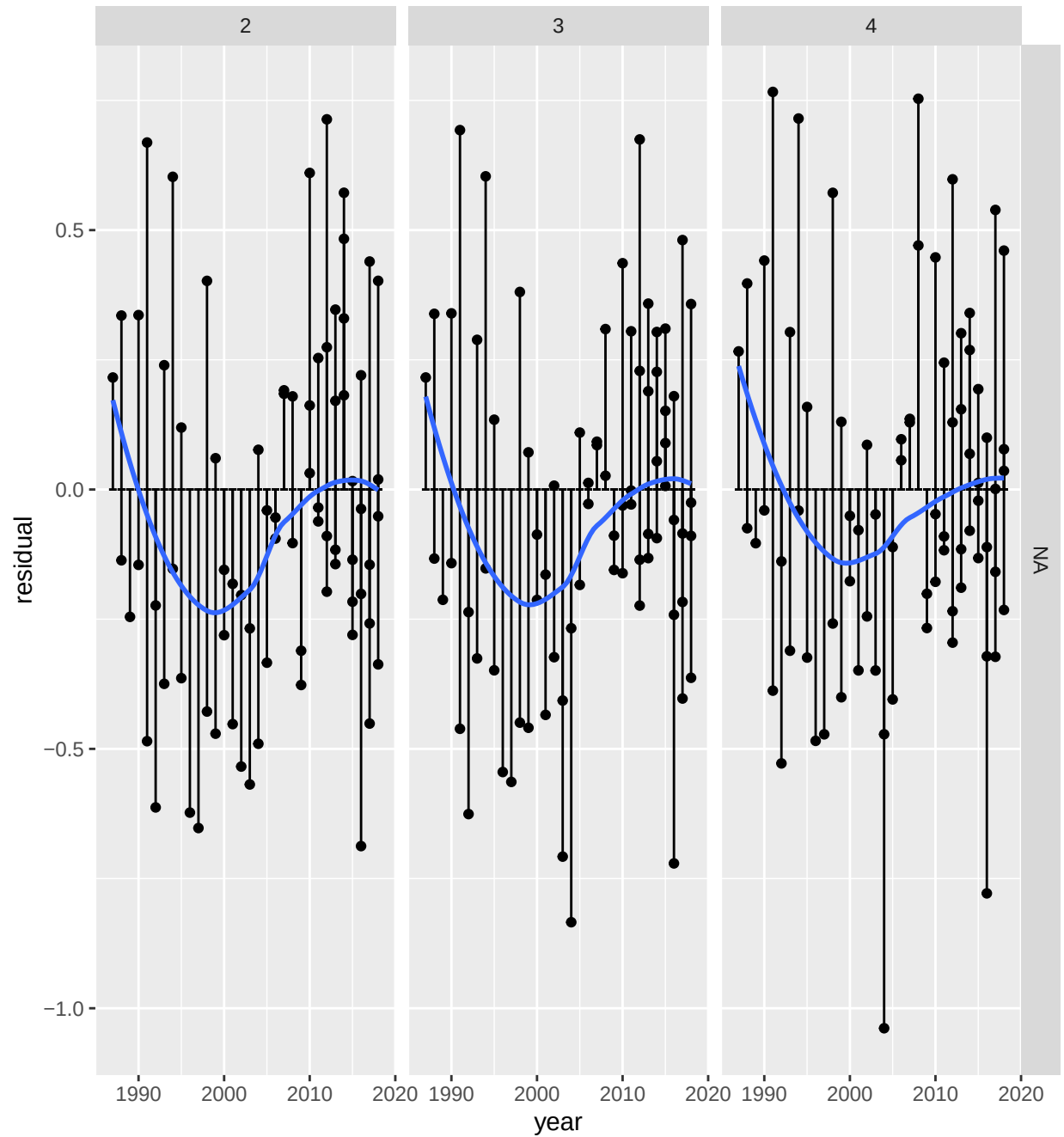


Figure 56. Residuals by fleet against year for the discard run assuming an M vector by age.

```
## `geom_smooth()` using formula 'y ~ x'
```

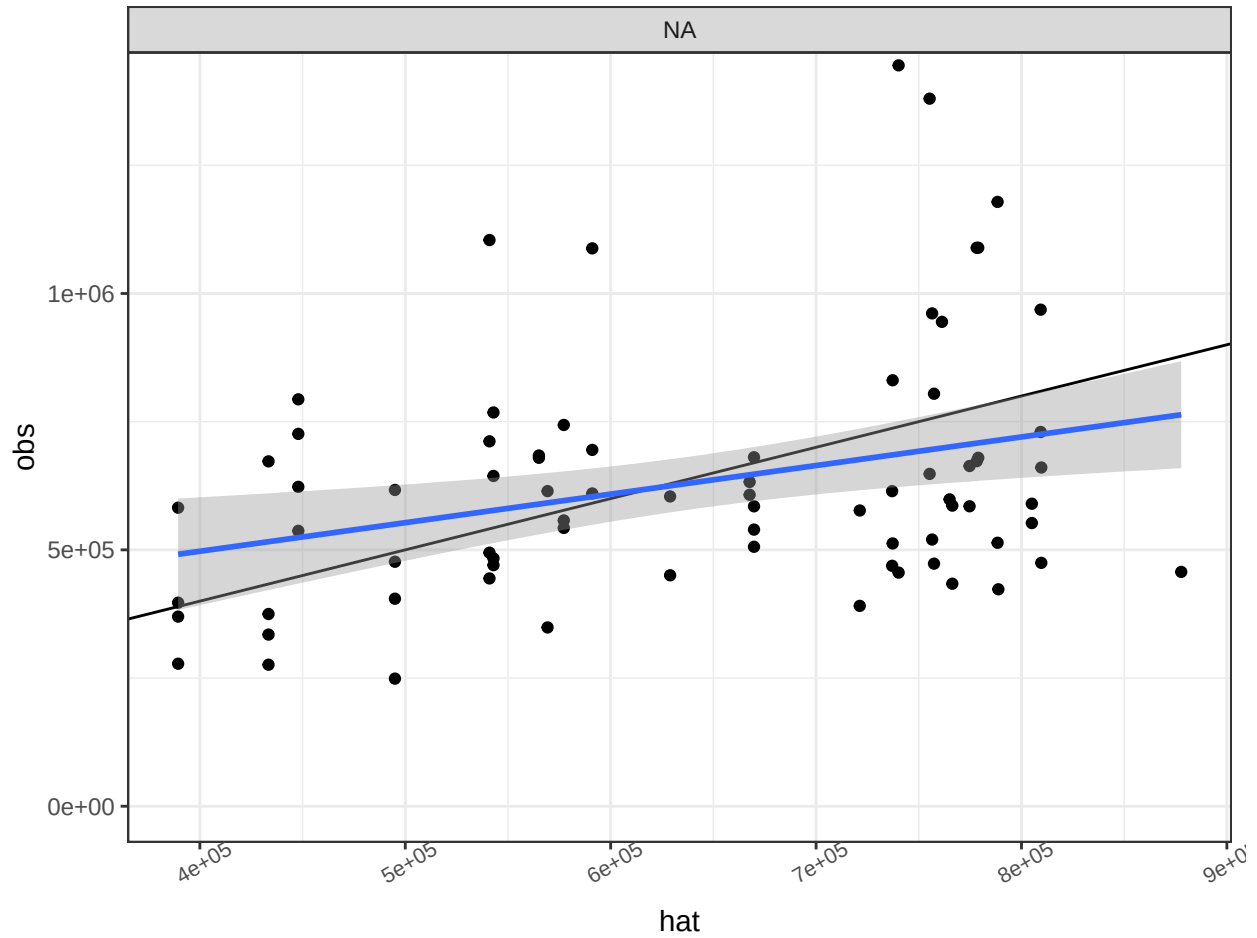


Figure 57. Calibration regression plots for age 2 for the discard run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

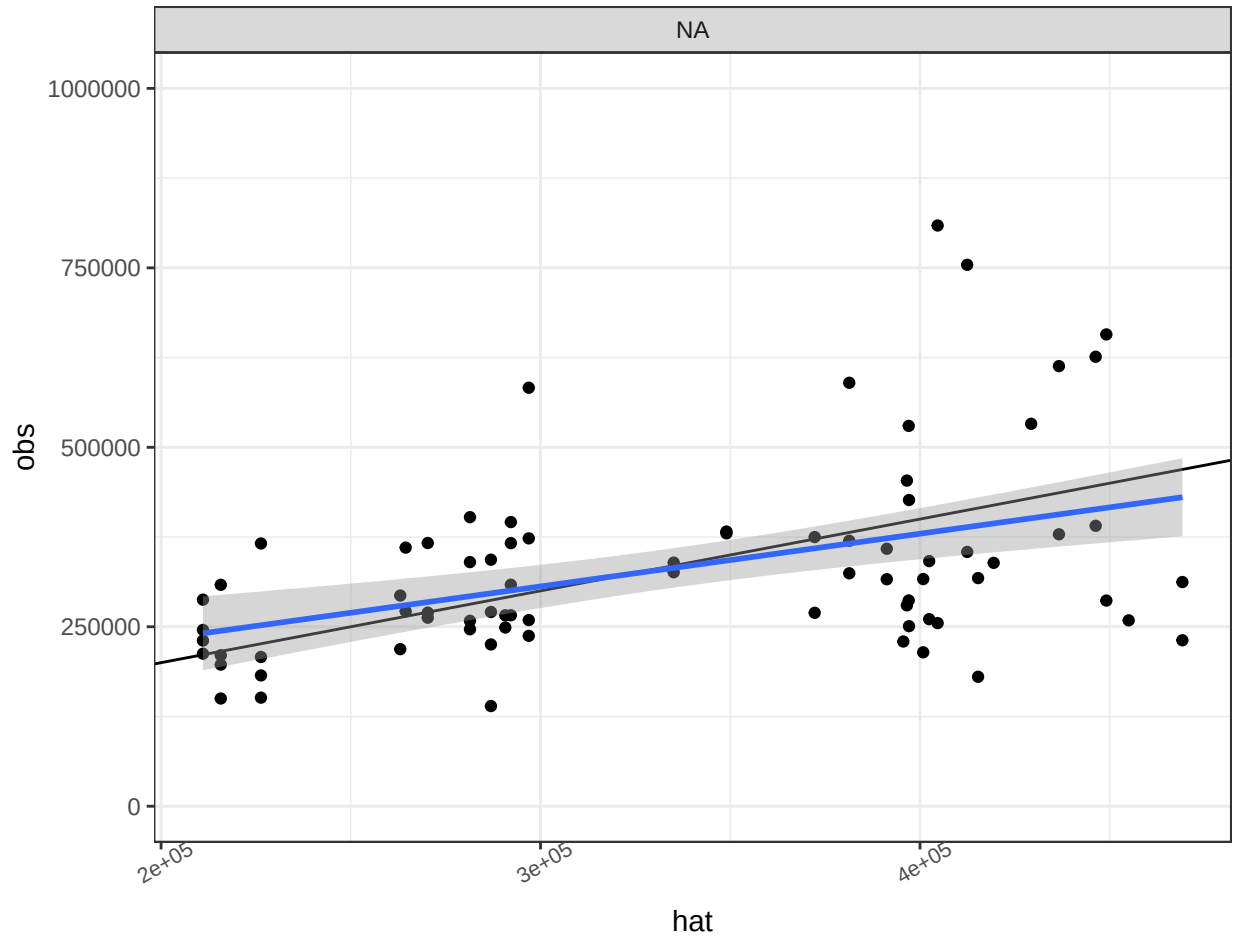


Figure 58. Calibration regression plots for age 3 for the discard run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

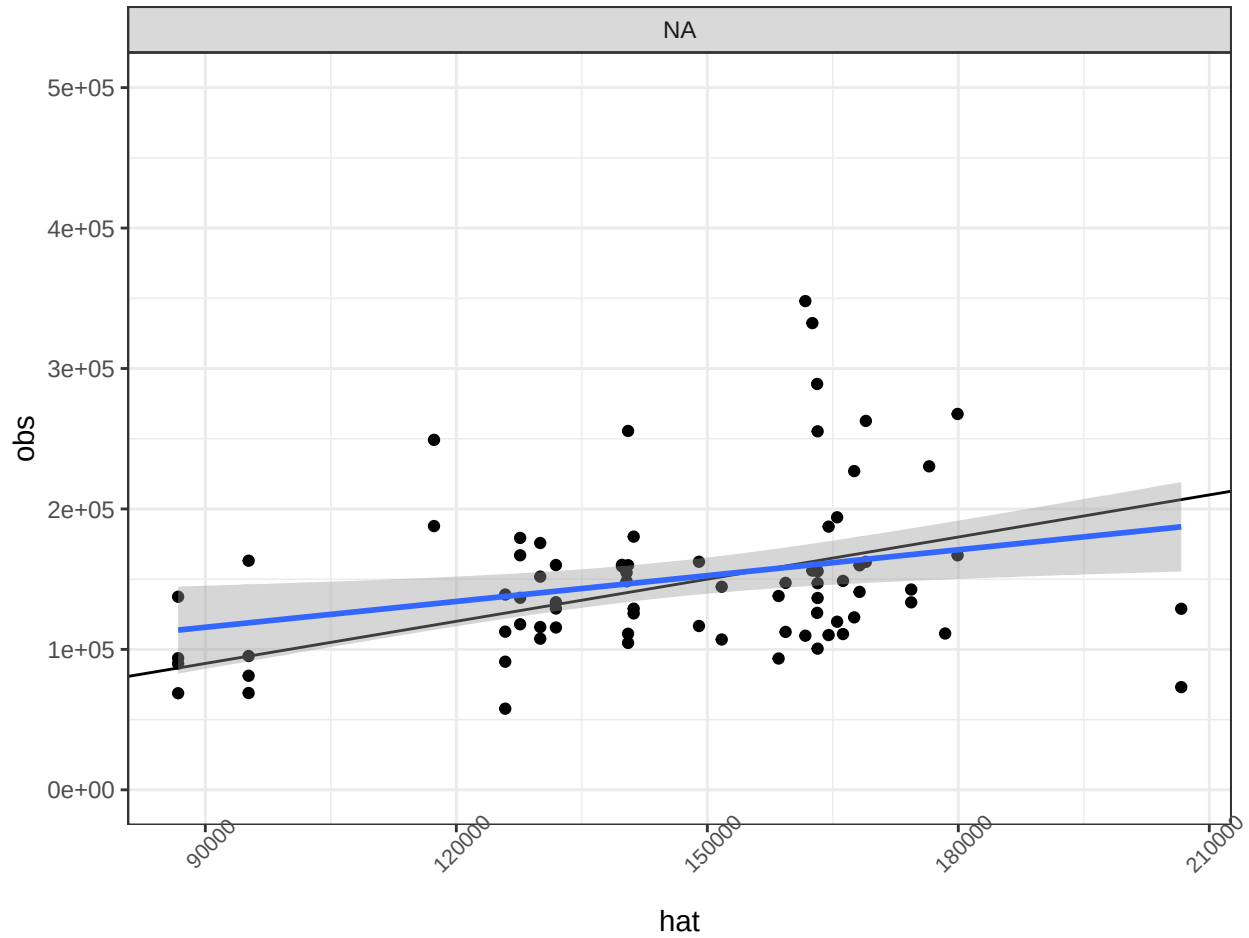


Figure 59. Calibration regression plots for age 4 for the discard run assuming an M vector by age. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

```
## `geom_smooth()` using formula 'y ~ x'
```

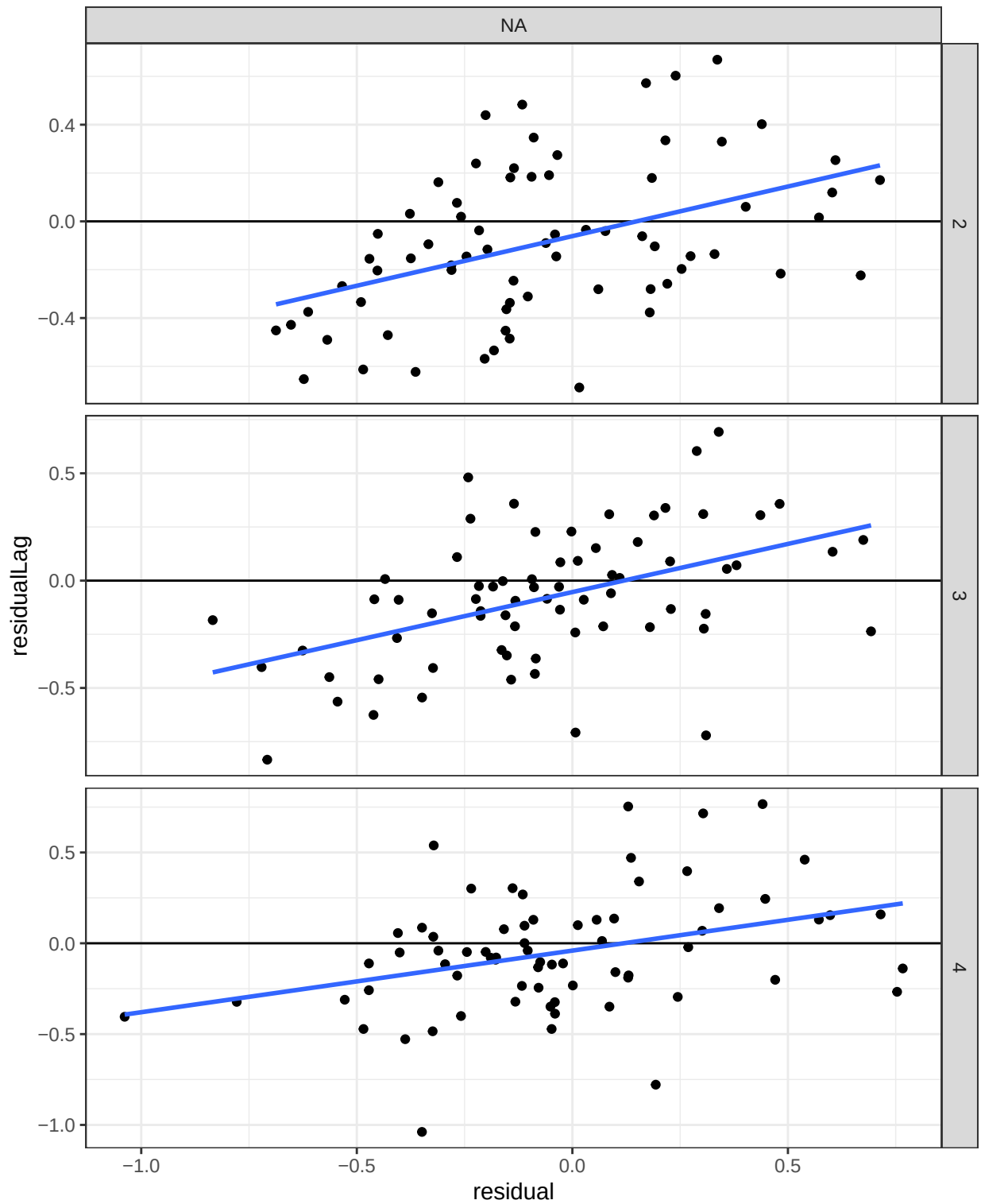


Figure 60. AR plots of lagged residuals for the discard run assuming an M vector by age. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

```
## `geom_smooth()` using formula 'y ~ x'
```

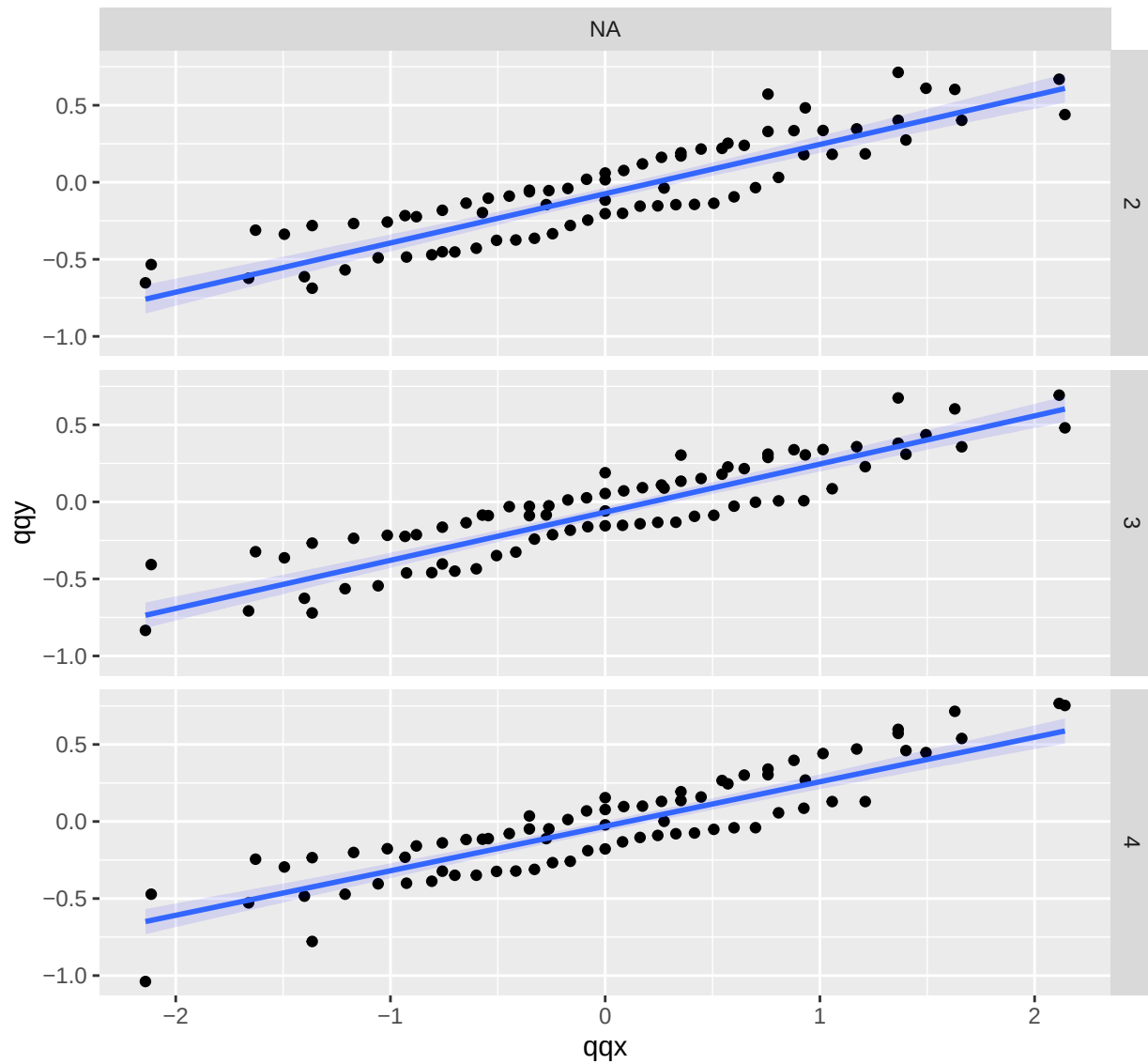


Figure 61. QQ plots to check for log normality (i.e. points and line should coincide) for the discard run assuming an M vector by age.

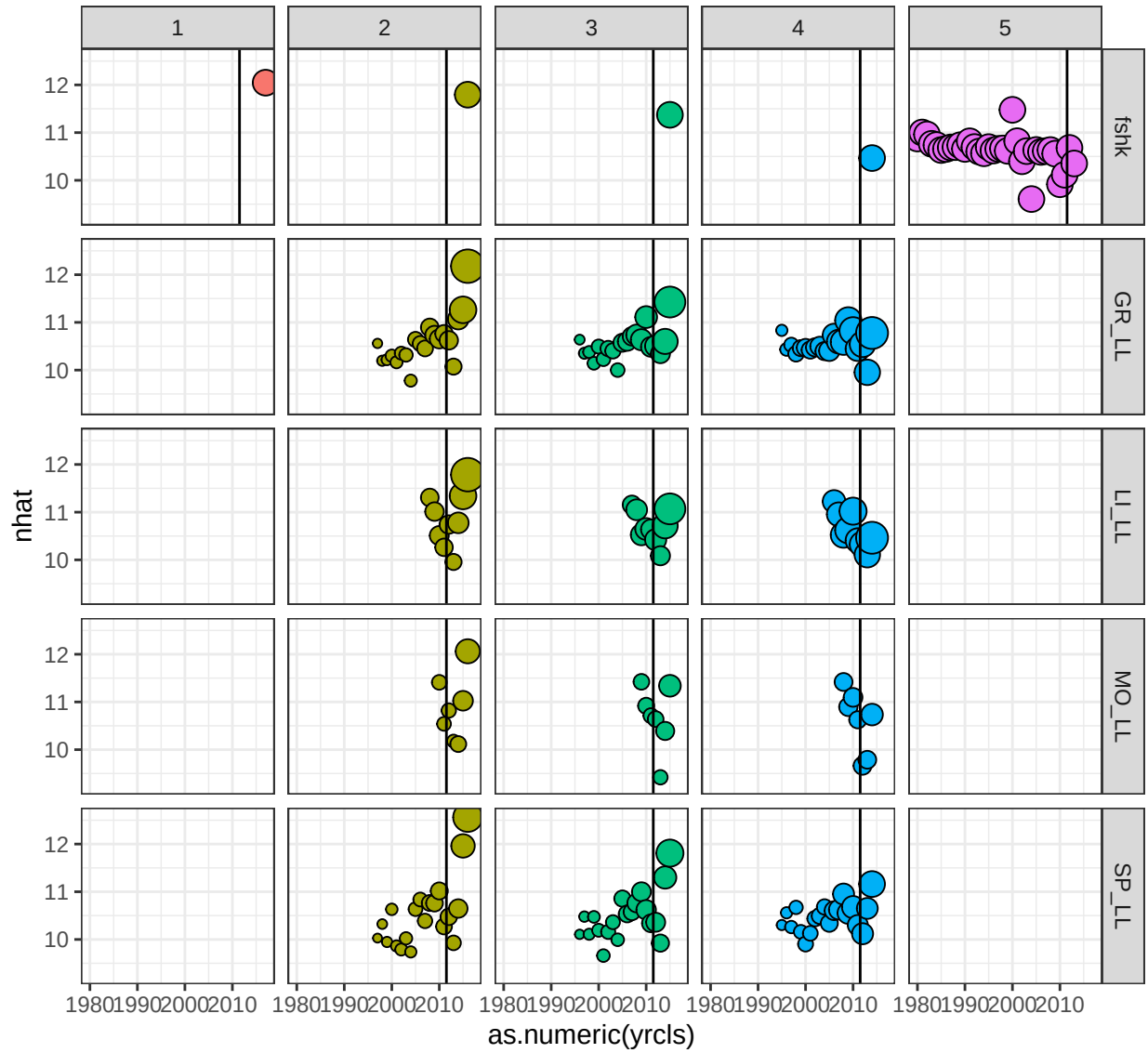


Figure 62. Weights for terminal year N_s for each CPUE observation and shrinkage to the mean $F(\text{fshk})$ for the discard run assuming an M vector by age. Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

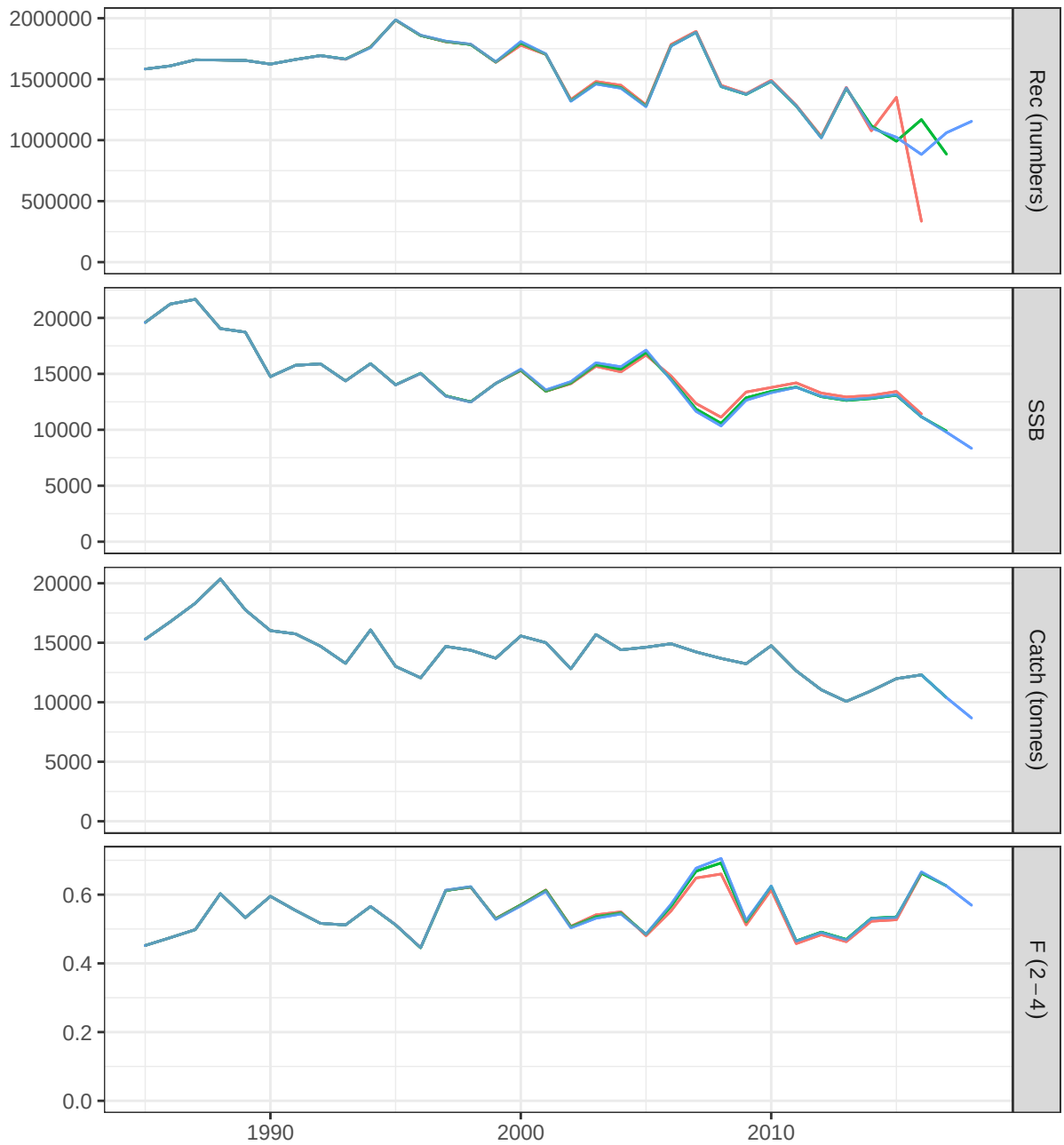


Figure 63. Retrospective plot for the discard run assuming an M vector by age.

f. Comparison among candidate and discard runs

Estimates of the two candidate and the two discard runs are plotted in **Figure 64**.

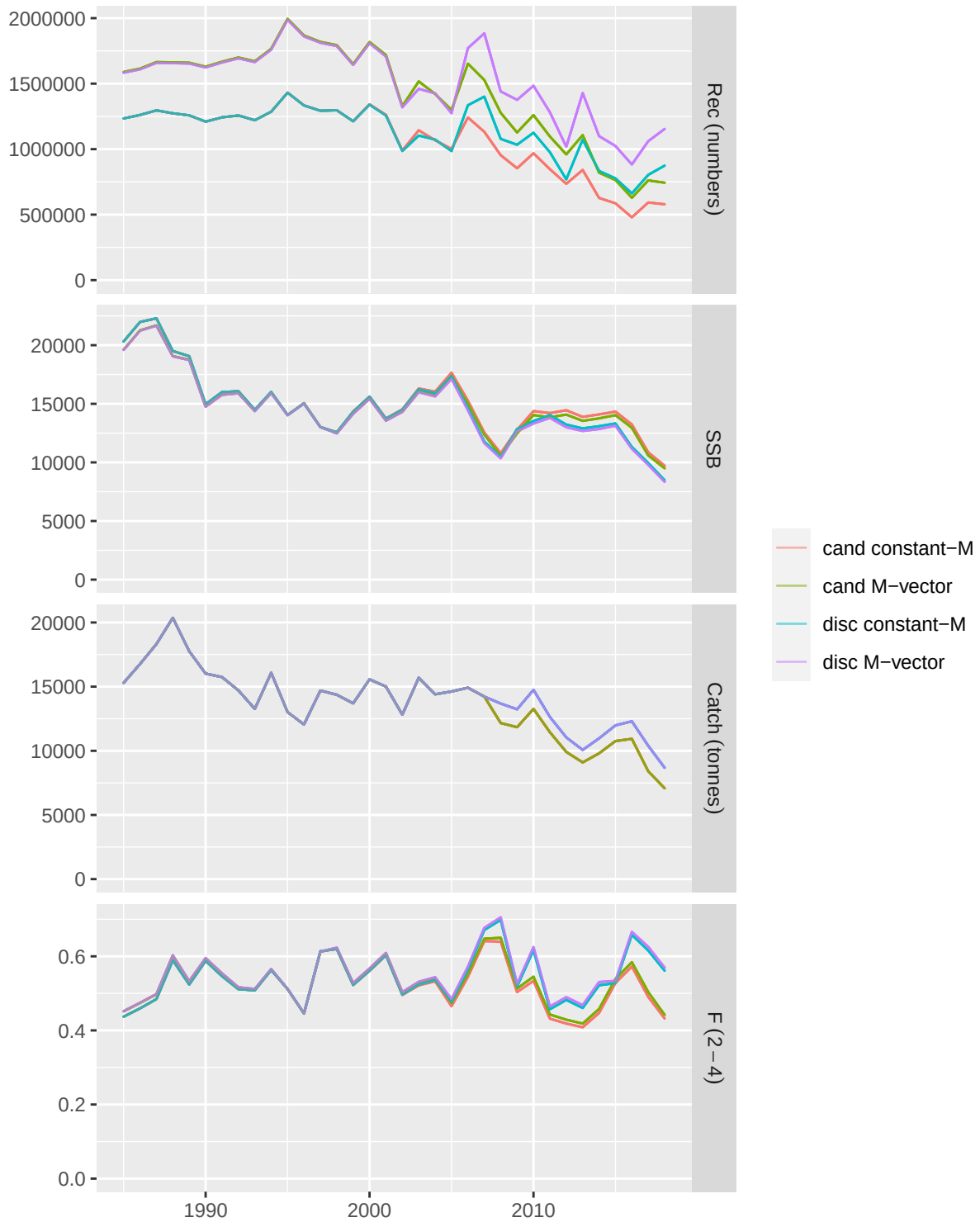


Figure 64. Comparison of estimates among candidate and discard runs.

g. Biological reference points

Estimates of the different XSA runs were used to evaluate the state of the stock and obtain biological reference points based upon yield per recruit and stock recruit equilibrium calculations, assuming a Stock-Recruitment Relationship (SRR) of the Beverton and Holt functional form. Given the lack of sufficient data for a proper fit an empirical relationship with high steepness value was developed. **Figures 65-69** illustrate the equilibrium curves, while the estimated reference points are shown in **Table 1**. Results indicate that the stock is overexploited and, depending on the run (candidate or discard), its current biomass is 4-6 times lower than SSB_{MSY} . Current F over F_{MSY} , ranges from 1.3-1.5 and 1.6-1.9 for the candidate and discard runs, respectively.

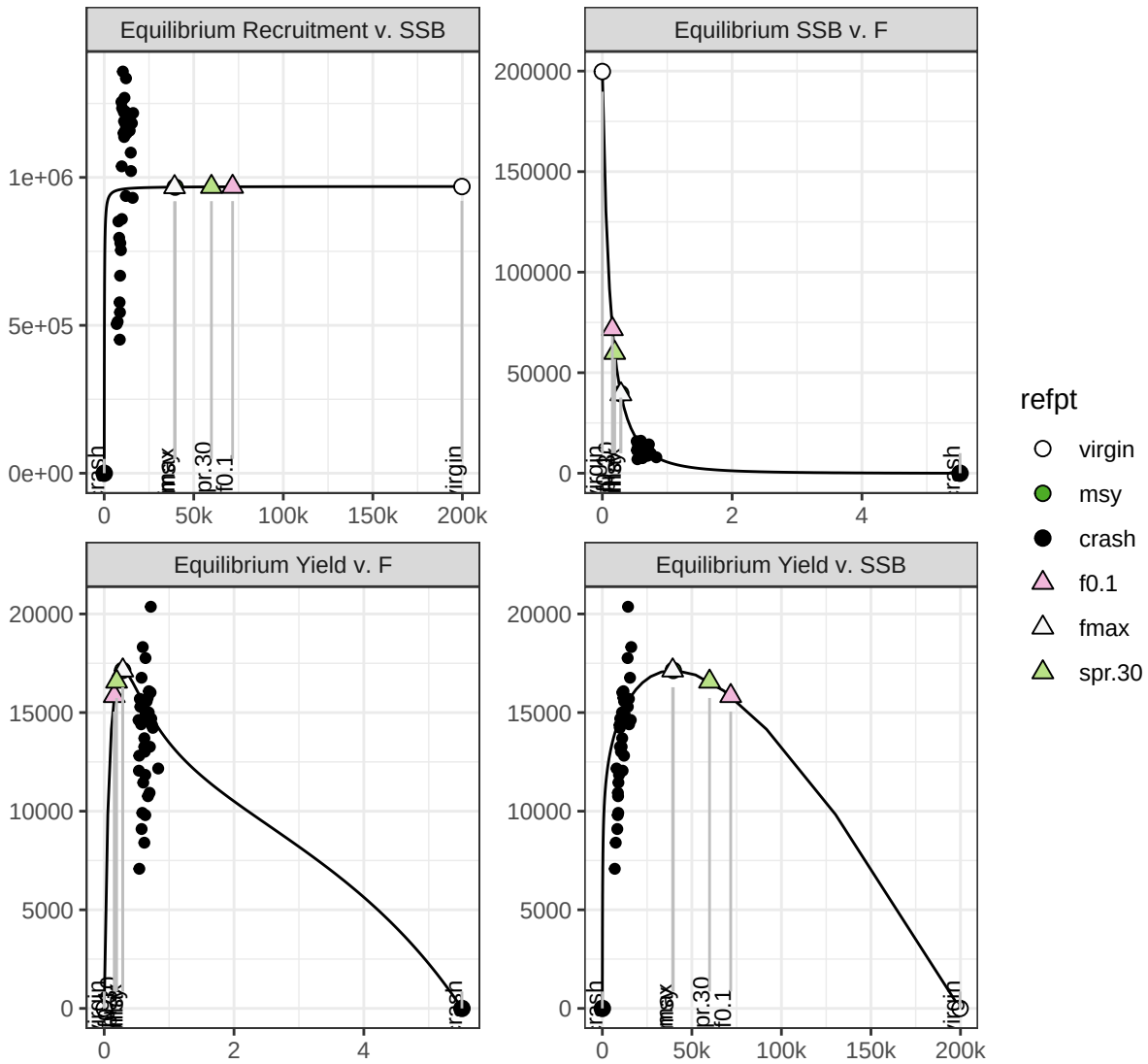


Figure 65. Equilibrium curves with data pairs based on the continuity run.

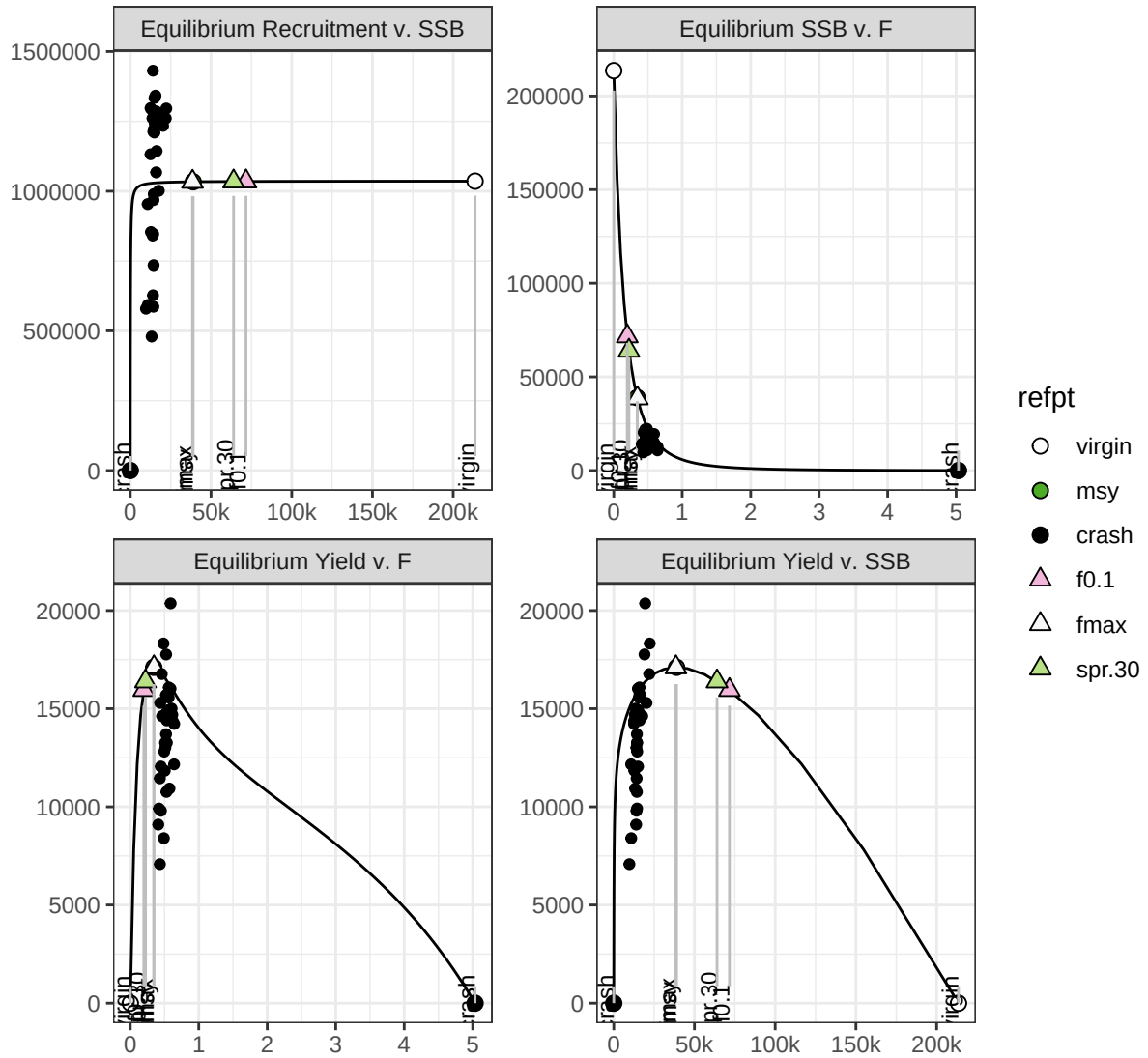


Figure 66. Equilibrium curves with data pairs based on the candidate run assuming constant M .

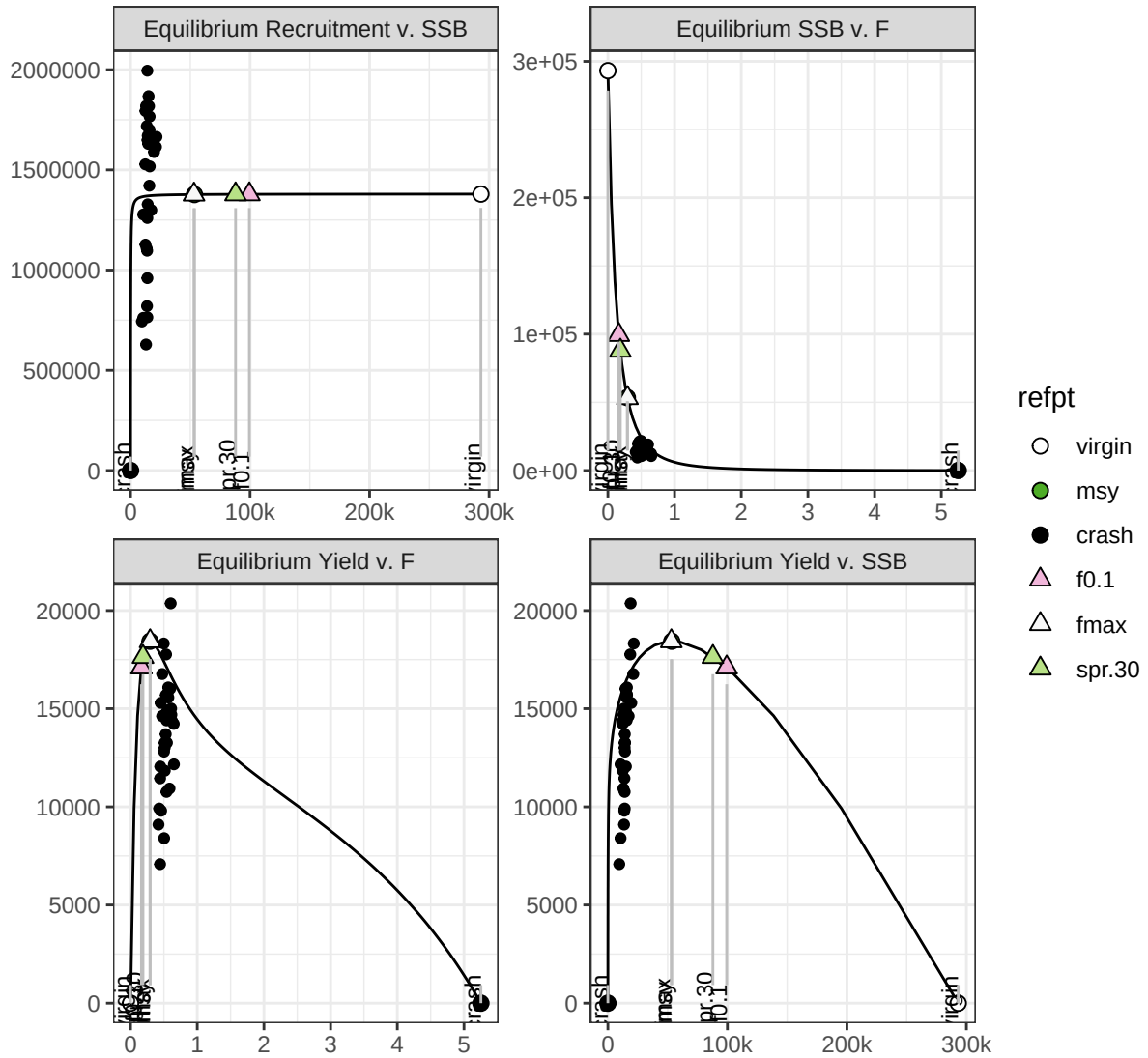


Figure 67. Equilibrium curves with data pairs based on the candidate run assuming an M vector.

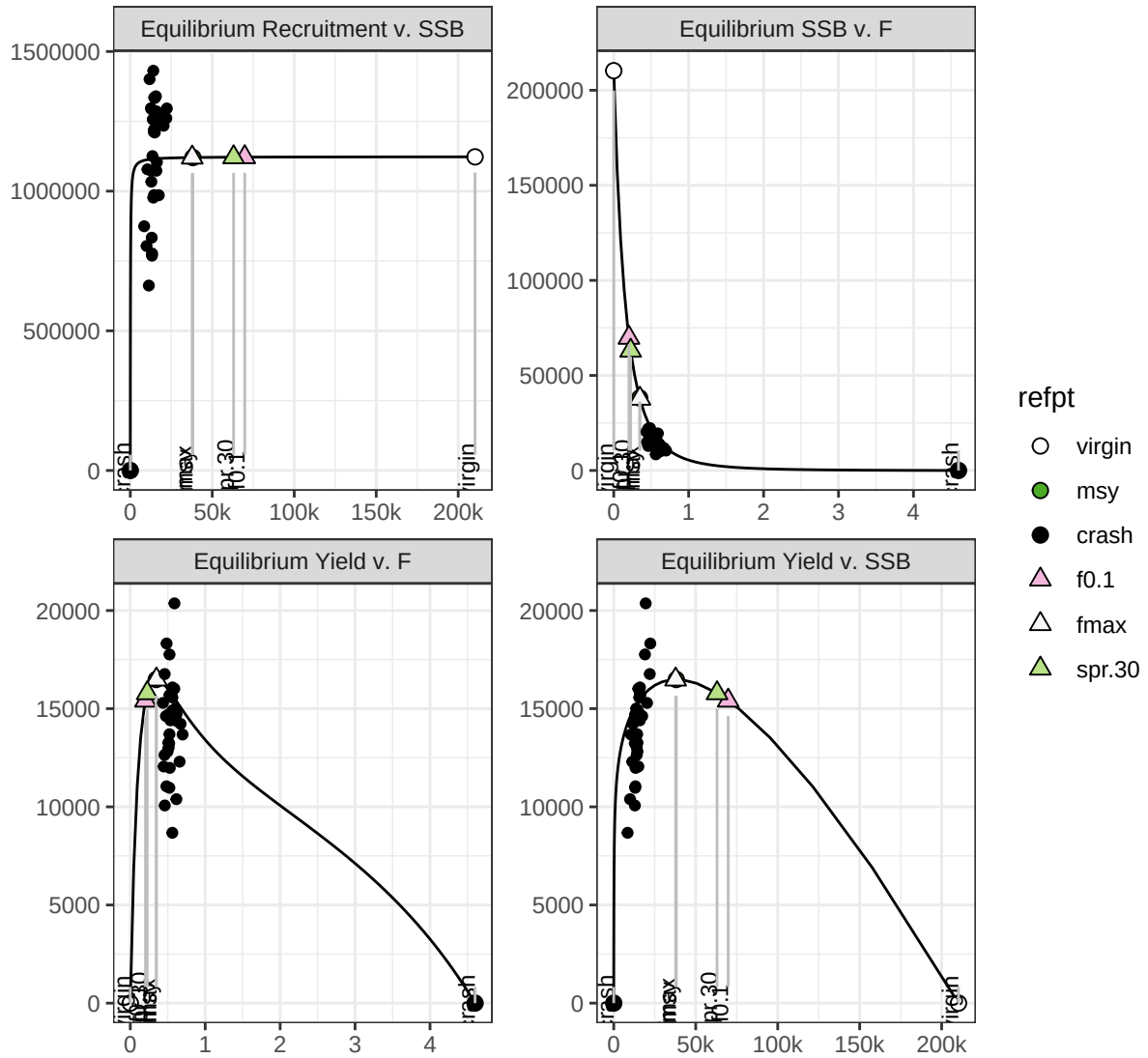


Figure 68. Equilibrium curves with data pairs based on the discard run assuming constant M.

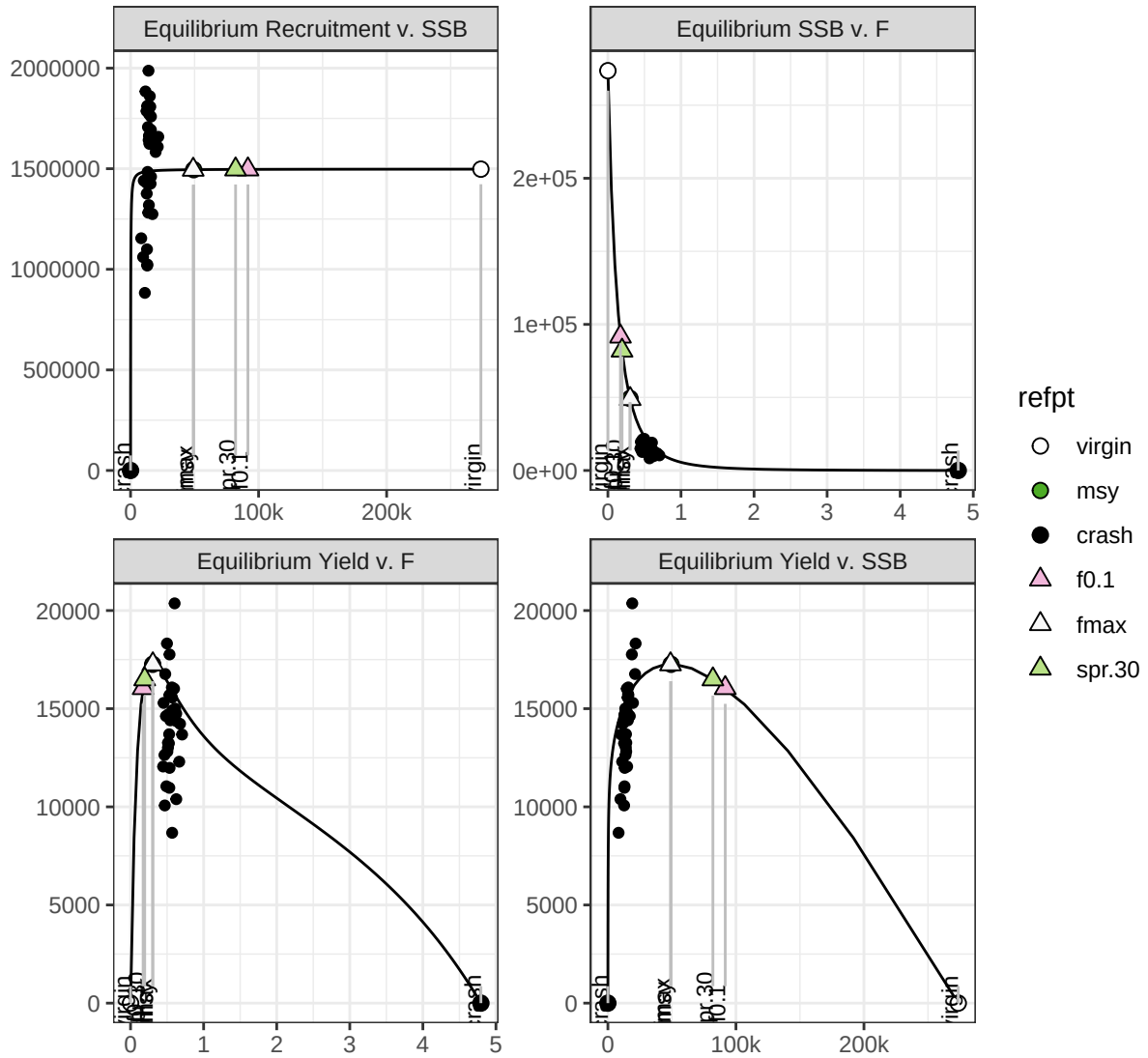


Figure 69. Equilibrium curves with data pairs based on the discard run assuming an M vector.

An object of class "FLPar"

```

      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 9.69e+05 2.00e+05 2.27e+05
    msy   2.84e-01 1.71e+04 9.67e+05 3.96e+04 6.28e+04
    f0.1  1.55e-01 1.58e+04 9.68e+05 7.17e+04 9.64e+04
    spr.30 1.91e-01 1.66e+04 9.68e+05 5.99e+04 8.41e+04
units:  NA

```

An object of class "FLPar"

```

      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.04e+06 2.14e+05 2.42e+05
    msy   3.43e-01 1.71e+04 1.03e+06 3.89e+04 6.27e+04
    f0.1  1.98e-01 1.60e+04 1.04e+06 7.17e+04 9.72e+04
    spr.30 2.23e-01 1.64e+04 1.03e+06 6.40e+04 8.92e+04
units:  NA

```

An object of class "FLPar"

```

      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.38e+06 2.93e+05 3.24e+05
    msy   2.92e-01 1.84e+04 1.38e+06 5.35e+04 7.97e+04
    f0.1  1.63e-01 1.71e+04 1.38e+06 9.94e+04 1.27e+05
    spr.30 1.87e-01 1.76e+04 1.38e+06 8.79e+04 1.15e+05
units:  NA

```

An object of class "FLPar"

```

      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.12e+06 2.10e+05 2.38e+05
    msy   3.46e-01 1.65e+04 1.12e+06 3.82e+04 6.12e+04
    f0.1  2.03e-01 1.54e+04 1.12e+06 6.98e+04 9.47e+04
    spr.30 2.26e-01 1.58e+04 1.12e+06 6.30e+04 8.76e+04
units:  NA

```

An object of class "FLPar"

```

      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.50e+06 2.74e+05 3.03e+05
    msy   3.03e-01 1.73e+04 1.50e+06 4.93e+04 7.44e+04
    f0.1  1.72e-01 1.60e+04 1.50e+06 9.17e+04 1.18e+05
    spr.30 1.94e-01 1.65e+04 1.50e+06 8.20e+04 1.09e+05
units:  NA

```

Table 1. Estimates of reference points. From top to bottom: continuity run, candidate run-constant M, candidate run-M vector, discard run-constant M, discard run-M vector.

h. Selection-at-age

Based on the XSA runs, **Figure 70** illustrates the fishery selection pattern. In all runs, apart from the continuity, full selection is achieved at age 3 and it is decreasing afterwards.

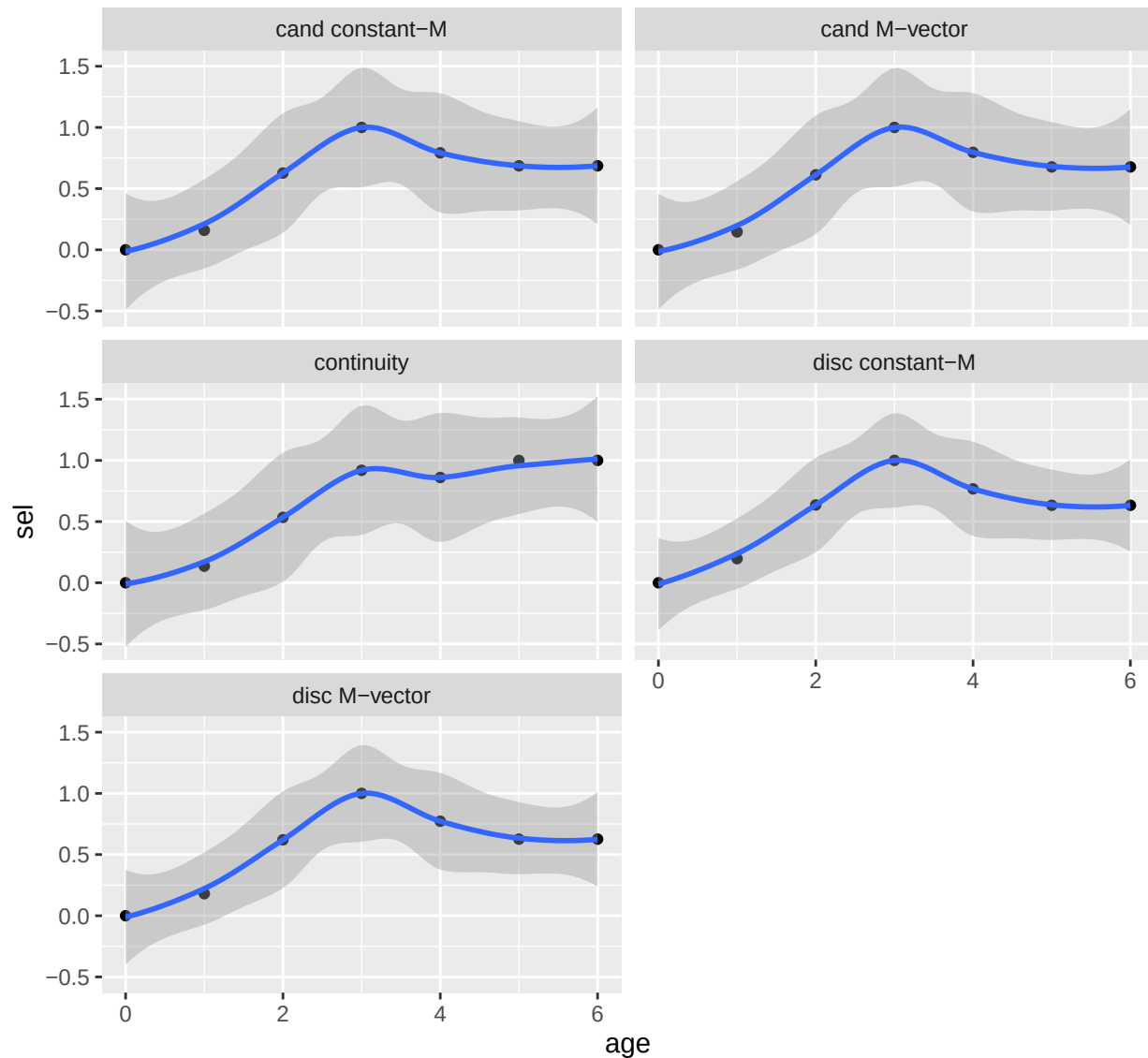


Figure 70. Estimates of selection by age from the different model runs.

Discussion

- Both candidate runs gave comparable results and provided slightly better diagnostics than the continuity run. Additionally, contrary to the continuity run, do not show any particular retrospective pattern, meaning that their results are much more robust to uncertainty with respect to the CPUE indices. Fit quality of the discard runs was comparable to that of the candidate ones.
- Discard runs show relatively higher recruitment and mortality levels than the candidate ones.
- Although catches have declined in the recent years and harvest rates are at rather lower levels than the previous decade, recruitment appears to follow declining trends. Therefore the robustness of the reference points, which practically assume constant recruitment around the average of the entire time

series, is questionable. Given, that the model runs do not allow proper fit of a stock-recruitment relationship, it might be better to assume constant recruitment levels close to those observed in the last 10-15 years.

- Alternatively, it could be examined the adoption of empirical reference points based on the biomass and mortality estimates derived from the assessments. For instance, -based on past works dealing with similar conditions (see Gabriel and Mace 1999 for a review)- the 10, 50, and 90 percentiles of the SSB and F time series obtained from the XSA runs could provide proxies for SSB_{MSY} and F_{MSY} .

References

- Gabriel W.L and Mace P.M 1999. A Review of Biological Reference Points in the Context of the Precautionary Approach Proceedings, 5th NMFS NSAW. 1999. NOAA Tech. Memo. NMFS-F/SPO-40.
- ICCAT, 2015. Report of the 2014 ICCAT Mediterranean swordfish stock assessment meeting (Heraklion, Greece, 21–25 July 2014). Collect. Vol. Sci. Pap. ICCAT, 71(5): 1870–1979.
- ICCAT, 2017. Report of the 2016 ICCAT Mediterranean swordfish stock assessment meeting (Casablanca, Morocco, 11–16 July 2016). Collect. Vol. Sci. Pap. ICCAT, 73(3): 1005–1096.
- ICCAT, 2020. REPORT OF THE 2020 ICCAT INTERSESSIONAL MEETING OF THE SWORDFISH SPECIES GROUP. <https://tinyurl.com/ycqfhjzl>.
- Kell, L.T., Mosqueira, I., Grosjean, P., Fromentin, J.M., Garcia, D., Hillary, R., Jardim, E., Mardle, S., Pastoors, M.A., Poos, J.J. and Scott, F., 2007. FLR: an open-source framework for the evaluation and development of management strategies. ICES Journal of Marine Science: Journal du Conseil, 64(4), pp.640-646.
- SCRS 2020/021. UPDATED STANDARDIZED SWORDFISH CATCH RATES FROM THE GREEK SURFACE LONGLINE FISHERIES OPERATING IN THE E. MEDITERRANEAN.
- SCRS 2020/026. UPDATED CATCH RATES OF SWORDFISH (*XIPHIAS GLADIUS*) CAUGHT BY THE MOROCCAN LONGLINE FLEET IN THE MEDITERRANEAN SEA, 2012-2019
- SCRS 2020/027. AN UPDATE OF THE SWORDFISH FISHERY IN THE LIGURIAN SEA (WESTERN MEDITERRANEAN) WITH A PRELIMINARY ATTEMPT TO STANDARDIZE THE MESOPELAGIC LONGLINE CPUEs.
- SCRS 2020/028. ESTIMATION OF UNDERSIZED MEDITERRANEAN SWORDFISH (*Xiphias gladius*) CATCHES BETWEEN 2008 – 2018 FOR THE LONGLINE MAIN FLEETS.
- SCRS 2020/043. Standardized Catch Rates for the Mediterranean Swordfish (*Xiphias gladius* Linnaeus, 1758) from the Spanish Longline Fishery. 1988-2018.
- Shepherd, J. (1999). Extended survivor's analysis: An improved method for the analysis of catch-at-age data and abundance indices. ICES Journal of Marine Science, 56(5), 584–591.